

Submission of project

Title: Securing Linux Servers Using Honeypots and IP Blocking

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Date: 06/10/2025

Objective

To deploy a honeypot and configure defensive mechanisms — including SSH hardening and automated IP blocking — to detect, analyze, and mitigate SSH brute-force attacks on public-facing Linux servers.

Executive summary

This project deploys a public-facing Linux honeypot and implements defensive controls to detect, analyze, and automatically mitigate SSH brute-force attacks. The honeypot (a CentOS virtual machine) runs a deliberately exposed service and collects attack telemetry. SSH hardening (changing the SSH port, adjusting SELinux, and restarting sshd) and automated IP blocking (scripts that parse failed login attempts and add firewall rich rules) were used to block abusive IPs after they exceeded a configurable threshold. The system produces human-readable reports of attackers and their countries and demonstrates that malicious SSH connection attempts are recorded and blocked.

Solution — implementation overview

Environment & Components

- A CentOS virtual machine was deployed to act as the honeypot (VM image installed in VMware). Web (Apache/httpd), FTP (vsftpd), and SSH services were installed/configured to simulate a public server for attackers to probe.
- A Kali Linux VM was used as an attacker machine to generate and verify brute-force SSH attempts.

Key configuration changes

1. SSH hardening

- Changed default SSH port from 22 to 2222 by editing `/etc/ssh/sshd_config` and restarting sshd.
- Updated SELinux to allow SSH on the new port (`semanage port -a -t ssh_port_t -p tcp 2222`) and restarted the service.

2. Honeypot / web interface

- Apache virtual host `testsite.mydomain.local` created at `/var/www/testsite.mydomain.local` and exposed port 80. A `/reports` directory was provided where automated reports are written (e.g., `/reports/ssh_failed_logins_report.txt`). SELinux file contexts were set for `/reports` and the log file with `chcon`.

3. Service hardening

- FTP write and local login disabled in `/etc/vsftpd/vsftpd.conf` (e.g., `local_enable=NO`, `write_enable=NO`) to reduce attack surface. sshd, httpd, and vsftpd were enabled with `systemctl enable` and started.

4. Firewall

- `firewalld` rules were added for required ports (80, 2222, 20, 21) and reloaded. Blocking of malicious IPs is done using `firewall-cmd --permanent --add-rich-rule="rule family='ipv4' source address='x.x.x.x' reject" then firewall-cmd --reload.`

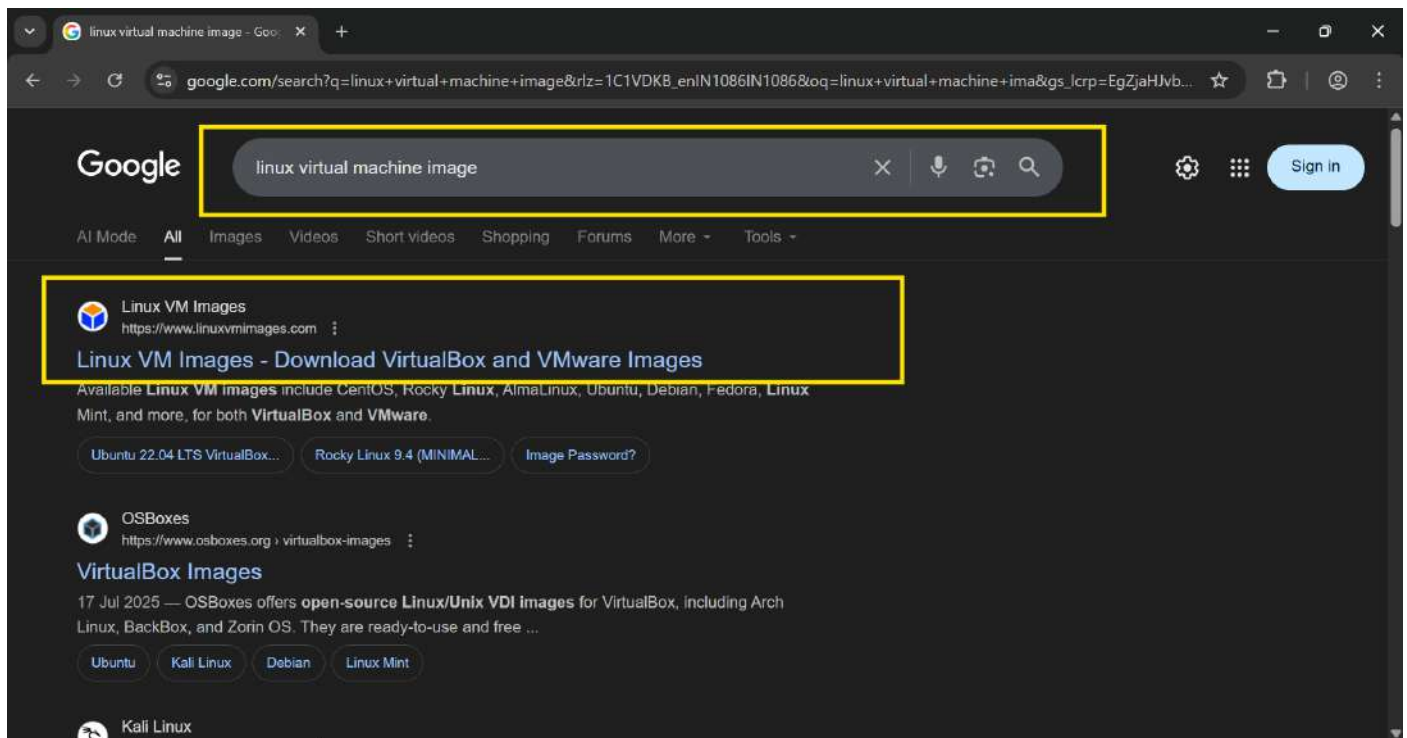
Detection & automation scripts

- **/root/b_track_ssh_daily** — a shell script scheduled via crontab -e (@daily /root/b_track_ssh_daily) to parse /var/log/secure*, extract unique IPs with failed SSH logins, run whois lookups to map countries and compose a 3.htm/report file which gets appended to /reports/ssh_failed_logins_report.txt. This script collects attacker IPs and produces a readable report for review.
- **/root/protect_ssh** — a blocking script that:
 1. Extracts IPs from failed SSH logins (from /var/log/secure),
 2. Counts failed attempts per IP,
 3. For IPs with attempts $\geq$ a threshold (script used 10 in this project; described threshold behavior can be adjusted), adds a permanent firewall rule to reject traffic from that IP,
 4. Reloads the firewall once after processing.The script is made executable and run manually (or could be added to cron) to enforce blocking.

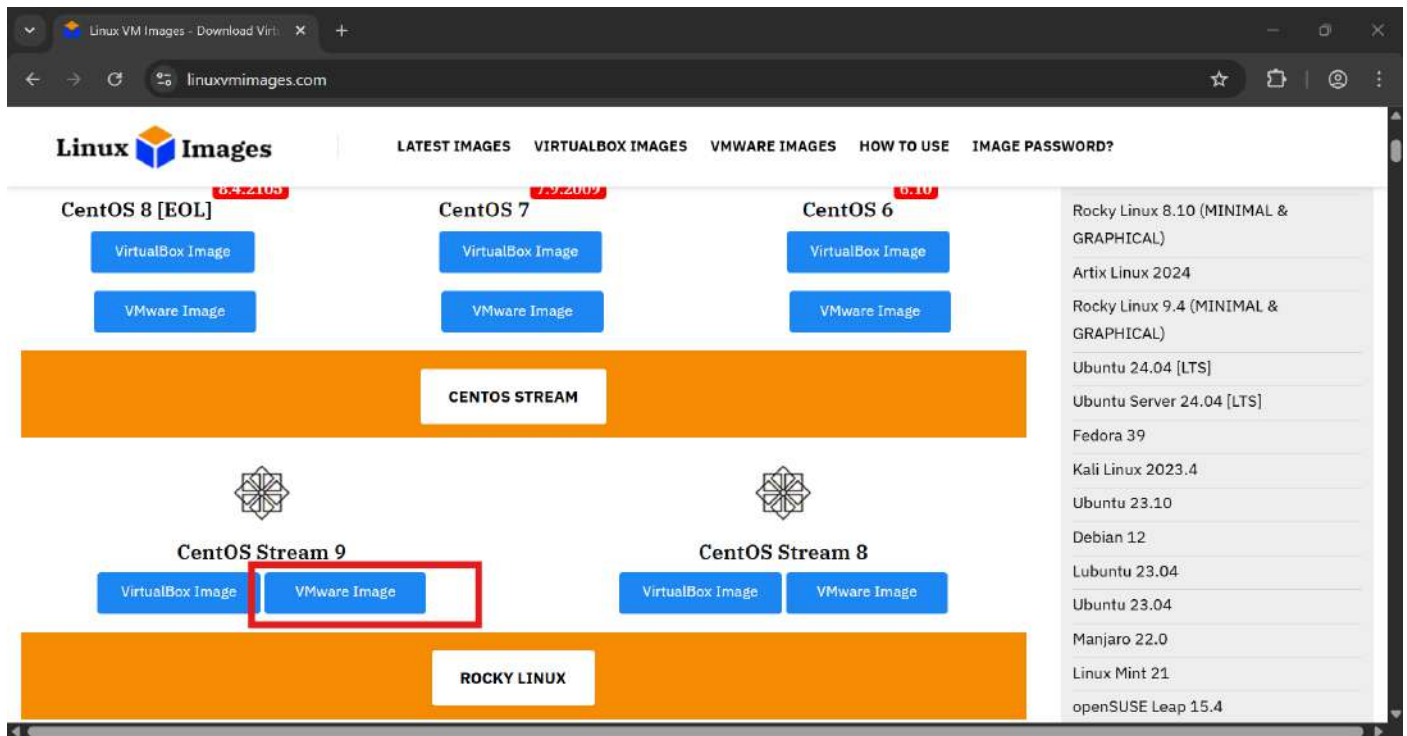
Task 1: Deployment of Honeypot

1.1 Installing CentOS

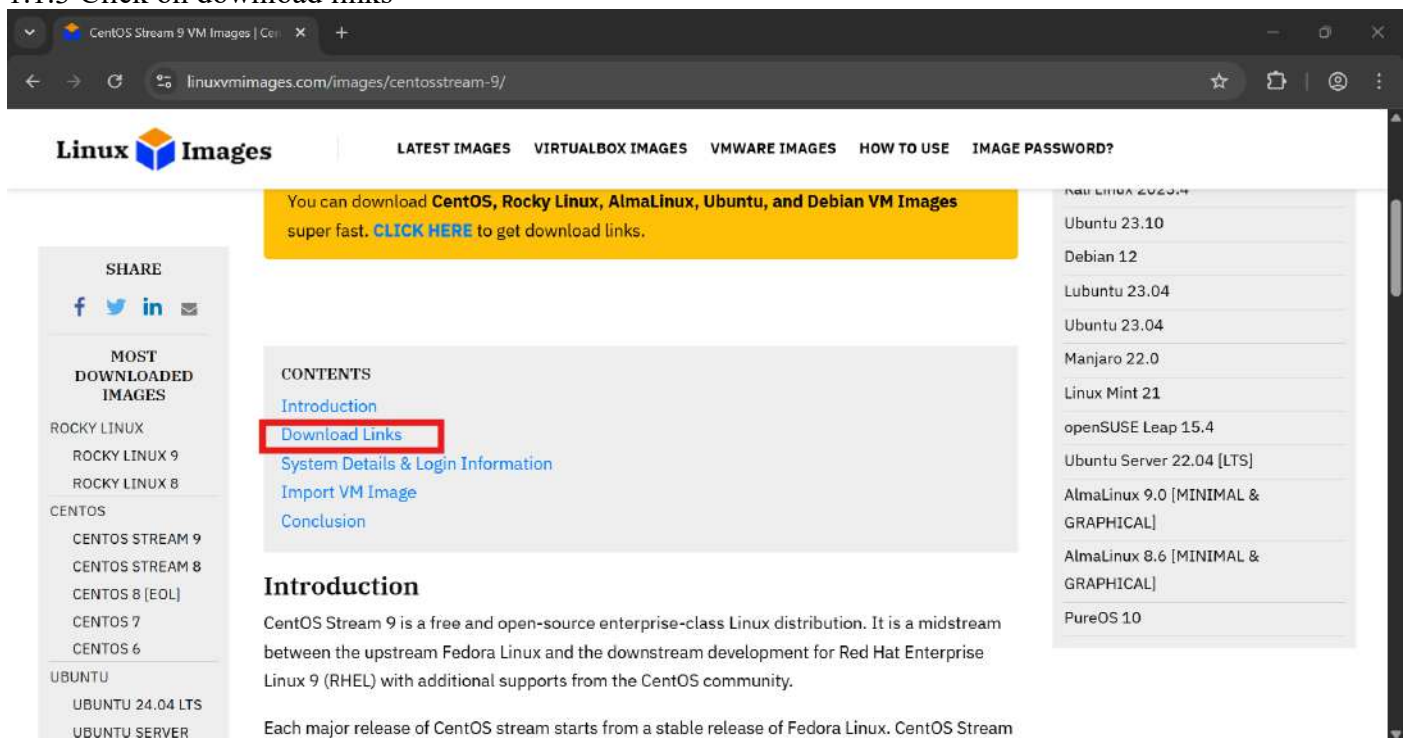
1.1.1 Opened the browser and searched for linux virtual machine image and clicked on [Linux VM Images - Download VirtualBox and VMware Images](#)



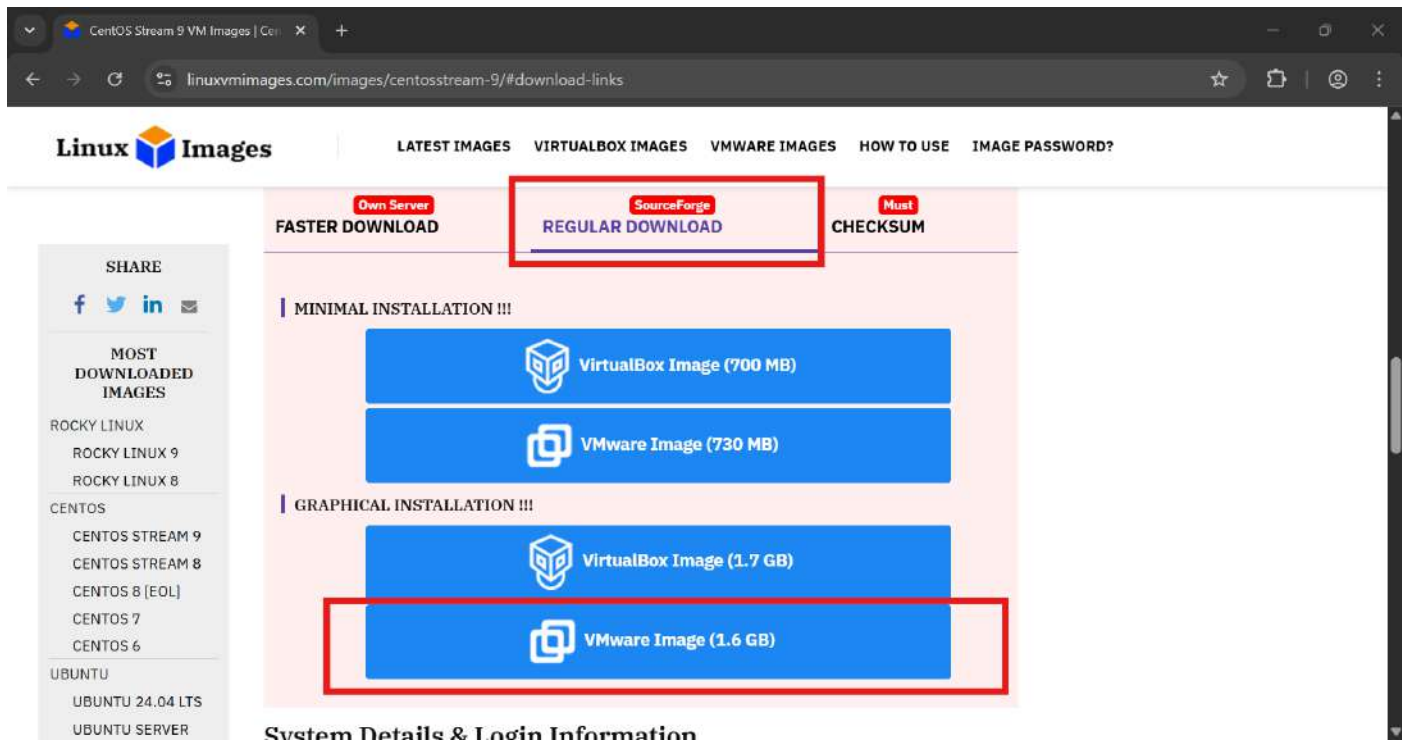
1.1.2 Click on VMware Image



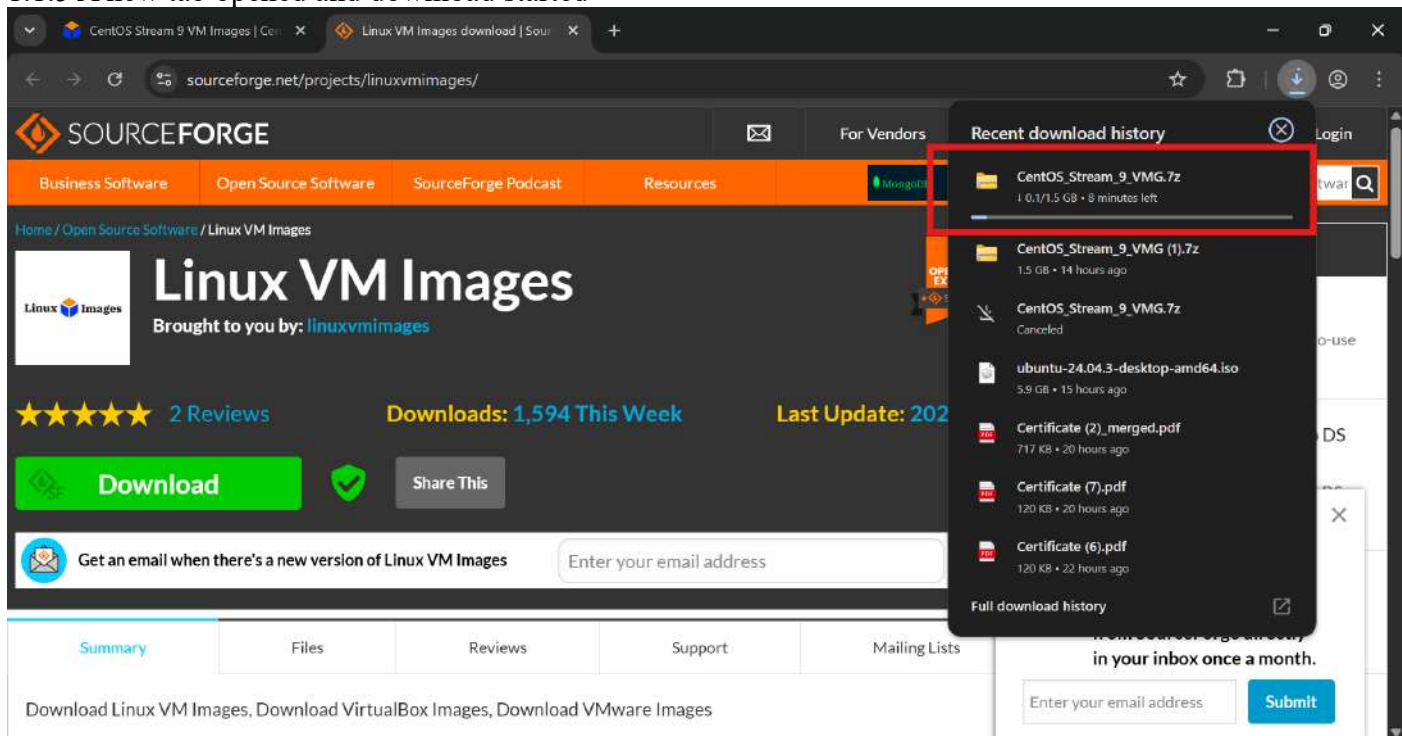
1.1.3 Click on download links



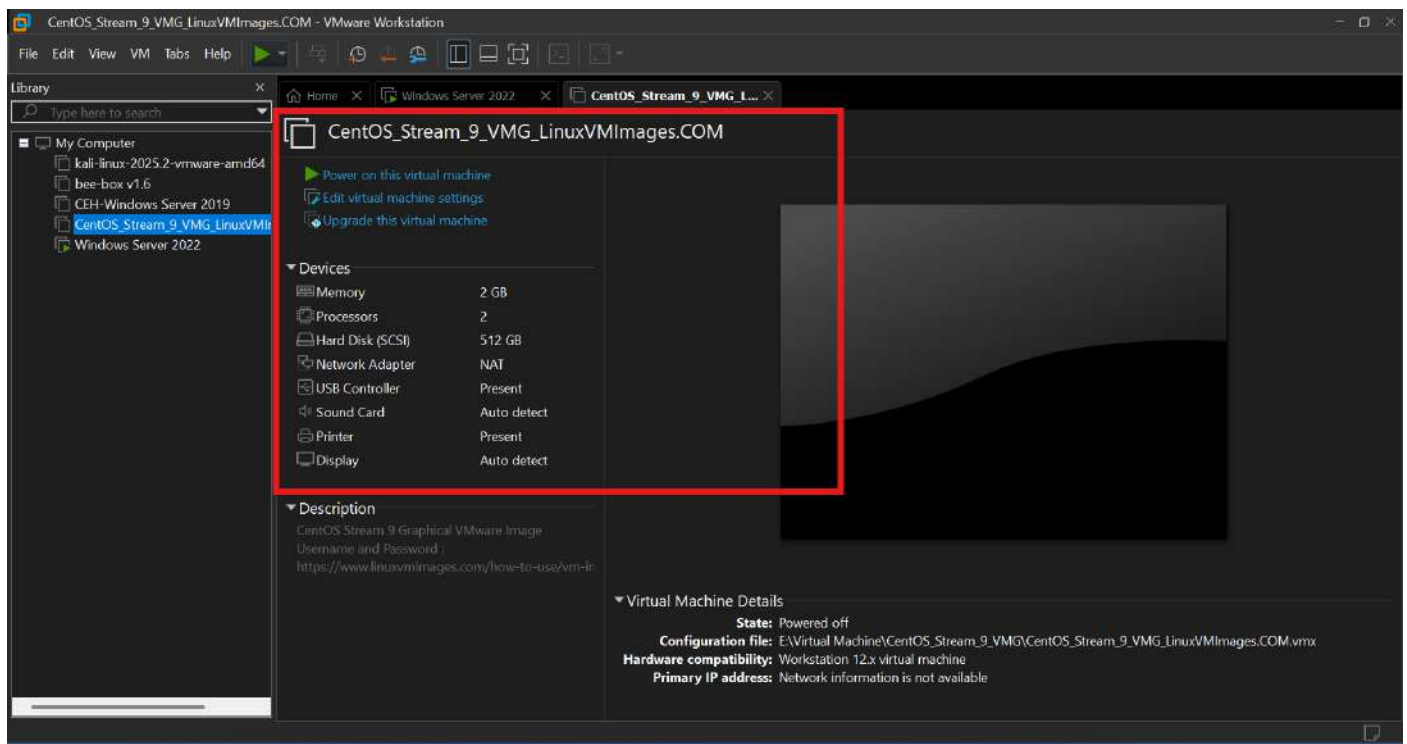
1.1.4 Click on Regular download and then VMware Image



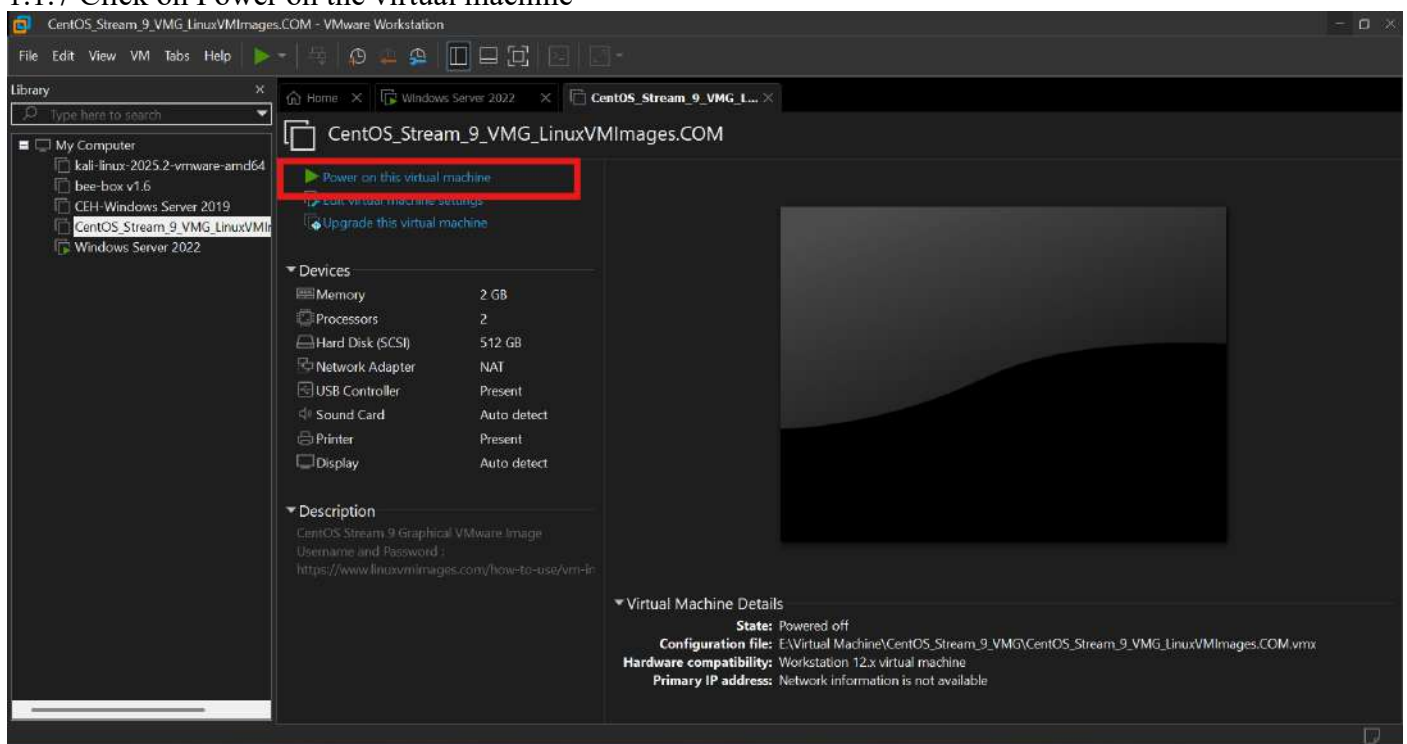
1.1.5 A new tab opened and download started



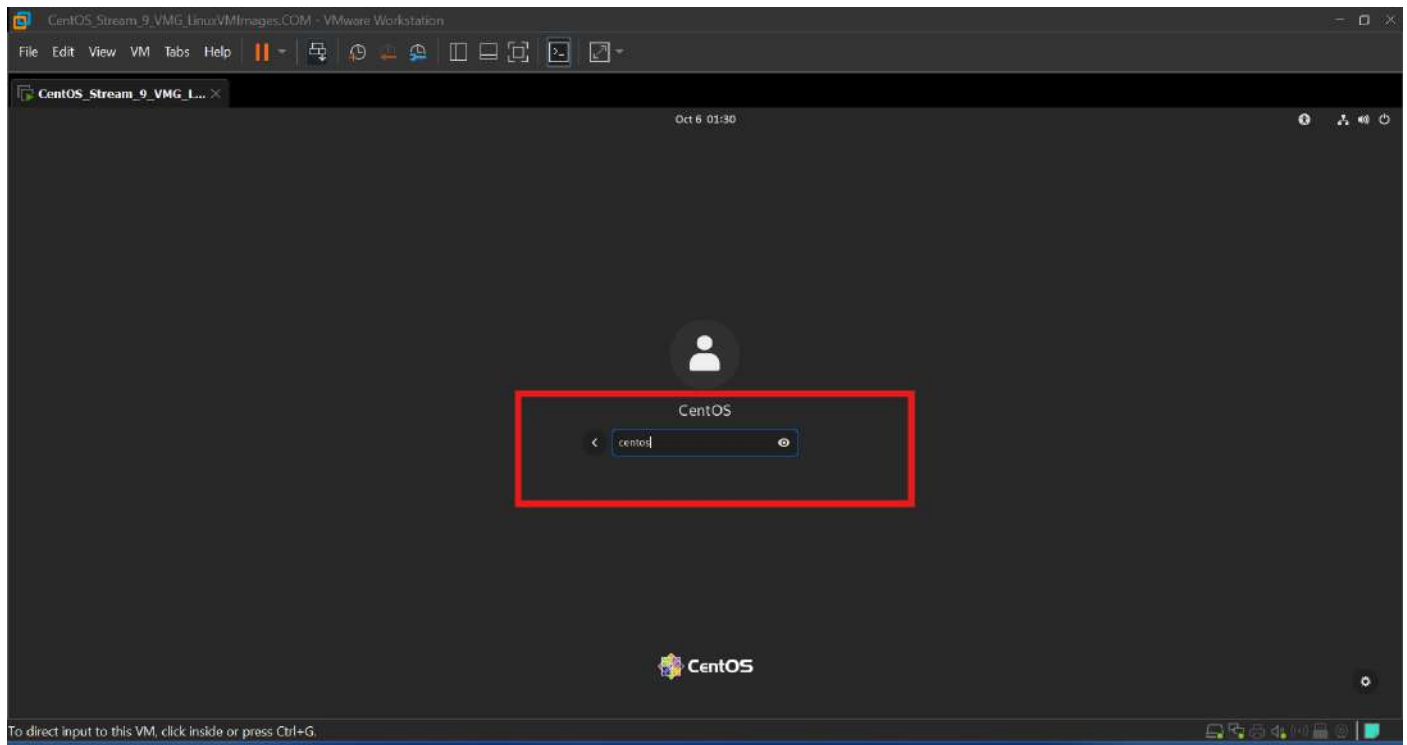
1.1.6 Extracted the file and installed the CentOS in VMware Workstation



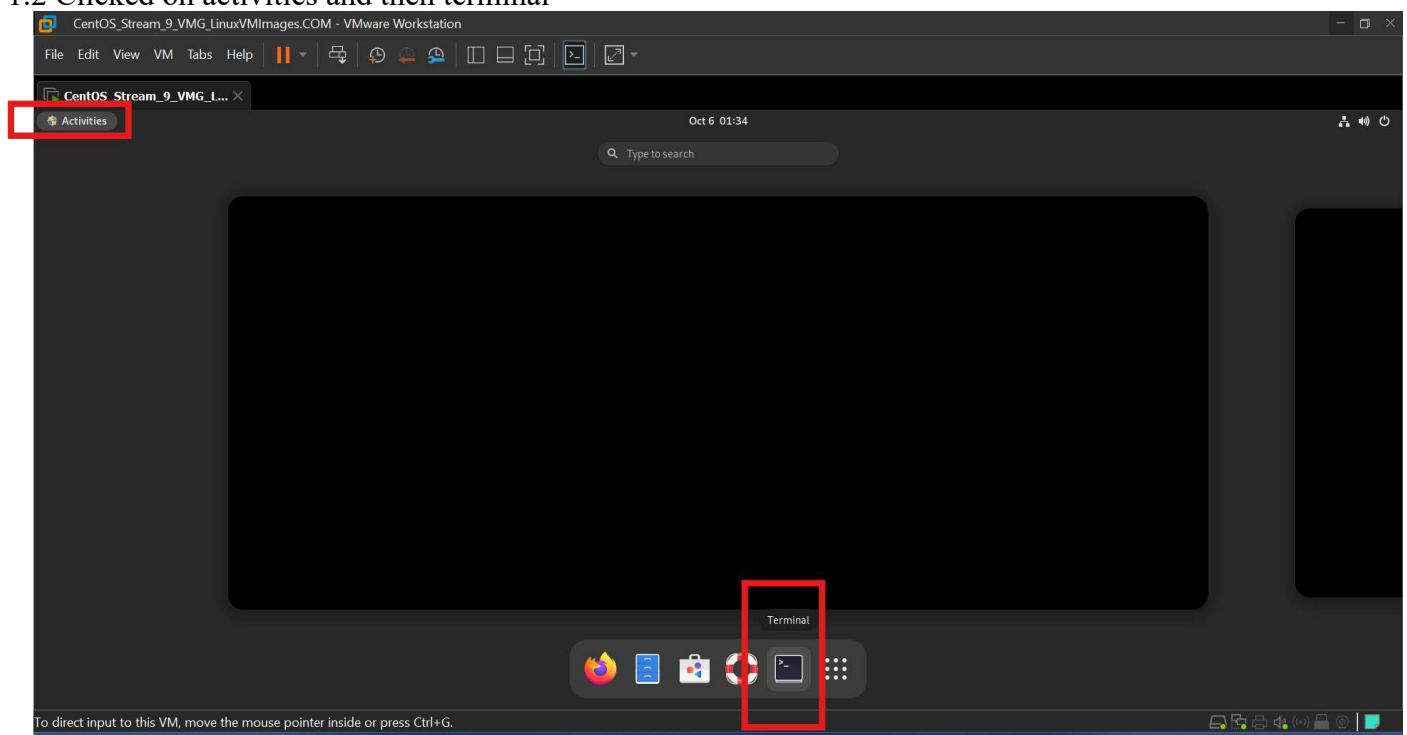
1.1.7 Click on Power on the virtual machine



1.1.8 Entered the password **centos** and login



1.2 Clicked on activities and then terminal



1.3 Used command: **sudo su** and then **yum update**


```
CentOS_Stream_9_VMG_LinuxVMImages.COM - VMware Workstation
File Edit View VM Tabs Help
CentOS_Stream_9_VMG_L... x
Activities Terminal Oct 6 01:41
root@centosstream9:/home/centos

[centos@centosstream9 ~]$ sudo su
[root@centosstream9 centos]# yum update
CentOS Stream 9 - BaseOS
CentOS Stream 9 - AppStream
Last metadata expiration check: 0:00:03 ago on Mon 06 Oct 2025 01:41:06 AM EDT.
Dependencies resolved.

=====
Package                                Architecture Version                                Repository                                Size
=====
Installing:
kernel                                x86_64      5.14.0-620.el9                                baseos                                59 k
Upgrading:
ModemManager                          x86_64      1.20.2-1.el9                                baseos                                1.3 M
ModemManager-glib                     x86_64      1.20.2-1.el9                                baseos                                335 k
NetworkManager                       x86_64      1:1.54.1-1.el9                              baseos                                2.4 M
NetworkManager-adsl                   x86_64      1:1.54.1-1.el9                              baseos                                33 k
NetworkManager-bluetooth              x86_64      1:1.54.1-1.el9                              baseos                                59 k
NetworkManager-config-server          noarch      1:1.54.1-1.el9                              baseos                                19 k
NetworkManager-libnm                  x86_64      1:1.54.1-1.el9                              baseos                                1.9 M
NetworkManager-team                   x86_64      1:1.54.1-1.el9                              baseos                                38 k
NetworkManager-tui                    x86_64      1:1.54.1-1.el9                              baseos                                248 k
NetworkManager-wifi                   x86_64      1:1.54.1-1.el9                              baseos                                82 k
NetworkManager-wwan                   x86_64      1:1.54.1-1.el9                              baseos                                66 k
PackageKit                            x86_64      1.2.6-1.el9                                appstream                             646 k
PackageKit-command-not-found           x86_64      1.2.6-1.el9                                appstream                             22 k
PackageKit-glib                        x86_64      1.2.6-1.el9                                appstream                             157 k
PackageKit-gstreamer-plugin            x86_64      1.2.6-1.el9                                appstream                             17 k
PackageKit-gtk3-module                 x86_64      1.2.6-1.el9                                appstream                             16 k
accountsservice                        x86_64      0.6.55-10.el9                              appstream                             130 k
=====

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
```

1.4 Installing openssh-server using the command: `yum install openssh-server`

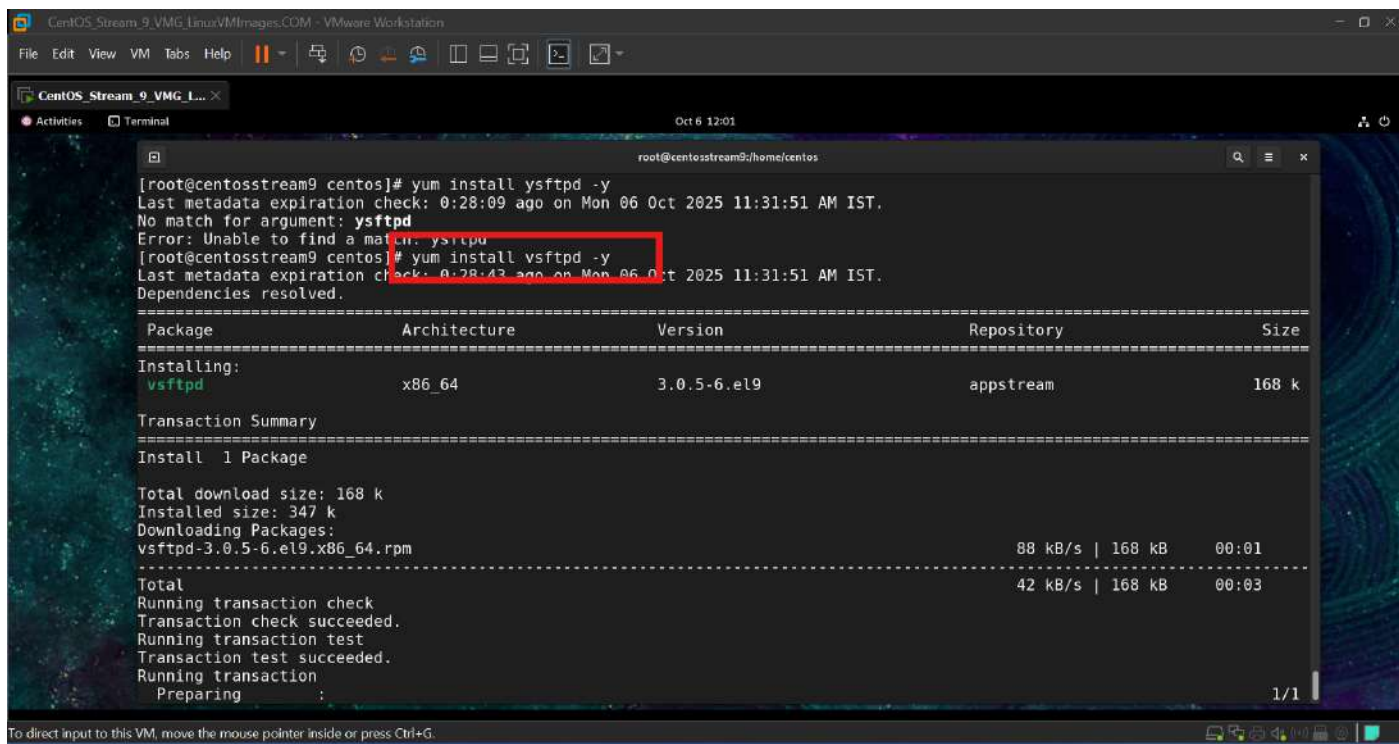
```
CentOS_Stream_9_VMG_LinuxVMImages.COM - VMware Workstation
File Edit View VM Tabs Help
CentOS_Stream_9_VMG_L... x
Activities Terminal Oct 6 11:55
root@centosstream9:/home/centos

evolution-data-server-ui-3.40.4-10.el9.x86_64
gnome-shell-extension-desktop-icons-40.7-28.el9.noarch
grub2-tools-extra-1:2.06-115.el9.x86_64
kbd-legacy-2.4.0-11.el9.noarch
kernel-core-5.14.0-620.el9.x86_64
kernel-modules-core-5.14.0-620.el9.x86_64
ledmon-libs-1.1.0-3.el9.x86_64
libatomic-11.5.0-11.el9.x86_64
libnvme-1.13-1.el9.x86_64
libproxy-webkitgtk4-0.4.15-35.el9.x86_64
liburing-2.5-1.el9.x86_64
libxcvt-0.1.2-2.el9.x86_64
netavark-2:1.16.0-1.el9.x86_64
passt-0*20250512.g8ec1341-2.el9.x86_64
pipewire-jack-audio-connection-kit-libs-1.0.1-1.el9.x86_64
python3-file-magic-5.39-16.el9.noarch
python3-pyyaml-5.4.1-6.el9.x86_64
systemd-boot-unsigned-252-55.el9.x86_64

gnome-shell-extension-background-logo-40.0-rc-4.el9.noarch
grub2-tools-efi-1:2.06-115.el9.x86_64
gstreamer1-plugins-bad-free-libs-1.22.12-4.el9.x86_64
kernel-5.14.0-620.el9.x86_64
kernel-modules-5.14.0-620.el9.x86_64
keyutils-1.6.3-1.el9.x86_64
libadwaita-1.4.2-2.el9.x86_64
libdecor-0.1.1-1.el9.x86_64
libproxy-0.4.15-35.el9.x86_64
libqrtr-glib-1.2.2-1.el9.x86_64
libva-2.22.0-1.el9.x86_64
llvm-filesystem-20.1.8-3.el9.x86_64
openssl-fips-provider-1:3.5.1-5.el9.x86_64
passt-selinux-0*20250512.g8ec1341-2.el9.noarch
python3-distro-1.5.0-7.el9.noarch
python3-gobject-base-noarch-3.40.1-6.el9.noarch
spirv-tools-libs-2025.2-1.el9.x86_64
xdg-desktop-portal-gnome-41.2-3.el9.x86_64

Complete!
[root@centosstream9 centos]#
[root@centosstream9 centos]#
[root@centosstream9 centos]# yum install openssh-server
Last metadata expiration check: 0:00:55 ago on Mon 06 Oct 2025 11:31:51 AM IST.
Package openssh-server-9.9p1-1.el9.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@centosstream9 centos]#
```

1.5 Installing vsftpd using the command: `yum install vsftpd -y`



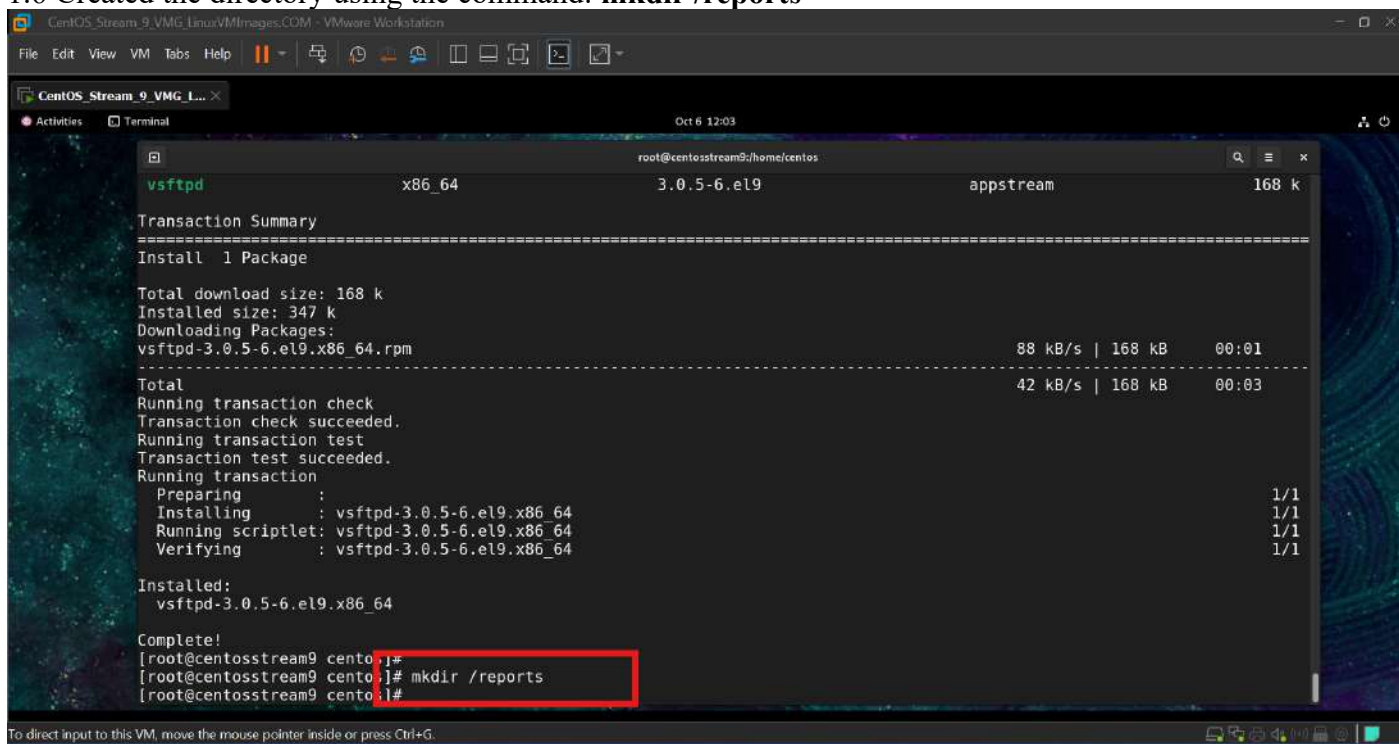
```
[root@centosstream9 centos]# yum install ysftpd -y
Last metadata expiration check: 0:28:09 ago on Mon 06 Oct 2025 11:31:51 AM IST.
No match for argument: ysftpd
Error: Unable to find a match: ysftpd
[root@centosstream9 centos]# yum install vsftpd -y
Last metadata expiration check: 0:28:43 ago on Mon 06 Oct 2025 11:31:51 AM IST.
Dependencies resolved.

=====
Package                        Architecture      Version           Repository        Size
=====
Installing:
vsftpd                        x86_64            3.0.5-6.el9       appstream          168 k

Transaction Summary
=====
Install 1 Package

Total download size: 168 k
Installed size: 347 k
Downloading Packages:
vsftpd-3.0.5-6.el9.x86_64.rpm                                88 kB/s | 168 kB    00:01
-----
Total
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing :
1/1
```

1.6 Created the directory using the command: **mkdir /reports**



```
vsftpd                        x86_64            3.0.5-6.el9       appstream          168 k

Transaction Summary
=====
Install 1 Package

Total download size: 168 k
Installed size: 347 k
Downloading Packages:
vsftpd-3.0.5-6.el9.x86_64.rpm                                88 kB/s | 168 kB    00:01
-----
Total
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing :
Installing : vsftpd-3.0.5-6.el9.x86_64
Running scriptlet: vsftpd-3.0.5-6.el9.x86_64
Verifying : vsftpd-3.0.5-6.el9.x86_64
1/1
1/1
1/1
1/1

Installed:
vsftpd-3.0.5-6.el9.x86_64

Complete!
[root@centosstream9 centos]#
[root@centosstream9 centos]# mkdir /reports
[root@centosstream9 centos]#
```

1.7 Installd the httpd using the command: **yum install httpd**


```
[root@centosstream9 centos]# mkdir /reports
[root@centosstream9 centos]# yum install httpd
Last metadata expiration check: 0:33:21 ago on Mon 06 Oct 2025 11:31:51 AM IST.
Dependencies resolved.
=====
Package                               Architecture      Version           Repository        Size
=====
Installing:
httpd                                 x86_64            2.4.62-7.el9      appstream         46 k
Installing dependencies:
apr                                   x86_64            1.7.0-12.el9      appstream         123 k
apr-util                             x86_64            1.6.1-23.el9      appstream         95 k
apr-util-bdb                         x86_64            1.6.1-23.el9      appstream         13 k
centos-logos-httpd                  noarch            90.8-3.el9        appstream         1.5 M
httpd-core                           x86_64            2.4.62-7.el9      appstream         1.5 M
httpd-filesystem                    noarch            2.4.62-7.el9      appstream         11 k
httpd-tools                          x86_64            2.4.62-7.el9      appstream         80 k
Installing weak dependencies:
apr-util-openssl                    x86_64            1.6.1-23.el9      appstream         15 k
mod_http2                           x86_64            2.0.26-5.el9      appstream         163 k
mod_lua                              x86_64            2.4.62-7.el9      appstream         58 k
=====
Transaction Summary
=====
Install 11 Packages

Total download size: 3.6 M
Installed size: 8.6 M
Is this ok [y/N]:
```

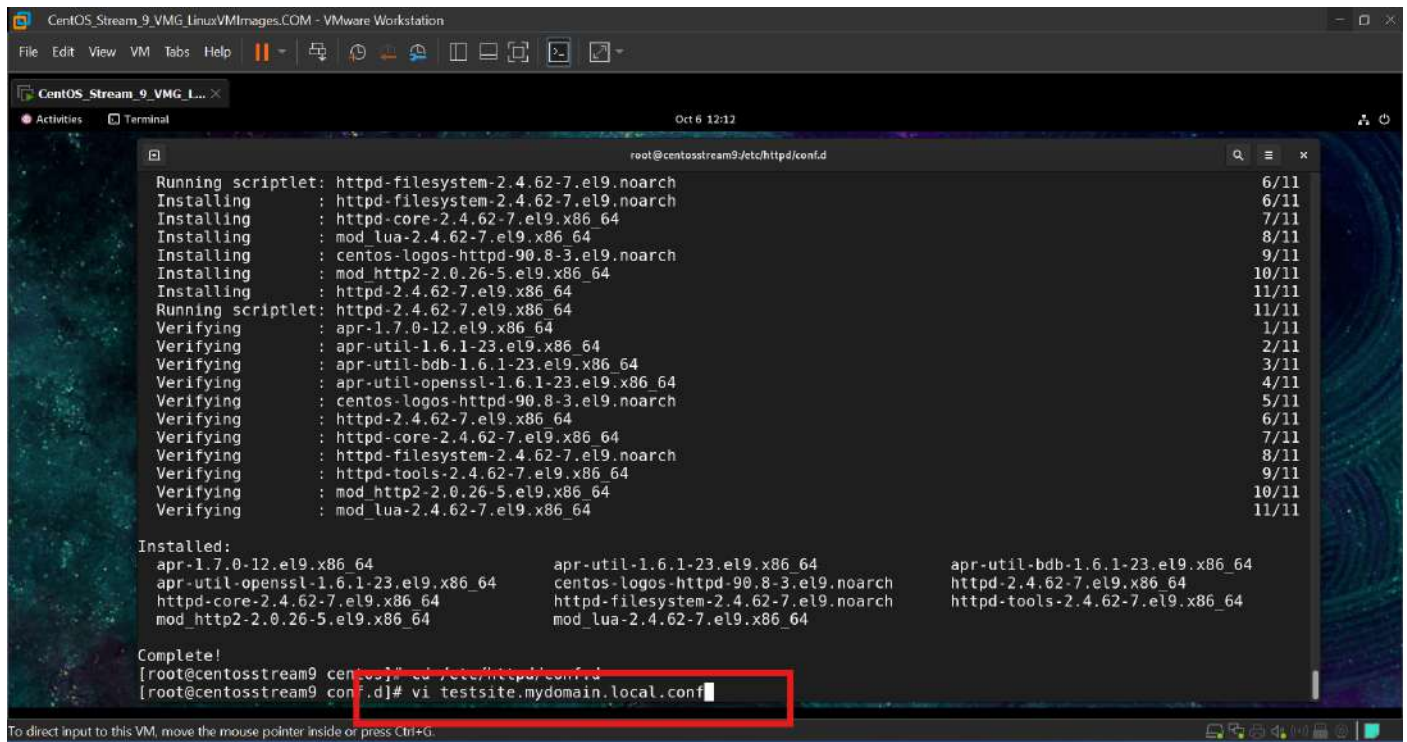
1.8 Changed the directory using the command: `cd /etc/httpd/conf.d`

```
Running scriptlet: httpd-filesystem-2.4.62-7.el9.noarch 6/11
Installing      : httpd-filesystem-2.4.62-7.el9.noarch 6/11
Installing      : httpd-core-2.4.62-7.el9.x86_64      7/11
Installing      : mod_lua-2.4.62-7.el9.x86_64         8/11
Installing      : centos-logos-httpd-90.8-3.el9.noarch 9/11
Installing      : mod_http2-2.0.26-5.el9.x86_64       10/11
Installing      : httpd-2.4.62-7.el9.x86_64           11/11
Running scriptlet: httpd-2.4.62-7.el9.x86_64          11/11
Verifying       : apr-1.7.0-12.el9.x86_64             1/11
Verifying       : apr-util-1.6.1-23.el9.x86_64        2/11
Verifying       : apr-util-bdb-1.6.1-23.el9.x86_64    3/11
Verifying       : apr-util-openssl-1.6.1-23.el9.x86_64 4/11
Verifying       : centos-logos-httpd-90.8-3.el9.noarch 5/11
Verifying       : httpd-2.4.62-7.el9.x86_64           6/11
Verifying       : httpd-core-2.4.62-7.el9.x86_64      7/11
Verifying       : httpd-filesystem-2.4.62-7.el9.noarch 8/11
Verifying       : httpd-tools-2.4.62-7.el9.x86_64     9/11
Verifying       : mod_http2-2.0.26-5.el9.x86_64       10/11
Verifying       : mod_lua-2.4.62-7.el9.x86_64         11/11

Installed:
apr-1.7.0-12.el9.x86_64      apr-util-1.6.1-23.el9.x86_64      apr-util-bdb-1.6.1-23.el9.x86_64
apr-util-openssl-1.6.1-23.el9.x86_64 centos-logos-httpd-90.8-3.el9.noarch httpd-2.4.62-7.el9.x86_64
httpd-core-2.4.62-7.el9.x86_64 httpd-filesystem-2.4.62-7.el9.noarch httpd-tools-2.4.62-7.el9.x86_64
mod_http2-2.0.26-5.el9.x86_64 mod_lua-2.4.62-7.el9.x86_64

Complete!
[root@centosstream9 centos]# cd /etc/httpd/conf.d
[root@centosstream9 conf.d]#
```

1.9 Opening the **vi** editor using the command: `vi testsite.mydomain.local.conf`



```
CentOS_Stream_9_VMG_LinuxVMImages.COM - VMware Workstation
File Edit View VM Tabs Help
CentOS_Stream_9_VMG_L... x
Activities Terminal Oct 6 12:12
root@centosstream9:/etc/httpd/conf.d

Running scriptlet: httpd-filesystem-2.4.62-7.el9.noarch 6/11
Installing      : httpd-filesystem-2.4.62-7.el9.noarch 6/11
Installing      : httpd-core-2.4.62-7.el9.x86_64      7/11
Installing      : mod_lua-2.4.62-7.el9.x86_64         8/11
Installing      : centos-logos-httpd-90.8-3.el9.noarch 9/11
Installing      : mod_http2-2.0.26-5.el9.x86_64       10/11
Installing      : httpd-2.4.62-7.el9.x86_64          11/11
Running scriptlet: httpd-2.4.62-7.el9.x86_64          11/11
Verifying       : apr-1.7.0-12.el9.x86_64            1/11
Verifying       : apr-util-1.6.1-23.el9.x86_64       2/11
Verifying       : apr-util-bdb-1.6.1-23.el9.x86_64  3/11
Verifying       : apr-util-openssl-1.6.1-23.el9.x86_64 4/11
Verifying       : centos-logos-httpd-90.8-3.el9.noarch 5/11
Verifying       : httpd-2.4.62-7.el9.x86_64         6/11
Verifying       : httpd-core-2.4.62-7.el9.x86_64    7/11
Verifying       : httpd-filesystem-2.4.62-7.el9.noarch 8/11
Verifying       : httpd-tools-2.4.62-7.el9.x86_64   9/11
Verifying       : mod_http2-2.0.26-5.el9.x86_64     10/11
Verifying       : mod_lua-2.4.62-7.el9.x86_64      11/11

Installed:
apr-1.7.0-12.el9.x86_64      apr-util-1.6.1-23.el9.x86_64      apr-util-bdb-1.6.1-23.el9.x86_64
apr-util-openssl-1.6.1-23.el9.x86_64  centos-logos-httpd-90.8-3.el9.noarch  httpd-2.4.62-7.el9.x86_64
httpd-core-2.4.62-7.el9.x86_64  httpd-filesystem-2.4.62-7.el9.noarch  httpd-tools-2.4.62-7.el9.x86_64
mod_http2-2.0.26-5.el9.x86_64  mod_lua-2.4.62-7.el9.x86_64

Complete!
[root@centosstream9 centos]# vi testsite.mydomain.local.conf
```

1.10 Pressed i to enter in editor mode and written this script
#domain=testsite.mydomain.local

<Directory "/var/www/testsite.mydomain.local">

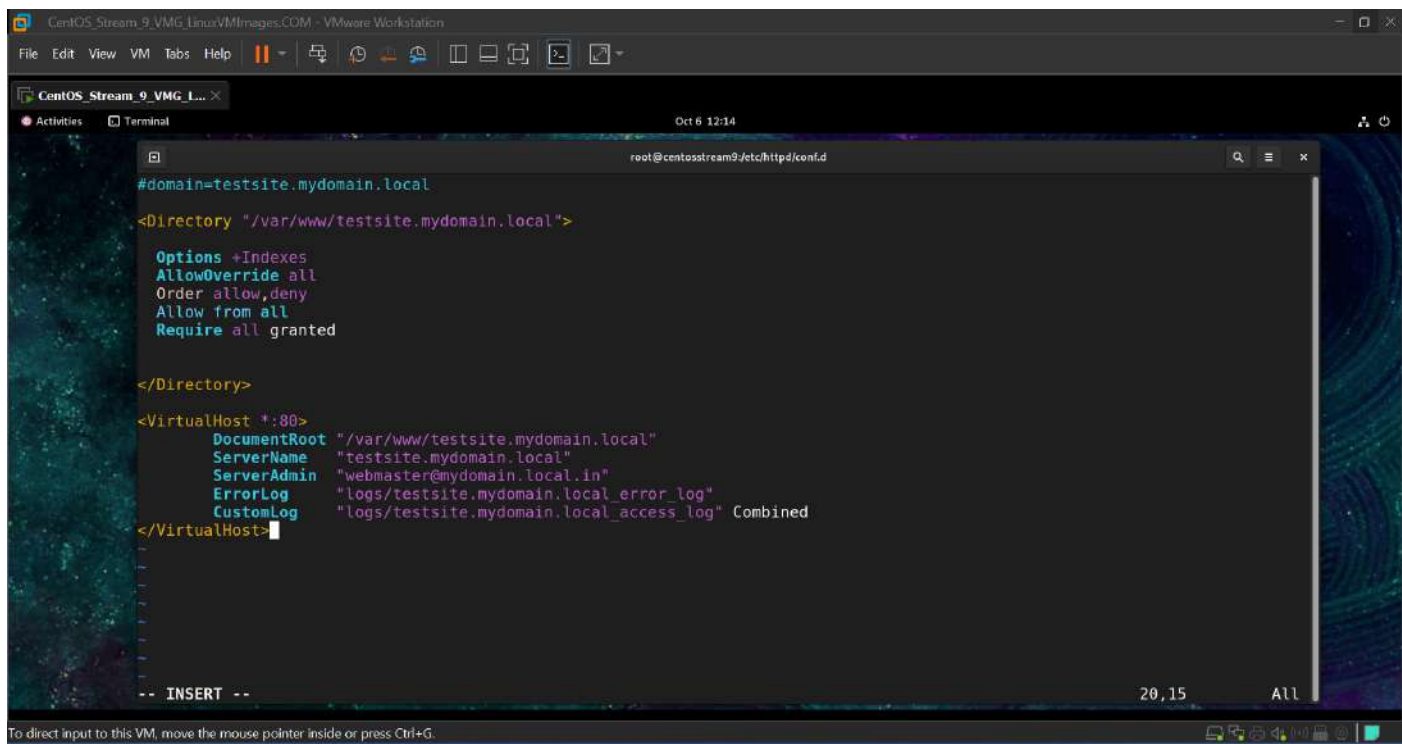
Options +Indexes
AllowOverride all
Order allow,deny
Allow from all
Require all granted

</Directory>

<VirtualHost *:80>

DocumentRoot "/var/www/testsite.mydomain.local"
ServerName "testsite.mydomain.local"
ServerAdmin "webmaster@mydomain.local.in"
ErrorLog "logs/testsite.mydomain.local_error_log"
CustomLog "logs/testsite.mydomain.local_access_log" Combined

</VirtualHost>



CentOS_Stream_9_VMG_LinuxVMImages.COM - VMware Workstation

File Edit View VM Tabs Help

CentOS_Stream_9_VMG_L... x

Activities Terminal

Oct 6 12:14

root@centosstream9:/etc/httpd/conf.d

```
#domain=testsite.mydomain.local

<Directory "/var/www/testsite.mydomain.local">

    Options +Indexes
    AllowOverride all
    Order allow,deny
    Allow from all
    Require all granted

</Directory>

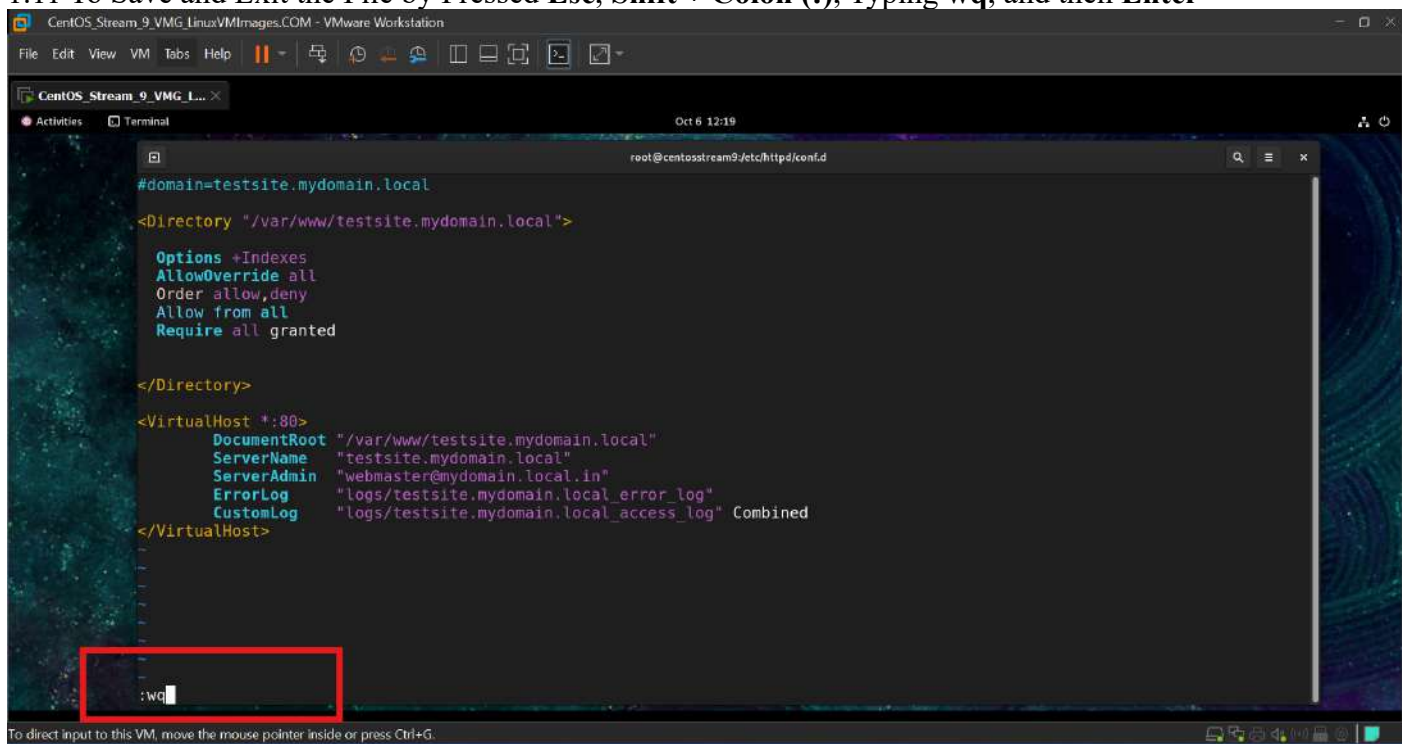
<VirtualHost *:80>
    DocumentRoot "/var/www/testsite.mydomain.local"
    ServerName      "testsite.mydomain.local"
    ServerAdmin     "webmaster@mydomain.local.in"
    ErrorLog        "logs/testsite.mydomain.local_error_log"
    CustomLog       "logs/testsite.mydomain.local_access_log" Combined
</VirtualHost>
```

-- INSERT --

20,15 All

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

1.11 To Save and Exit the File by Pressed **Esc**, **Shift + Colon (:)**, Typing **wq**, and then **Enter**



CentOS_Stream_9_VMG_LinuxVMImages.COM - VMware Workstation

File Edit View VM Tabs Help

CentOS_Stream_9_VMG_L... x

Activities Terminal

Oct 6 12:19

root@centosstream9:/etc/httpd/conf.d

```
#domain=testsite.mydomain.local

<Directory "/var/www/testsite.mydomain.local">

    Options +Indexes
    AllowOverride all
    Order allow,deny
    Allow from all
    Require all granted

</Directory>

<VirtualHost *:80>
    DocumentRoot "/var/www/testsite.mydomain.local"
    ServerName      "testsite.mydomain.local"
    ServerAdmin     "webmaster@mydomain.local.in"
    ErrorLog        "logs/testsite.mydomain.local_error_log"
    CustomLog       "logs/testsite.mydomain.local_access_log" Combined
</VirtualHost>
```

:wq

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

1.12 Changed the directory using the command: **cd /var/www**


```
CentOS_Stream_9_VMG_LinuxVMImages.COM - VMware Workstation
File Edit View VM Tabs Help
CentOS_Stream_9_VMG_L... x
Activities Terminal Oct 6 12:20
root@centosstream9:/var/www

Installing      : mod_lua-2.4.62-7.el9.x86_64      8/11
Installing      : centos-logos-httpd-90.8-3.el9.noarch 9/11
Installing      : mod_http2-2.0.26-5.el9.x86_64    10/11
Installing      : httpd-2.4.62-7.el9.x86_64       11/11
Running scriptlet: httpd-2.4.62-7.el9.x86_64      11/11
Verifying       : apr-1.7.0-12.el9.x86_64         1/11
Verifying       : apr-util-1.6.1-23.el9.x86_64    2/11
Verifying       : apr-util-bdb-1.6.1-23.el9.x86_64 3/11
Verifying       : apr-util-openssl-1.6.1-23.el9.x86_64 4/11
Verifying       : centos-logos-httpd-90.8-3.el9.noarch 5/11
Verifying       : httpd-2.4.62-7.el9.x86_64       6/11
Verifying       : httpd-core-2.4.62-7.el9.x86_64   7/11
Verifying       : httpd-filesystem-2.4.62-7.el9.noarch 8/11
Verifying       : httpd-tools-2.4.62-7.el9.x86_64 9/11
Verifying       : mod_http2-2.0.26-5.el9.x86_64    10/11
Verifying       : mod_lua-2.4.62-7.el9.x86_64     11/11

Installed:
apr-1.7.0-12.el9.x86_64      apr-util-1.6.1-23.el9.x86_64      apr-util-bdb-1.6.1-23.el9.x86_64
apr-util-openssl-1.6.1-23.el9.x86_64  centos-logos-httpd-90.8-3.el9.noarch  httpd-2.4.62-7.el9.x86_64
httpd-core-2.4.62-7.el9.x86_64  httpd-filesystem-2.4.62-7.el9.noarch  httpd-tools-2.4.62-7.el9.x86_64
mod_http2-2.0.26-5.el9.x86_64  mod_lua-2.4.62-7.el9.x86_64

Complete!
[root@centosstream9 centos]# cd /etc/httpd/conf.d
[root@centosstream9 conf.d]# vi testsite.mydomain.local.conf
[root@centosstream9 conf.d]# vi testsite.mydomain.local.conf
[root@centosstream9 conf.d]# cd /var/www
[root@centosstream9 www]#
```

1.13 Created a new directory inside www using the command: **mkdir testsite.mydomain.local**

```
CentOS_Stream_9_VMG_LinuxVMImages.COM - VMware Workstation
File Edit View VM Tabs Help
CentOS_Stream_9_VMG_L... x
Activities Terminal Oct 6 12:22
root@centosstream9:/var/www

Installing      : mod_lua-2.4.62-7.el9.x86_64      8/11
Installing      : centos-logos-httpd-90.8-3.el9.noarch 9/11
Installing      : mod_http2-2.0.26-5.el9.x86_64    10/11
Installing      : httpd-2.4.62-7.el9.x86_64       11/11
Running scriptlet: httpd-2.4.62-7.el9.x86_64      11/11
Verifying       : apr-1.7.0-12.el9.x86_64         1/11
Verifying       : apr-util-1.6.1-23.el9.x86_64    2/11
Verifying       : apr-util-bdb-1.6.1-23.el9.x86_64 3/11
Verifying       : apr-util-openssl-1.6.1-23.el9.x86_64 4/11
Verifying       : centos-logos-httpd-90.8-3.el9.noarch 5/11
Verifying       : httpd-2.4.62-7.el9.x86_64       6/11
Verifying       : httpd-core-2.4.62-7.el9.x86_64   7/11
Verifying       : httpd-filesystem-2.4.62-7.el9.noarch 8/11
Verifying       : httpd-tools-2.4.62-7.el9.x86_64 9/11
Verifying       : mod_http2-2.0.26-5.el9.x86_64    10/11
Verifying       : mod_lua-2.4.62-7.el9.x86_64     11/11

Installed:
apr-1.7.0-12.el9.x86_64      apr-util-1.6.1-23.el9.x86_64      apr-util-bdb-1.6.1-23.el9.x86_64
apr-util-openssl-1.6.1-23.el9.x86_64  centos-logos-httpd-90.8-3.el9.noarch  httpd-2.4.62-7.el9.x86_64
httpd-core-2.4.62-7.el9.x86_64  httpd-filesystem-2.4.62-7.el9.noarch  httpd-tools-2.4.62-7.el9.x86_64
mod_http2-2.0.26-5.el9.x86_64  mod_lua-2.4.62-7.el9.x86_64

Complete!
[root@centosstream9 centos]# cd /etc/httpd/conf.d
[root@centosstream9 conf.d]# vi testsite.mydomain.local.conf
[root@centosstream9 conf.d]# vi testsite.mydomain.local.conf
[root@centosstream9 conf.d]# cd /var/www
[root@centosstream9 www]# mkdir testsite.mydomain.local
```

1.14 Changed the directory using the command: **cd testsite.mydomain.local**

```
CentOS_Stream_9_VMG_LinuxVMImages.COM - VMware Workstation
File Edit View VM Tabs Help
CentOS_Stream_9_VMG_L... x
Activities Terminal Oct 6 12:25
root@centosstream9:/var/www/testsite.mydomain.local

Installing      : httpd-2.4.62-7.el9.x86_64 11/11
Running scriptlet: httpd-2.4.62-7.el9.x86_64 11/11
Verifying      : apr-1.7.0-12.el9.x86_64 1/11
Verifying      : apr-util-1.6.1-23.el9.x86_64 2/11
Verifying      : apr-util-bdb-1.6.1-23.el9.x86_64 3/11
Verifying      : apr-util-openssl-1.6.1-23.el9.x86_64 4/11
Verifying      : centos-logos-httpd-90.8-3.el9.noarch 5/11
Verifying      : httpd-2.4.62-7.el9.x86_64 6/11
Verifying      : httpd-core-2.4.62-7.el9.x86_64 7/11
Verifying      : httpd-filesystem-2.4.62-7.el9.noarch 8/11
Verifying      : httpd-tools-2.4.62-7.el9.x86_64 9/11
Verifying      : mod_http2-2.0.26-5.el9.x86_64 10/11
Verifying      : mod_lua-2.4.62-7.el9.x86_64 11/11

Installed:
apr-1.7.0-12.el9.x86_64      apr-util-1.6.1-23.el9.x86_64      apr-util-bdb-1.6.1-23.el9.x86_64
apr-util-openssl-1.6.1-23.el9.x86_64  centos-logos-httpd-90.8-3.el9.noarch  httpd-2.4.62-7.el9.x86_64
httpd-core-2.4.62-7.el9.x86_64      httpd-filesystem-2.4.62-7.el9.noarch  httpd-tools-2.4.62-7.el9.x86_64
mod_http2-2.0.26-5.el9.x86_64      mod_lua-2.4.62-7.el9.x86_64

Complete!
[root@centosstream9 centos]# cd /etc/httpd/conf.d
[root@centosstream9 conf.d]# vi testsite.mydomain.local.conf
[root@centosstream9 conf.d]# cd /var/www
[root@centosstream9 www]# mkdir testsite.mydomain.local
[root@centosstream9 www]# cd testsite.mydomain.local
[root@centosstream9 testsite.mydomain.local]#
```

1.15 Opened index.html file in vi editor using command: **vi index.html**

```
CentOS_Stream_9_VMG_LinuxVMImages.COM - VMware Workstation
File Edit View VM Tabs Help
CentOS_Stream_9_VMG_L... x
Activities Terminal Oct 6 12:30
root@centosstream9:/var/www/testsite.mydomain.local

Installing      : httpd-2.4.62-7.el9.x86_64 11/11
Running scriptlet: httpd-2.4.62-7.el9.x86_64 11/11
Verifying      : apr-1.7.0-12.el9.x86_64 1/11
Verifying      : apr-util-1.6.1-23.el9.x86_64 2/11
Verifying      : apr-util-bdb-1.6.1-23.el9.x86_64 3/11
Verifying      : apr-util-openssl-1.6.1-23.el9.x86_64 4/11
Verifying      : centos-logos-httpd-90.8-3.el9.noarch 5/11
Verifying      : httpd-2.4.62-7.el9.x86_64 6/11
Verifying      : httpd-core-2.4.62-7.el9.x86_64 7/11
Verifying      : httpd-filesystem-2.4.62-7.el9.noarch 8/11
Verifying      : httpd-tools-2.4.62-7.el9.x86_64 9/11
Verifying      : mod_http2-2.0.26-5.el9.x86_64 10/11
Verifying      : mod_lua-2.4.62-7.el9.x86_64 11/11

Installed:
apr-1.7.0-12.el9.x86_64      apr-util-1.6.1-23.el9.x86_64      apr-util-bdb-1.6.1-23.el9.x86_64
apr-util-openssl-1.6.1-23.el9.x86_64  centos-logos-httpd-90.8-3.el9.noarch  httpd-2.4.62-7.el9.x86_64
httpd-core-2.4.62-7.el9.x86_64      httpd-filesystem-2.4.62-7.el9.noarch  httpd-tools-2.4.62-7.el9.x86_64
mod_http2-2.0.26-5.el9.x86_64      mod_lua-2.4.62-7.el9.x86_64

Complete!
[root@centosstream9 centos]# cd /etc/httpd/conf.d
[root@centosstream9 conf.d]# vi testsite.mydomain.local.conf
[root@centosstream9 conf.d]# vi testsite.mydomain.local.conf
[root@centosstream9 conf.d]# cd /var/www
[root@centosstream9 www]# mkdir testsite.mydomain.local
[root@centosstream9 www]# cd testsite.mydomain.local
[root@centosstream9 testsite.mydomain.local]# vi index.html
```

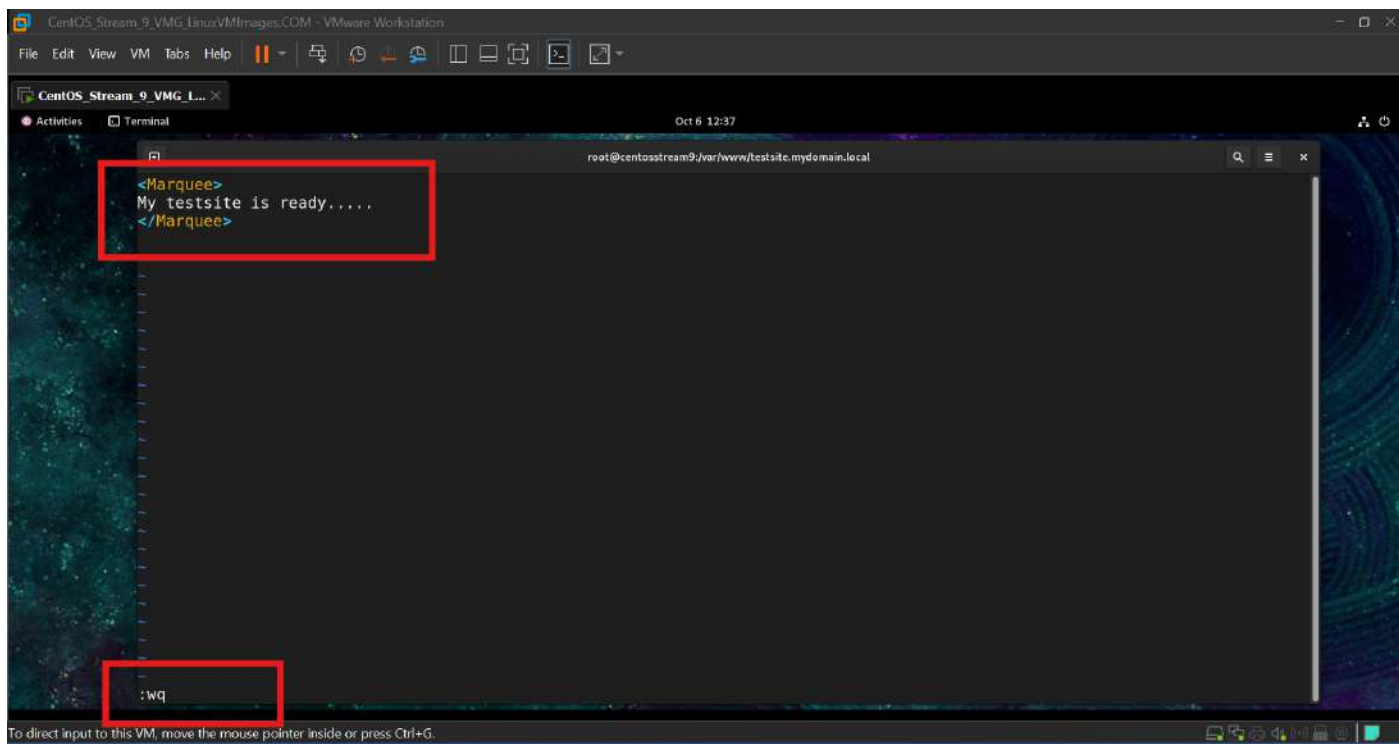
1.16 pressed i to enter in editor mode and write the script

<Marquee>

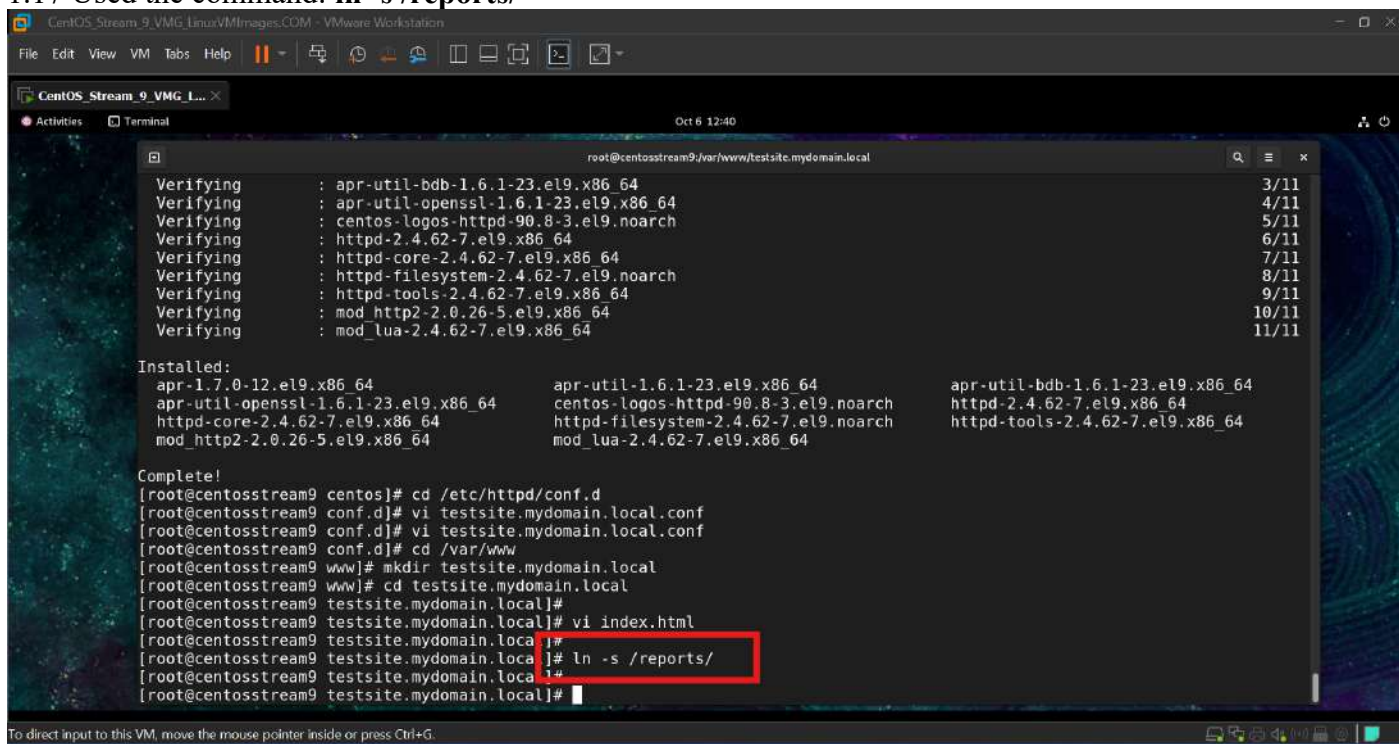
My testsite is ready.....

</Marquee>

Then pressed **Esc** and then **shift + :** and the **wq** to save the file.



1.17 Used the command: `ln -s /reports/`



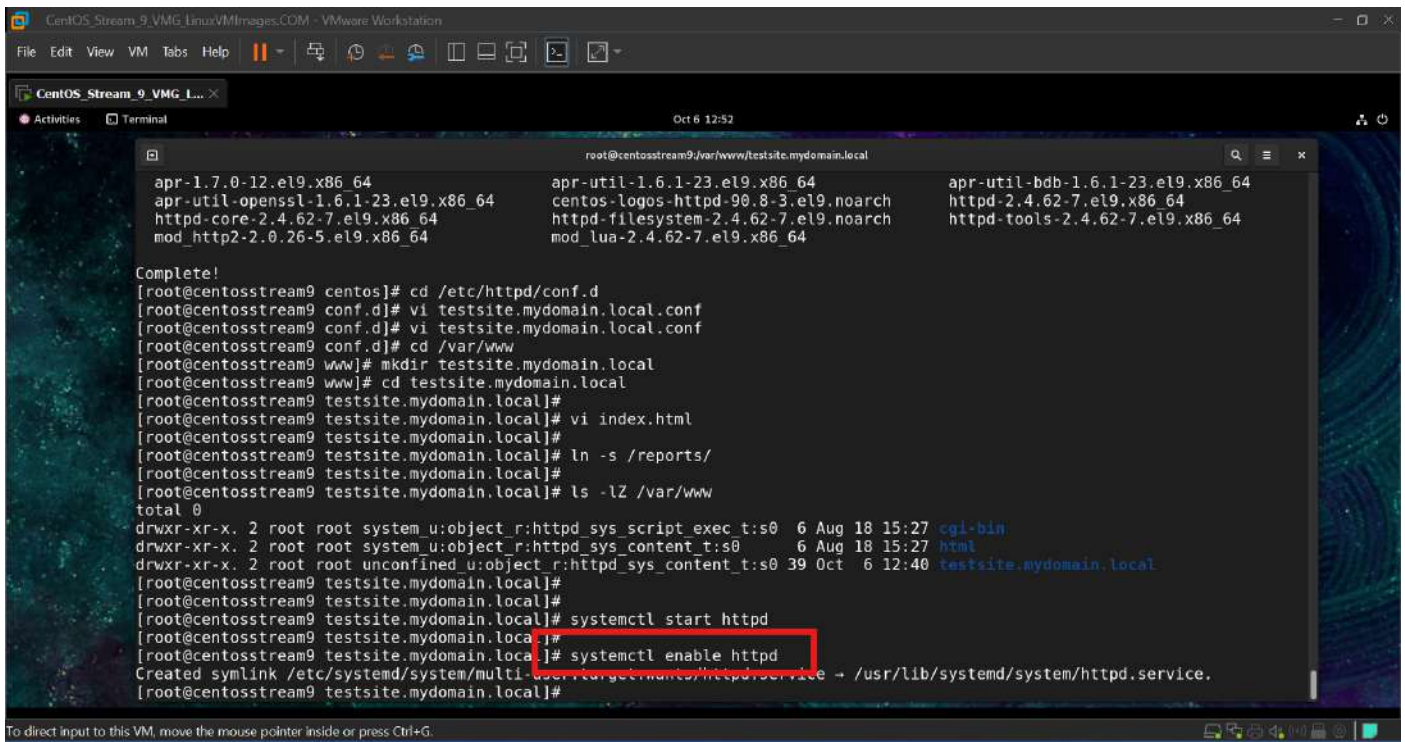
1.18 Used the command: `ls -lZ /var/www`

```
CentOS_Stream_9_VMG_LinuxVMImages.COM - VMware Workstation
File Edit View VM Tabs Help
CentOS_Stream_9_VMG_L... x
Activities Terminal Oct 6 12:44
root@centosstream9:/var/www/testsite.mydomain.local
Verifying : httpd-filesystem-2.4.62-7.el9.noarch 8/11
Verifying : httpd-tools-2.4.62-7.el9.x86_64 9/11
Verifying : mod_http2-2.0.26-5.el9.x86_64 10/11
Verifying : mod_lua-2.4.62-7.el9.x86_64 11/11
Installed:
apr-1.7.0-12.el9.x86_64 apr-util-1.6.1-23.el9.x86_64 apr-util-bdb-1.6.1-23.el9.x86_64
apr-util-openssl-1.6.1-23.el9.x86_64 centos-logos-httpd-90.8-3.el9.noarch httpd-2.4.62-7.el9.x86_64
httpd-core-2.4.62-7.el9.x86_64 httpd-filesystem-2.4.62-7.el9.noarch httpd-tools-2.4.62-7.el9.x86_64
mod_http2-2.0.26-5.el9.x86_64 mod_lua-2.4.62-7.el9.x86_64
Complete!
[root@centosstream9 centos]# cd /etc/httpd/conf.d
[root@centosstream9 conf.d]# vi testsite.mydomain.local.conf
[root@centosstream9 conf.d]# vi testsite.mydomain.local.conf
[root@centosstream9 conf.d]# cd /var/www
[root@centosstream9 www]# mkdir testsite.mydomain.local
[root@centosstream9 www]# cd testsite.mydomain.local
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# vi index.html
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# ln -s /reports/
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# ls -lZ /var/www
total 0
drwxr-xr-x. 2 root root system_u:object_r:httpd_sys_script_exec_t:s0 6 Aug 18 15:27 cgi-bin
drwxr-xr-x. 2 root root system_u:object_r:httpd_sys_content_t:s0 6 Aug 18 15:27 html
drwxr-xr-x. 2 root root unconfined_u:object_r:httpd_sys_content_t:s0 39 Oct 6 12:40 testsite.mydomain.local
[root@centosstream9 testsite.mydomain.local]#
```

1.19 Starting the Apache HTTP Server (httpd) using the command: **systemctl start httpd**

```
CentOS_Stream_9_VMG_LinuxVMImages.COM - VMware Workstation
File Edit View VM Tabs Help
CentOS_Stream_9_VMG_L... x
Activities Terminal Oct 6 12:51
root@centosstream9:/var/www/testsite.mydomain.local
Installed:
apr-1.7.0-12.el9.x86_64 apr-util-1.6.1-23.el9.x86_64 apr-util-bdb-1.6.1-23.el9.x86_64
apr-util-openssl-1.6.1-23.el9.x86_64 centos-logos-httpd-90.8-3.el9.noarch httpd-2.4.62-7.el9.x86_64
httpd-core-2.4.62-7.el9.x86_64 httpd-filesystem-2.4.62-7.el9.noarch httpd-tools-2.4.62-7.el9.x86_64
mod_http2-2.0.26-5.el9.x86_64 mod_lua-2.4.62-7.el9.x86_64
Complete!
[root@centosstream9 centos]# cd /etc/httpd/conf.d
[root@centosstream9 conf.d]# vi testsite.mydomain.local.conf
[root@centosstream9 conf.d]# vi testsite.mydomain.local.conf
[root@centosstream9 conf.d]# cd /var/www
[root@centosstream9 www]# mkdir testsite.mydomain.local
[root@centosstream9 www]# cd testsite.mydomain.local
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# vi index.html
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# ln -s /reports/
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# ls -lZ /var/www
total 0
drwxr-xr-x. 2 root root system_u:object_r:httpd_sys_script_exec_t:s0 6 Aug 18 15:27 cgi-bin
drwxr-xr-x. 2 root root system_u:object_r:httpd_sys_content_t:s0 6 Aug 18 15:27 html
drwxr-xr-x. 2 root root unconfined_u:object_r:httpd_sys_content_t:s0 39 Oct 6 12:40 testsite.mydomain.local
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# systemctl start httpd
[root@centosstream9 testsite.mydomain.local]#
```

1.20 Enabling the httpd server using command: **systemctl enable httpd**



```
CentOS_Stream_9_VMG_LinuxVMImages.COM - VMware Workstation
File Edit View VM Tabs Help
CentOS_Stream_9_VMG_L... x
Activities Terminal
Oct 6 12:52
root@centosstream9:/var/www/testsite.mydomain.local

apr-1.7.0-12.el9.x86_64
apr-util-openssl-1.6.1-23.el9.x86_64
httpd-core-2.4.62-7.el9.x86_64
mod_http2-2.0.26-5.el9.x86_64
apr-util-1.6.1-23.el9.x86_64
centos-logos-httpd-90.8-3.el9.noarch
httpd-filesystem-2.4.62-7.el9.noarch
mod_lua-2.4.62-7.el9.x86_64
apr-util-bdb-1.6.1-23.el9.x86_64
httpd-2.4.62-7.el9.x86_64
httpd-tools-2.4.62-7.el9.x86_64

Complete!
[root@centosstream9 centos]# cd /etc/httpd/conf.d
[root@centosstream9 conf.d]# vi testsite.mydomain.local.conf
[root@centosstream9 conf.d]# vi testsite.mydomain.local.conf
[root@centosstream9 conf.d]# cd /var/www
[root@centosstream9 www]# mkdir testsite.mydomain.local
[root@centosstream9 www]# cd testsite.mydomain.local
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# vi index.html
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# ln -s /reports/
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# ls -lZ /var/www
total 0
drwxr-xr-x. 2 root root system_u:object_r:httpd_sys_script_exec_t:s0 6 Aug 18 15:27 cgi-bin
drwxr-xr-x. 2 root root system_u:object_r:httpd_sys_content_t:s0 6 Aug 18 15:27 html
drwxr-xr-x. 2 root root unconfined_u:object_r:httpd_sys_content_t:s0 39 Oct 6 12:40 testsite.mydomain.local
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# systemctl start httpd
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[root@centosstream9 testsite.mydomain.local]#
```

1.21 Configured the firewall

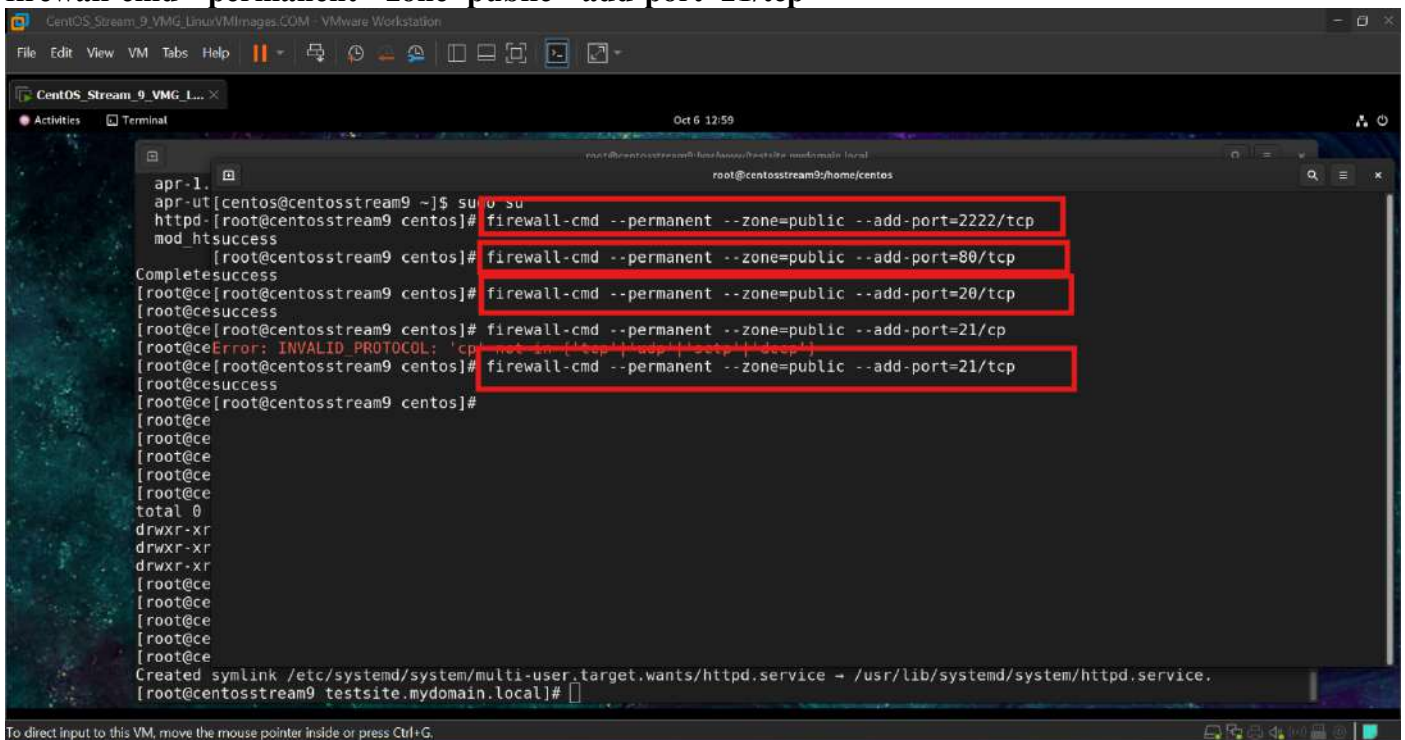
Opened the new terminal and used the command:

firewall-cmd --permanent --zone=public --add-port=2222/tcp

firewall-cmd --permanent --zone=public --add-port=80/tcp

firewall-cmd --permanent --zone=public --add-port=20/tcp

firewall-cmd --permanent --zone=public --add-port=21/tcp



```
CentOS_Stream_9_VMG_LinuxVMImages.COM - VMware Workstation
File Edit View VM Tabs Help
CentOS_Stream_9_VMG_L... x
Activities Terminal
Oct 6 12:59
root@centosstream9:/home/centos

apr-1.
apr-util-
httpd-
mod_

[centos@centosstream9 ~]$ su - su
[centos@centosstream9 ~]$ firewall-cmd --permanent --zone=public --add-port=2222/tcp
[centos@centosstream9 ~]$ firewall-cmd --permanent --zone=public --add-port=80/tcp
[centos@centosstream9 ~]$ firewall-cmd --permanent --zone=public --add-port=20/tcp
[centos@centosstream9 ~]$ firewall-cmd --permanent --zone=public --add-port=21/tcp
[centos@centosstream9 ~]$ firewall-cmd --permanent --zone=public --add-port=21/tcp
[centos@centosstream9 ~]$
```

1.22 Reloading the firewall configuration using the command: **firewall-cmd --reload**

The screenshot shows a VMware Workstation interface with a single virtual machine named 'CentOS_Stream_9_VMG_L...' running. The terminal window is active, showing a root shell session. The user has added several firewall rules to allow HTTP and HTTPS traffic on ports 2222, 80, and 21. The command 'firewall-cmd --reload' is highlighted with a red box, indicating the final step to apply the changes. The terminal output shows the successful execution of these commands.

```

apr-1. [root@centosstream9:~/home/centos]
apr-ut[centos@centosstream9 ~]$ sudo su
httpd-[root@centosstream9 centos]# firewall-cmd --permanent --zone=public --add-port=2222/tcp
mod_httpsuccess
[root@centosstream9 centos]# firewall-cmd --permanent --zone=public --add-port=80/tcp
Completesuccess
[root@ce[root@centosstream9 centos]# firewall-cmd --permanent --zone=public --add-port=21/tcp
[root@cesuccess
[root@ce[root@centosstream9 centos]# firewall-cmd --permanent --zone=public --add-port=21/cp
[root@ceError: INVALID_PROTOCOL: 'cp' not in {'tcp','udp','sctp','dccp'}
[root@ce[root@centosstream9 centos]# firewall-cmd --permanent --zone=public --add-port=21/tcp
[root@cesuccess
[root@ce[root@centosstream9 centos]# firewall-cmd --reload
[root@cesuccess
[root@ce[root@centosstream9 centos]#
[root@ce
[root@ce
[root@ce
total 0
drwxr-xr
drwxr-xr
drwxr-xr
[root@ce
[root@ce
[root@ce
[root@ce
[root@ce
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service -> /usr/lib/systemd/system/httpd.service.
[root@centosstream9 testsite.mydomain.local]#
  
```

1.23 Opened the hosts file using command: vi /etc/hosts
From previous terminal

```

centos-logos-httpd-90.8-3.el9.noarch      httpd-2.4.62-7.el9.x86_64
httpd-core-2.4.62-7.el9.x86_64            httpdfilesystem-2.4.62-7.el9.noarch      httpd-tools-2.4.62-7.el9.x86_64
mod_http2-2.0.26-5.el9.x86_64            mod_lua-2.4.62-7.el9.x86_64

Complete!
[root@centosstream9 ~]# cd /etc/httpd/conf.d
[root@centosstream9 conf.d]# vi testsite.mydomain.local.conf
[root@centosstream9 conf.d]# vi testsite.mydomain.local.conf
[root@centosstream9 conf.d]# cd /var/www
[root@centosstream9 www]# mkdir testsite.mydomain.local
[root@centosstream9 www]# cd testsite.mydomain.local
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# vi index.html
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# ln -s /reports/
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# ls -lZ /var/www
total 0
drwxr-xr-x. 2 root root system u:object r:httpd_sys_script_exec t:s0  6 Aug 18 15:27 cgi-bin
drwxr-xr-x. 2 root root system u:object r:httpd_sys_content t:s0      6 Aug 18 15:27 html
drwxr-xr-x. 2 root root unconfined u:object r:httpd_sys_content t:s0 39 Oct  6 12:40 testsite.mydomain.local
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# systemctl start httpd
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service -> /usr/lib/systemd/system/httpd.service.
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# vi /etc/hosts

```

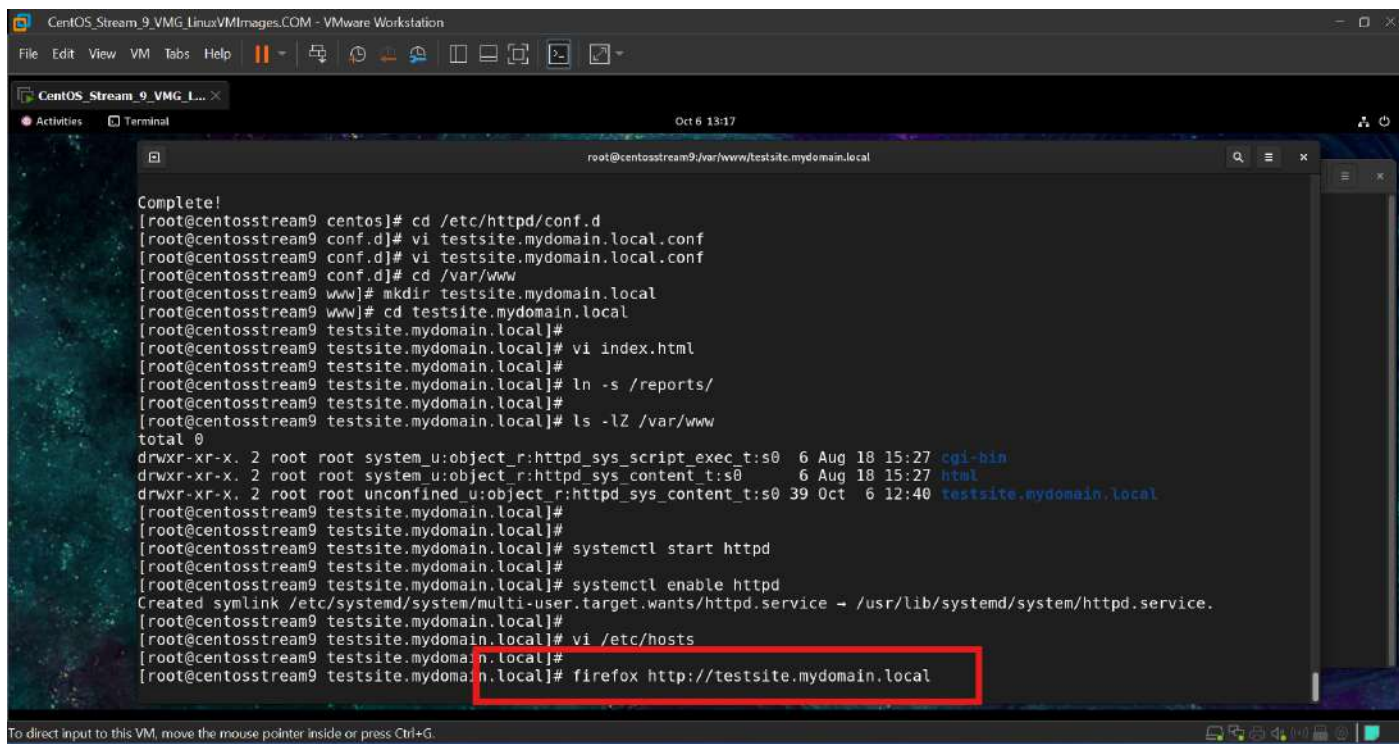
1.24 To enter in editor mode Pressed i

1.25 Typed the script:

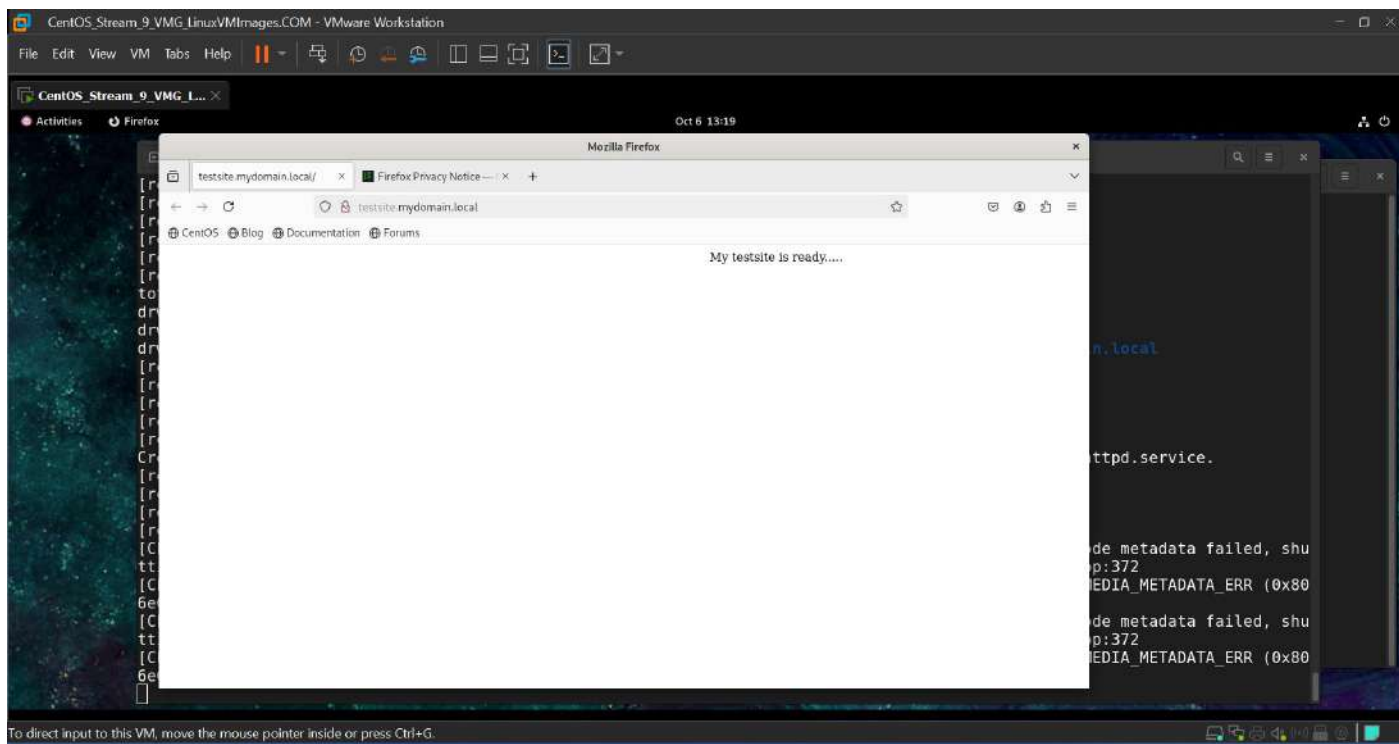
127.0.0.1 testsite.mydomain.local

For Saving and Exit the File by **Pressed Esc, Shift + Colon (:), Typed wq**, and then Enter

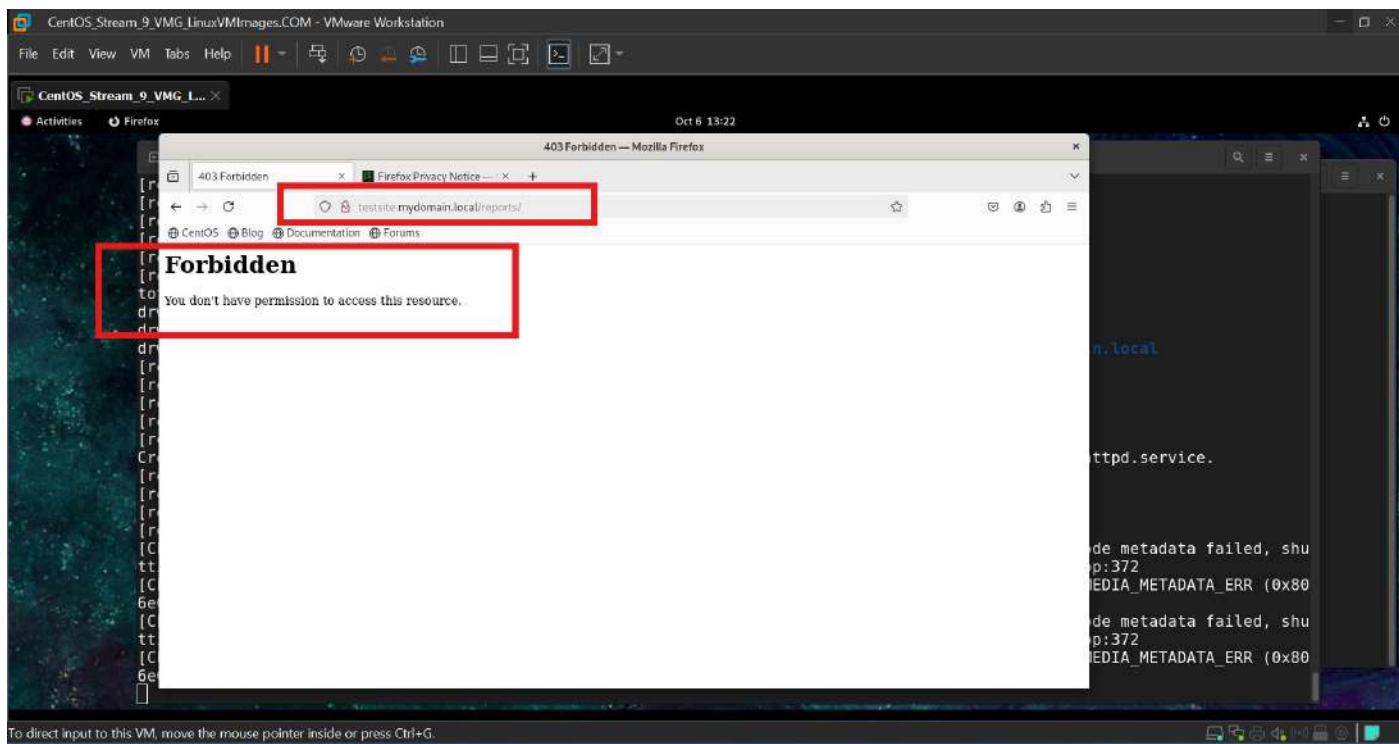
1.26 Opening this “<http://testsite.mydomain.local>” website using the command: **firefox**
<http://testsite.mydomain.local>



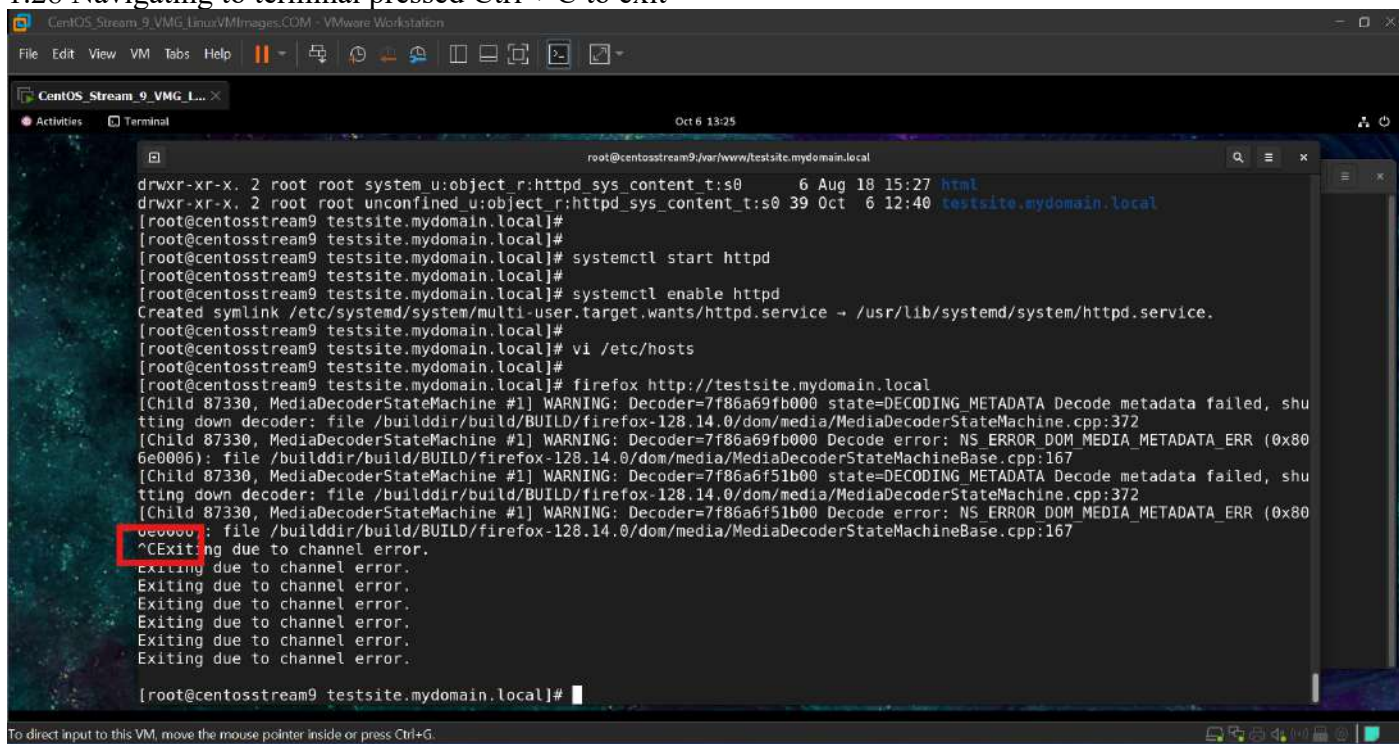
```
Complete!
[root@centosstream9 centos]# cd /etc/httpd/conf.d
[root@centosstream9 conf.d]# vi testsite.mydomain.local.conf
[root@centosstream9 conf.d]# vi testsite.mydomain.local.conf
[root@centosstream9 conf.d]# cd /var/www
[root@centosstream9 www]# mkdir testsite.mydomain.local
[root@centosstream9 www]# cd testsite.mydomain.local
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# vi index.html
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# ln -s /reports/
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# ls -lZ /var/www
total 0
drwxr-xr-x. 2 root root system_u:object_r:httpd_sys_script_exec_t:s0  6 Aug 18 15:27 cgi-bin
drwxr-xr-x. 2 root root system_u:object_r:httpd_sys_content_t:s0      6 Aug 18 15:27 html
drwxr-xr-x. 2 root root unconfined_u:object_r:httpd_sys_content_t:s0 39 Oct  6 12:40 testsite.mydomain.local
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# systemctl start httpd
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service - /usr/lib/systemd/system/httpd.service.
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# vi /etc/hosts
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# firefox http://testsite.mydomain.local
```



1.27 Navigating to this website <http://testsite.mydomain.local/reports>



1.28 Navigating to terminal pressed Ctrl + C to exit



1.29 Using the command: setenforce 1

```
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# systemctl start httpd
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service -> /usr/lib/systemd/system/httpd.service.
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# vi /etc/hosts
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# firefox http://testsite.mydomain.local
[Child 87330, MediaDecoderStateMachine #1] WARNING: Decoder=7f86a69fb000 state=DECODING_METADATA Decode metadata failed, shutting down decoder: file /builddir/build/BUILD/firefox-128.14.0/dom/media/MediaDecoderStateMachine.cpp:372
[Child 87330, MediaDecoderStateMachine #1] WARNING: Decoder=7f86a69fb000 Decode error: NS_ERROR_DOM_MEDIA_METADATA_ERR (0x806e0006): file /builddir/build/BUILD/firefox-128.14.0/dom/media/MediaDecoderStateMachineBase.cpp:167
[Child 87330, MediaDecoderStateMachine #1] WARNING: Decoder=7f86a6f51b00 state=DECODING_METADATA Decode metadata failed, shutting down decoder: file /builddir/build/BUILD/firefox-128.14.0/dom/media/MediaDecoderStateMachine.cpp:372
[Child 87330, MediaDecoderStateMachine #1] WARNING: Decoder=7f86a6f51b00 Decode error: NS_ERROR_DOM_MEDIA_METADATA_ERR (0x806e0006): file /builddir/build/BUILD/firefox-128.14.0/dom/media/MediaDecoderStateMachineBase.cpp:167
^CExiting due to channel error.
Exiting due to channel error.
Exiting due to channel error.
Exiting due to channel error.
Exiting due to channel error.
Exiting due to channel error.
Exiting due to channel error.

[root@centosstream9 testsite.mydomain.local]# setenforce 1
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]#
```

1.30 Used the command: **getenforce**

```
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# systemctl start httpd
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service -> /usr/lib/systemd/system/httpd.service.
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# vi /etc/hosts
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# firefox http://testsite.mydomain.local
[Child 87330, MediaDecoderStateMachine #1] WARNING: Decoder=7f86a69fb000 state=DECODING_METADATA Decode metadata failed, shutting down decoder: file /builddir/build/BUILD/firefox-128.14.0/dom/media/MediaDecoderStateMachine.cpp:372
[Child 87330, MediaDecoderStateMachine #1] WARNING: Decoder=7f86a69fb000 Decode error: NS_ERROR_DOM_MEDIA_METADATA_ERR (0x806e0006): file /builddir/build/BUILD/firefox-128.14.0/dom/media/MediaDecoderStateMachineBase.cpp:167
[Child 87330, MediaDecoderStateMachine #1] WARNING: Decoder=7f86a6f51b00 state=DECODING_METADATA Decode metadata failed, shutting down decoder: file /builddir/build/BUILD/firefox-128.14.0/dom/media/MediaDecoderStateMachine.cpp:372
[Child 87330, MediaDecoderStateMachine #1] WARNING: Decoder=7f86a6f51b00 Decode error: NS_ERROR_DOM_MEDIA_METADATA_ERR (0x806e0006): file /builddir/build/BUILD/firefox-128.14.0/dom/media/MediaDecoderStateMachineBase.cpp:167
^CExiting due to channel error.
Exiting due to channel error.
Exiting due to channel error.
Exiting due to channel error.
Exiting due to channel error.
Exiting due to channel error.
Exiting due to channel error.

[root@centosstream9 testsite.mydomain.local]# setenforce 1
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# getenforce
Enforcing
[root@centosstream9 testsite.mydomain.local]#
```

1.31 Used the command: **chcon unconfined_u:object_r:httpd_sys_content_t:s0 /reports/**

1.32 Use the command: **chcon unconfined_u:object_r:httpd_sys_content_t:s0 /reports/101.log**
It shows "No such file or directory," so we first need to create the 101.log file.

1.33 Creating a file using the command: **touch /reports/101.log**


```
CentOS_Stream_9_VMG LinuxVMImages.COM - VMware Workstation
File Edit View VM Tabs Help
CentOS_Stream_9_VMG_L... x
Activities Terminal
Oct 6 13:44
root@centosstream9:/var/www/testsite.mydomain.local

[root@centosstream9 testsite.mydomain.local]# firefox http://testsite.mydomain.local
[Child 87330, MediaDecoderStateMachine #1] WARNING: Decoder=7f86a69fb000 state=DECODING_METADATA Decode metadata failed, shutting down decoder: file /builddir/build/BUILD/firefox-128.14.0/dom/media/MediaDecoderStateMachine.cpp:372
[Child 87330, MediaDecoderStateMachine #1] WARNING: Decoder=7f86a69fb000 Decode error: NS_ERROR_DOM_MEDIA_METADATA_ERR (0x806e0006): file /builddir/build/BUILD/firefox-128.14.0/dom/media/MediaDecoderStateMachineBase.cpp:167
[Child 87330, MediaDecoderStateMachine #1] WARNING: Decoder=7f86a6f51b00 state=DECODING_METADATA Decode metadata failed, shutting down decoder: file /builddir/build/BUILD/firefox-128.14.0/dom/media/MediaDecoderStateMachine.cpp:372
[Child 87330, MediaDecoderStateMachine #1] WARNING: Decoder=7f86a6f51b00 Decode error: NS_ERROR_DOM_MEDIA_METADATA_ERR (0x806e0006): file /builddir/build/BUILD/firefox-128.14.0/dom/media/MediaDecoderStateMachineBase.cpp:167
^CExiting due to channel error.
Exiting due to channel error.
Exiting due to channel error.
Exiting due to channel error.
Exiting due to channel error.
Exiting due to channel error.

[root@centosstream9 testsite.mydomain.local]# setenforce 1
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# getenforce
Enforcing
[root@centosstream9 testsite.mydomain.local]# chcon unconfined_u:object_r:httpd_sys_content_t:s0 /reports/
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# chcon unconfined_u:object_r:httpd_sys_content_t:s0 /reports/101.log
chcon: cannot access '/reports/101.log': No such file or directory
[root@centosstream9 testsite.mydomain.local]# touch /reports/101.log
[root@centosstream9 testsite.mydomain.local]#
```

1.34 Now running the command will not give error:
chcon unconfined u:object r:httpd_sys_content t:s0 /reports/101.log

```
CentOS_Stream_9_VMG LinuxVMImages.COM - VMware Workstation
File Edit View VM Tabs Help
CentOS_Stream_9_VMG_L... x
Activities Terminal
Oct 6 13:46
root@centosstream9:/var/www/testsite.mydomain.local

tting down decoder: file /builddir/build/BUILD/firefox-128.14.0/dom/media/MediaDecoderStateMachine.cpp:372
[Child 87330, MediaDecoderStateMachine #1] WARNING: Decoder=7f86a69fb000 Decode error: NS_ERROR_DOM_MEDIA_METADATA_ERR (0x806e0006): file /builddir/build/BUILD/firefox-128.14.0/dom/media/MediaDecoderStateMachineBase.cpp:167
[Child 87330, MediaDecoderStateMachine #1] WARNING: Decoder=7f86a6f51b00 state=DECODING_METADATA Decode metadata failed, shutting down decoder: file /builddir/build/BUILD/firefox-128.14.0/dom/media/MediaDecoderStateMachine.cpp:372
[Child 87330, MediaDecoderStateMachine #1] WARNING: Decoder=7f86a6f51b00 Decode error: NS_ERROR_DOM_MEDIA_METADATA_ERR (0x806e0006): file /builddir/build/BUILD/firefox-128.14.0/dom/media/MediaDecoderStateMachineBase.cpp:167
^CExiting due to channel error.
Exiting due to channel error.
Exiting due to channel error.
Exiting due to channel error.
Exiting due to channel error.
Exiting due to channel error.

[root@centosstream9 testsite.mydomain.local]# setenforce 1
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# getenforce
Enforcing
[root@centosstream9 testsite.mydomain.local]# chcon unconfined_u:object_r:httpd_sys_content_t:s0 /reports/
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# chcon unconfined_u:object_r:httpd_sys_content_t:s0 /reports/101.log
chcon: cannot access '/reports/101.log': No such file or directory
[root@centosstream9 testsite.mydomain.local]# touch /reports/101.log
[root@centosstream9 testsite.mydomain.local]# chcon unconfined u:object r:httpd_sys_content t:s0 /reports/101.log
[root@centosstream9 testsite.mydomain.local]#
```

1.35 Opening this “<http://testsite.mydomain.local/reports>” website using the command:
firefox <http://testsite.mydomain.local/reports>

CentOS_Stream_9_VMG_LinuxVMImages.COM - VMware Workstation

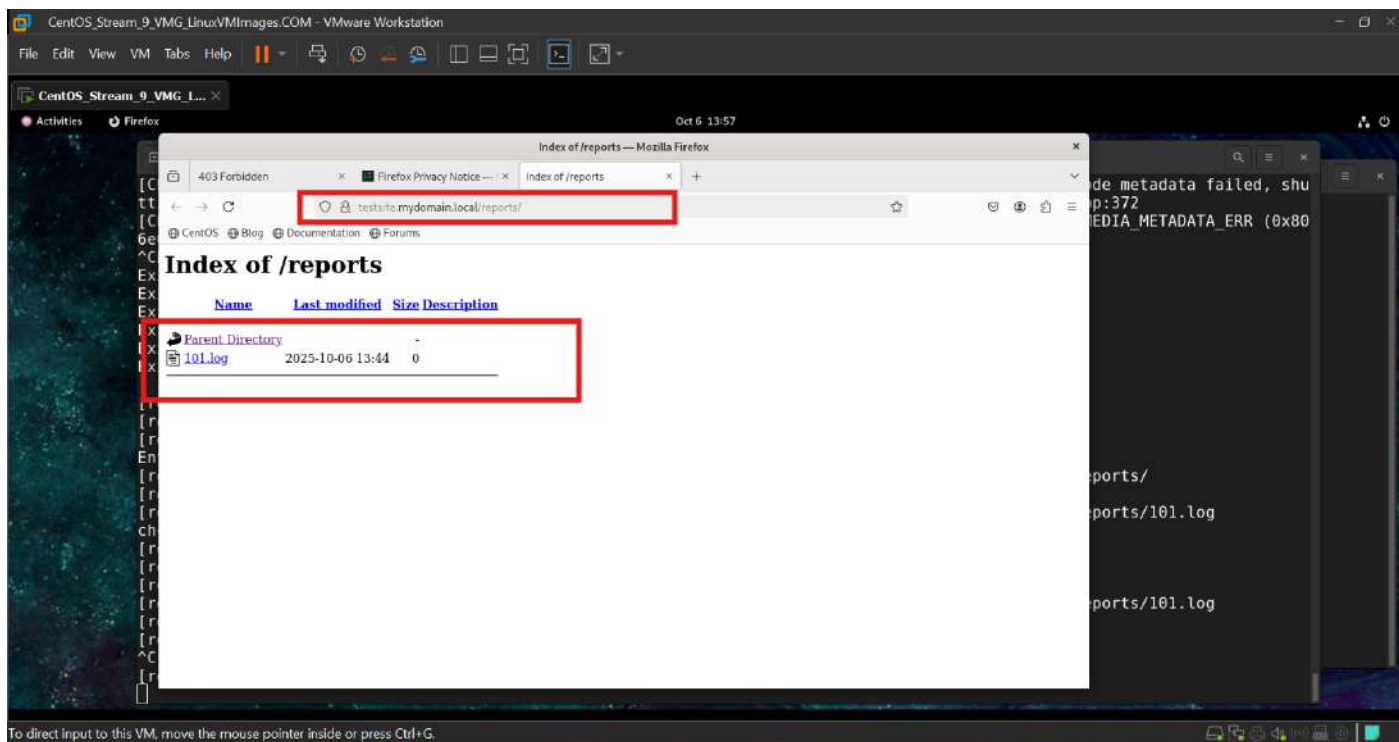
CentOS_Stream_9_VMG_1... x

Activities Terminal Oct 6 13:51

```
root@centosstream9:/var/www/testsite.mydomain.local
6e0006): file /builddir/build/BUILD/firefox-128.14.0/dom/media/MediaDecoderStateMachineBase.cpp:167
[Child 87330, MediaDecoderStateMachine #1] WARNING: Decoder=7f86a6f51b00 state=DECODING METADATA Decode metadata failed, shutting down decoder: file /builddir/build/BUILD/firefox-128.14.0/dom/media/MediaDecoderStateMachine.cpp:372
[Child 87330, MediaDecoderStateMachine #1] WARNING: Decoder=7f86a6f51b00 Decode error: NS_ERROR_DOM_MEDIA_METADATA_ERR (0x806e0006): file /builddir/build/BUILD/firefox-128.14.0/dom/media/MediaDecoderStateMachineBase.cpp:167
^CExiting due to channel error.
Exiting due to channel error.
Exiting due to channel error.
Exiting due to channel error.
Exiting due to channel error.
Exiting due to channel error.
Exiting due to channel error.

[root@centosstream9 testsite.mydomain.local]# setenforce 1
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# getenforce
Enforcing
[root@centosstream9 testsite.mydomain.local]# chcon unconfined_u:object_r:httpd_sys_content_t:s0 /reports/
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# chcon unconfined_u:object_r:httpd_sys_content_t:s0 /reports/101.log
chcon: cannot access '/reports/101.log': No such file or directory
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# touch /reports/101.log
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# chcon unconfined_u:object_r:httpd_sys_content_t:s0 /reports/101.log
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# firefox http://testsite.mydomain.local
^C
[root@centosstream9 testsite.mydomain.local]# firefox http://testsite.mydomain.local/reports
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.



1.36 Editing the vsftpd.conf file using the command: `vi /etc/vsftpd/vsftpd.conf`


```
CentOS_Stream_9_VMG_LinuxVMImages.COM - VMware Workstation
File Edit View VM Tabs Help
CentOS_Stream_9_VMG_L... x
Activities Terminal
Oct 6 14:12
root@centosstream9:/var/www/testsite.mydomain.local

[root@centosstream9 testsite.mydomain.local]# setenforce 1
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# getenforce
Enforcing
[root@centosstream9 testsite.mydomain.local]# chcon unconfined_u:object_r:httpd_sys_content_t:s0 /reports/
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# chcon unconfined_u:object_r:httpd_sys_content_t:s0 /reports/101.log
chcon: cannot access '/reports/101.log': No such file or directory
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# touch /reports/101.log
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# chcon unconfined_u:object_r:httpd_sys_content_t:s0 /reports/101.log
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# firefox http://testsite.mydomain.local
^C
[root@centosstream9 testsite.mydomain.local]# firefox http://testsite.mydomain.local/reports
^C
Exiting due to channel error.
Exiting due to channel error.
Exiting due to channel error.
Exiting due to channel error.
Crash Annotation GraphicsCriticalError: [[C0][GFX1-]: CompositorBridgeChild receives IPC close with reason=AbnormalShutdown
(t=488.261) Exiting due to channel error.

[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# vi /etc/vsftpd/vsftpd.conf
[root@centosstream9 testsite.mydomain.local]#
```

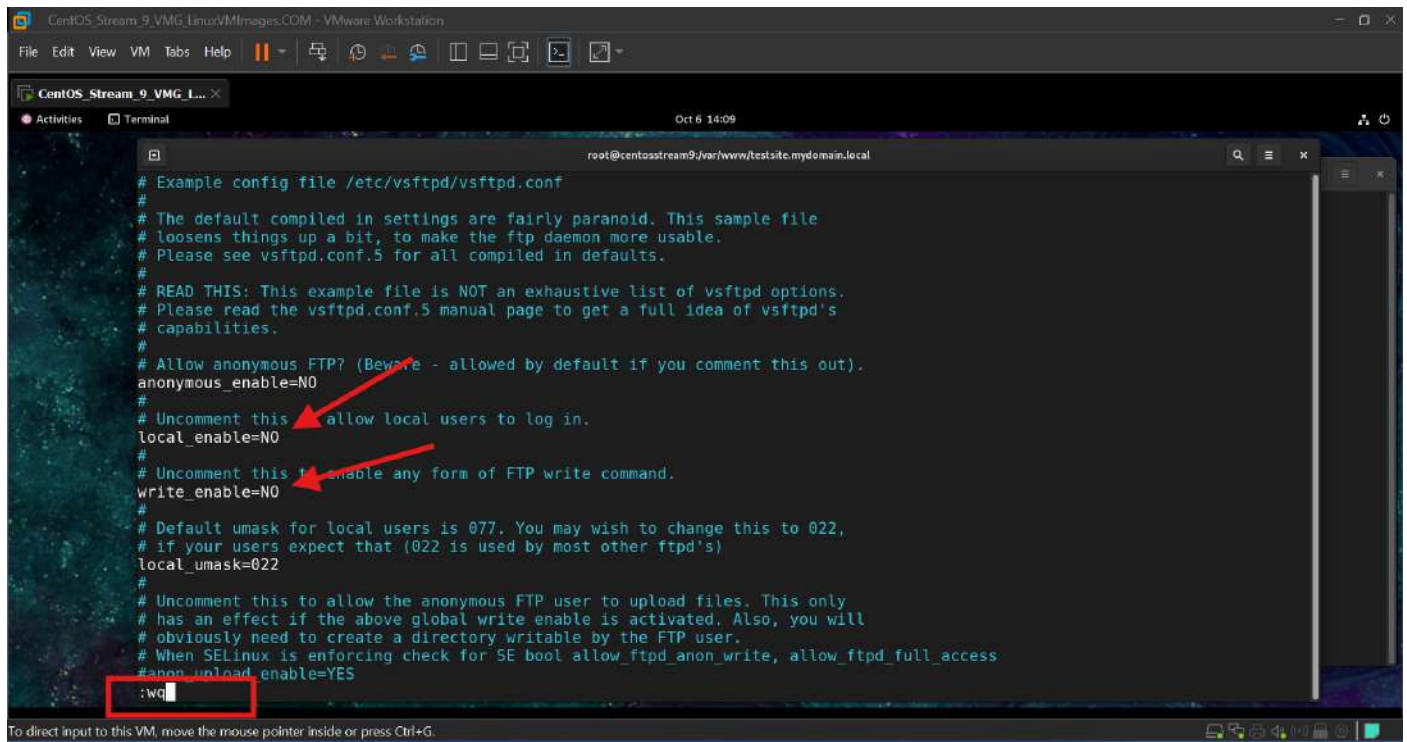
1.37 Modifying the FTP configuration and disabled local user login and FTP write commands, change the following lines in the /etc/vsftpd/vsftpd.conf file:

- **local_enable=NO**
- **write_enable=NO**

```
CentOS_Stream_9_VMG_LinuxVMImages.COM - VMware Workstation
File Edit View VM Tabs Help
CentOS_Stream_9_VMG_L... x
Activities Terminal
Oct 6 14:04
root@centosstream9:/var/www/testsite.mydomain.local

# Example config file /etc/vsftpd/vsftpd.conf
#
# The default compiled in settings are fairly paranoid. This sample file
# loosens things up a bit, to make the ftp daemon more usable.
# Please see vsftpd.conf.5 for all compiled in defaults.
#
# READ THIS: This example file is NOT an exhaustive list of vsftpd options.
# Please read the vsftpd.conf.5 manual page to get a full idea of vsftpd's
# capabilities.
#
# Allow anonymous FTP? (Beware - allowed by default if you comment this out).
anonymous_enable=NO
#
# Uncomment this to allow local users to log in.
local_enable=YES
#
# Uncomment this to enable any form of FTP write command.
write_enable=YES
#
# Default umask for local users is 077. You may wish to change this to 022,
# if your users expect that (022 is used by most other ftpd's)
local_umask=022
#
# Uncomment this to allow the anonymous FTP user to upload files. This only
# has an effect if the above global write enable is activated. Also, you will
# obviously need to create a directory writable by the FTP user.
# When SELinux is enforcing check for SE bool allow_ftpd_anon_write, allow_ftpd_full_access
#anon_upload_enable=YES
```

1.38 Pressed i, edited the script and Then pressed **Esc** and then **shift + :** and the **wq** to save the file.



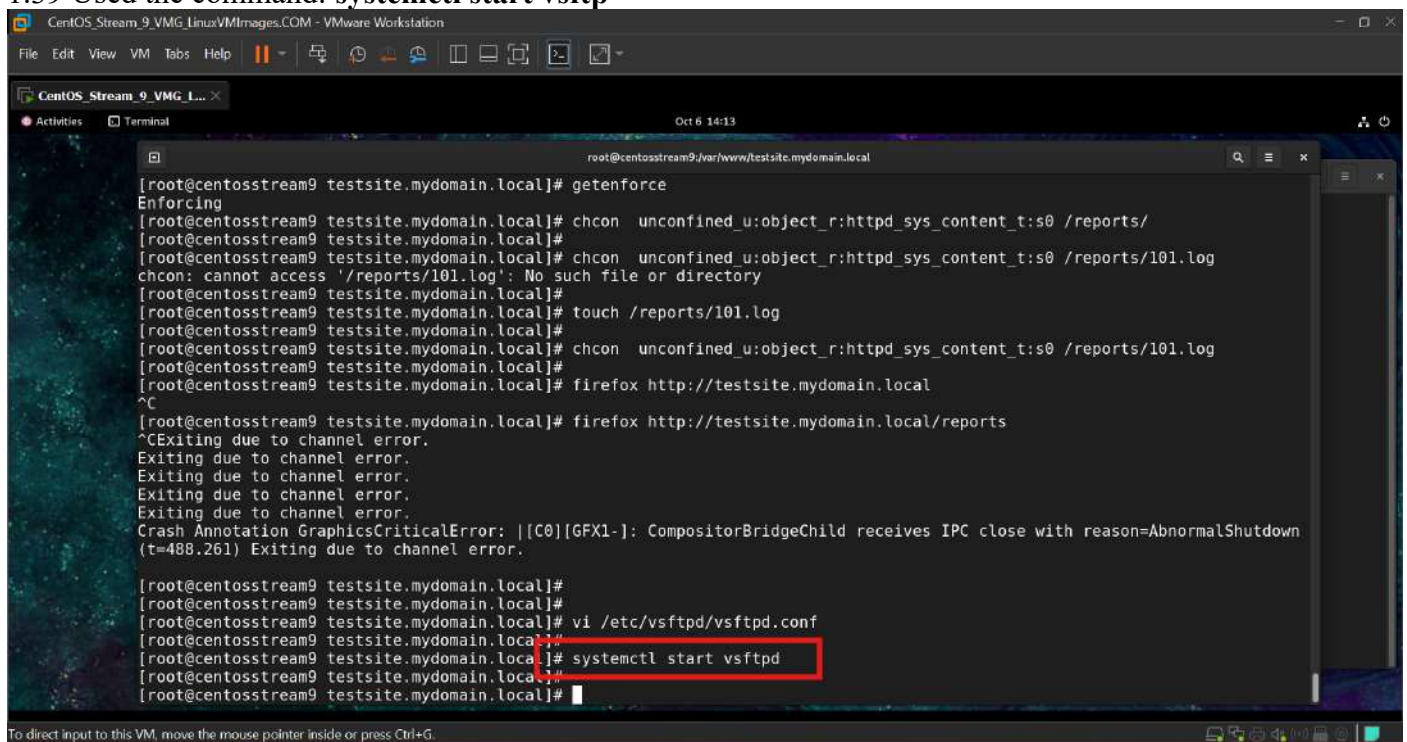
```
CentOS_Stream_9_VMG_LinuxVMImages.COM - VMware Workstation
File Edit View VM Tabs Help
CentOS_Stream_9_VMG_L... x
Activities Terminal
Oct 6 14:09
root@centosstream9:/var/www/testsite.mydomain.local

# Example config file /etc/vsftpd/vsftpd.conf
#
# The default compiled in settings are fairly paranoid. This sample file
# loosens things up a bit, to make the ftp daemon more usable.
# Please see vsftpd.conf.5 for all compiled in defaults.
#
# READ THIS: This example file is NOT an exhaustive list of vsftpd options.
# Please read the vsftpd.conf.5 manual page to get a full idea of vsftpd's
# capabilities.
#
# Allow anonymous FTP? (Beware - allowed by default if you comment this out).
anonymous_enable=NO
#
# Uncomment this to allow local users to log in.
local_enable=NO
#
# Uncomment this to enable any form of FTP write command.
write_enable=NO
#
# Default umask for local users is 077. You may wish to change this to 022,
# if your users expect that (022 is used by most other ftpd's)
local_umask=022
#
# Uncomment this to allow the anonymous FTP user to upload files. This only
# has an effect if the above global write enable is activated. Also, you will
# obviously need to create a directory writable by the FTP user.
# When SELinux is enforcing check for SE bool allow_ftpd_anon_write, allow_ftpd_full_access
#anon_upload_enable=YES

:wq
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

1.39 Used the command: **systemctl start vsftpd**



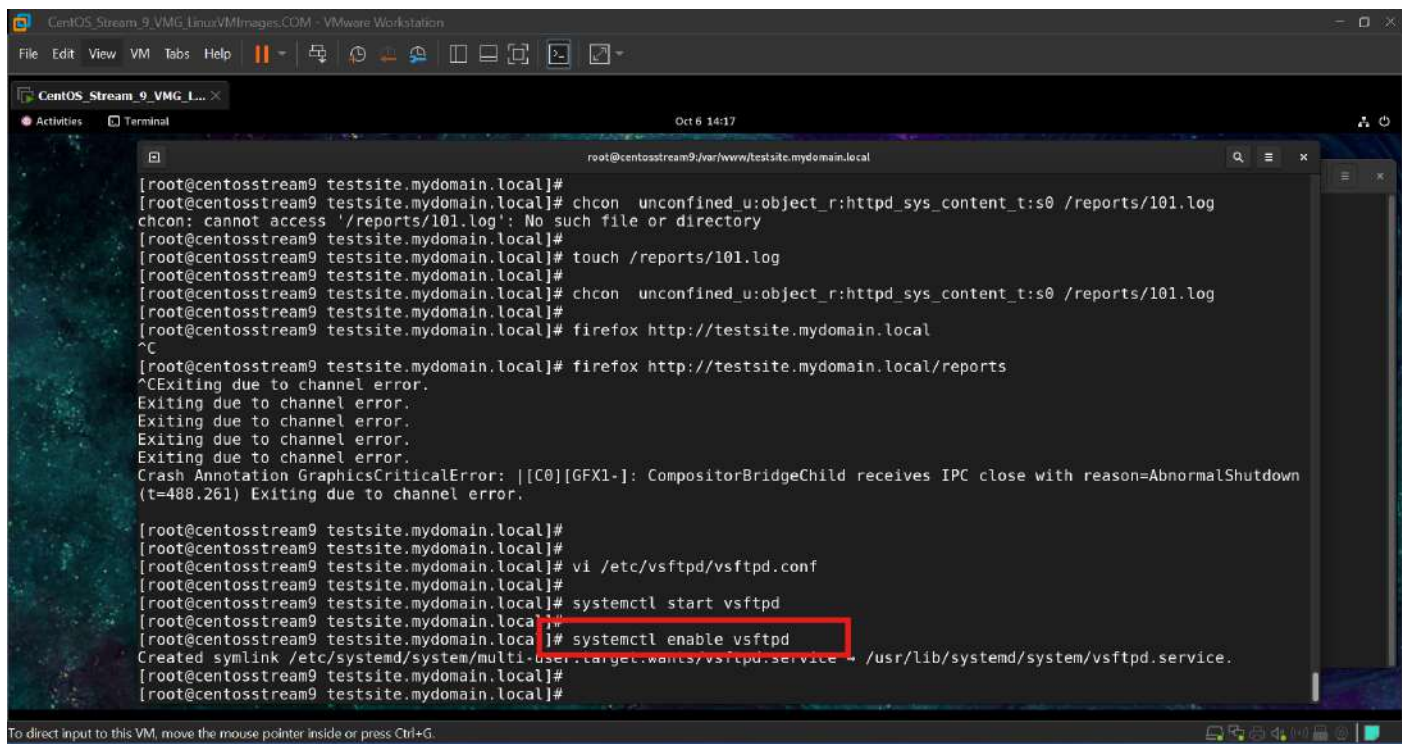
```
CentOS_Stream_9_VMG_LinuxVMImages.COM - VMware Workstation
File Edit View VM Tabs Help
CentOS_Stream_9_VMG_L... x
Activities Terminal
Oct 6 14:13
root@centosstream9:/var/www/testsite.mydomain.local

[root@centosstream9 testsite.mydomain.local]# getenforce
Enforcing
[root@centosstream9 testsite.mydomain.local]# chcon unconfined_u:object_r:httpd_sys_content_t:s0 /reports/
[root@centosstream9 testsite.mydomain.local]# chcon unconfined_u:object_r:httpd_sys_content_t:s0 /reports/101.log
chcon: cannot access '/reports/101.log': No such file or directory
[root@centosstream9 testsite.mydomain.local]# touch /reports/101.log
[root@centosstream9 testsite.mydomain.local]# chcon unconfined_u:object_r:httpd_sys_content_t:s0 /reports/101.log
[root@centosstream9 testsite.mydomain.local]# firefox http://testsite.mydomain.local
^C
[root@centosstream9 testsite.mydomain.local]# firefox http://testsite.mydomain.local/reports
^CExiting due to channel error.
Exiting due to channel error.
Exiting due to channel error.
Exiting due to channel error.
Exiting due to channel error.
Crash Annotation GraphicsCriticalError: [[C0][GFX1-]: CompositorBridgeChild receives IPC close with reason=AbnormalShutdown
(t=488.261) Exiting due to channel error.

[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# vi /etc/vsftpd/vsftpd.conf
[root@centosstream9 testsite.mydomain.local]# systemctl start vsftpd
[root@centosstream9 testsite.mydomain.local]#
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

1.40 Used the command: **systemctl enable vsftpd**



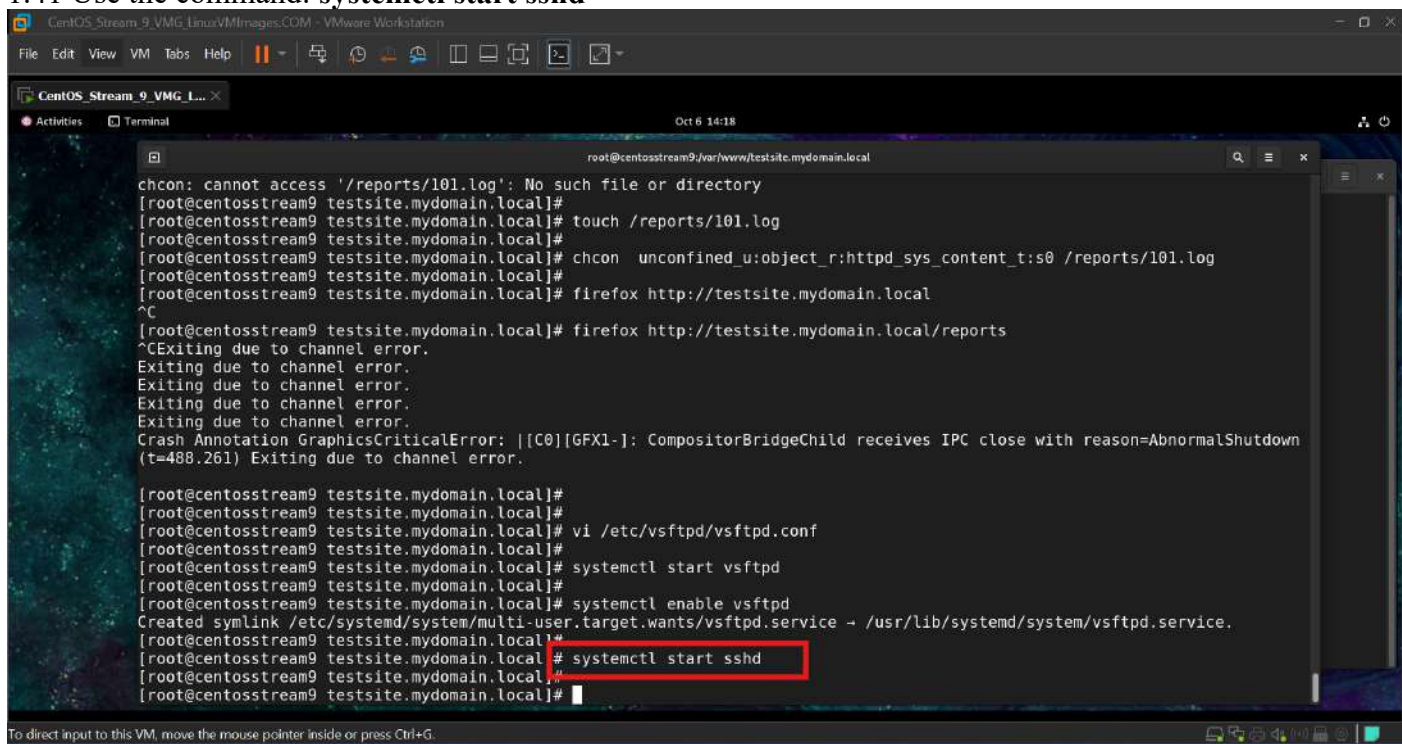
```
CentOS_Stream_9_VMG_LinuxVMImages.COM - VMware Workstation
File Edit View VM Tabs Help
CentOS_Stream_9_VMG_L... x
Activities Terminal
Oct 6 14:17
root@centosstream9:/var/www/testsite.mydomain.local

[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# chcon unconfined_u:object_r:httpd_sys_content_t:s0 /reports/101.log
chcon: cannot access '/reports/101.log': No such file or directory
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# touch /reports/101.log
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# chcon unconfined_u:object_r:httpd_sys_content_t:s0 /reports/101.log
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# firefox http://testsite.mydomain.local
^C
[root@centosstream9 testsite.mydomain.local]# firefox http://testsite.mydomain.local/reports
^CExiting due to channel error.
Exiting due to channel error.
Exiting due to channel error.
Exiting due to channel error.
Exiting due to channel error.
Crash Annotation GraphicsCriticalError: [[C0][GFX1-]: CompositorBridgeChild receives IPC close with reason=AbnormalShutdown
(t=488.261) Exiting due to channel error.

[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# vi /etc/vsftpd/vsftpd.conf
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# systemctl start vsftpd
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# systemctl enable vsftpd
Created symlink /etc/systemd/system/multi-user.target.wants/vsftpd.service -> /usr/lib/systemd/system/vsftpd.service.
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]#
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

1.41 Use the command: **systemctl start sshd**



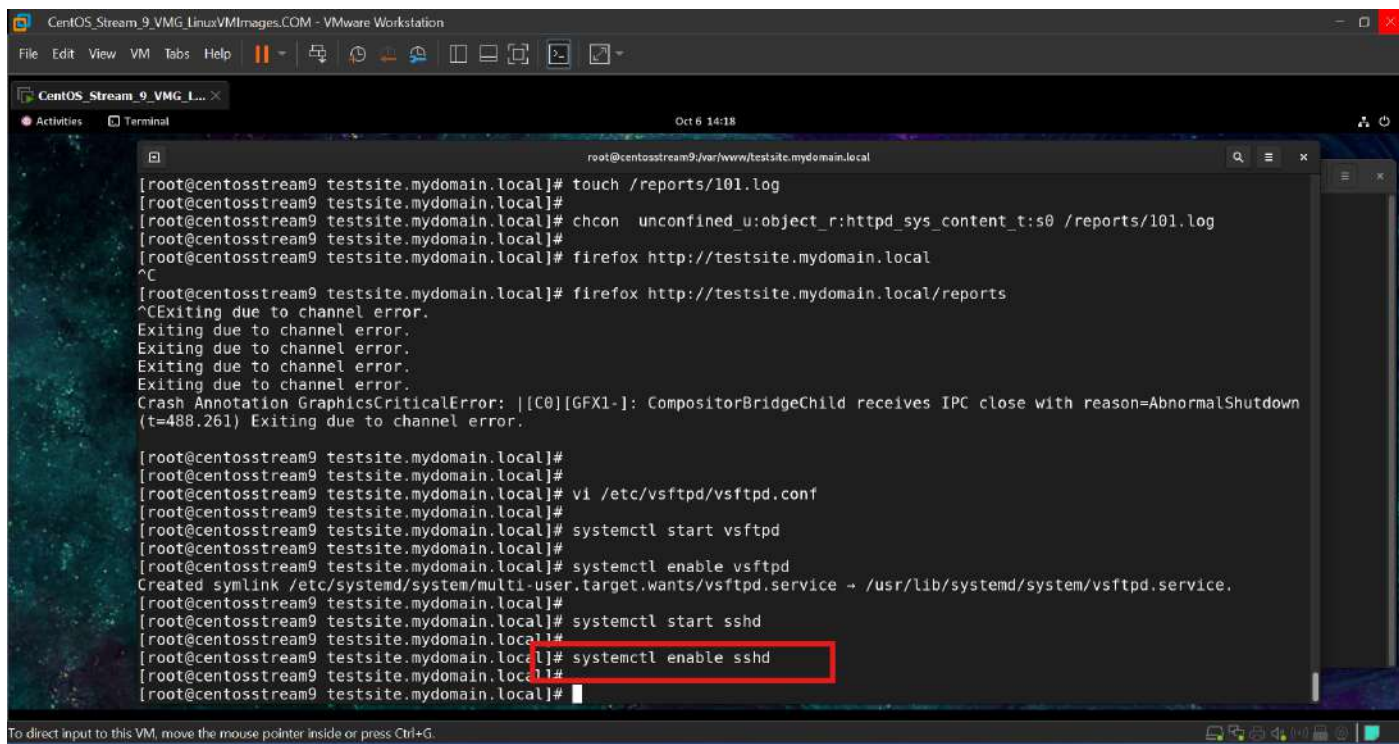
```
CentOS_Stream_9_VMG_LinuxVMImages.COM - VMware Workstation
File Edit View VM Tabs Help
CentOS_Stream_9_VMG_L... x
Activities Terminal
Oct 6 14:18
root@centosstream9:/var/www/testsite.mydomain.local

chcon: cannot access '/reports/101.log': No such file or directory
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# touch /reports/101.log
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# chcon unconfined_u:object_r:httpd_sys_content_t:s0 /reports/101.log
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# firefox http://testsite.mydomain.local
^C
[root@centosstream9 testsite.mydomain.local]# firefox http://testsite.mydomain.local/reports
^CExiting due to channel error.
Exiting due to channel error.
Exiting due to channel error.
Exiting due to channel error.
Exiting due to channel error.
Crash Annotation GraphicsCriticalError: [[C0][GFX1-]: CompositorBridgeChild receives IPC close with reason=AbnormalShutdown
(t=488.261) Exiting due to channel error.

[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# vi /etc/vsftpd/vsftpd.conf
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# systemctl start vsftpd
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# systemctl enable vsftpd
Created symlink /etc/systemd/system/multi-user.target.wants/vsftpd.service -> /usr/lib/systemd/system/vsftpd.service.
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# systemctl start sshd
[root@centosstream9 testsite.mydomain.local]#
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

1.42 Use the command: **systemctl enable sshd**



CentOS_Stream_9_VMG_LinuxVMImages.COM - VMware Workstation

File Edit View VM Tabs Help

CentOS_Stream_9_VMG_L... x

Activities Terminal Oct 6 14:18

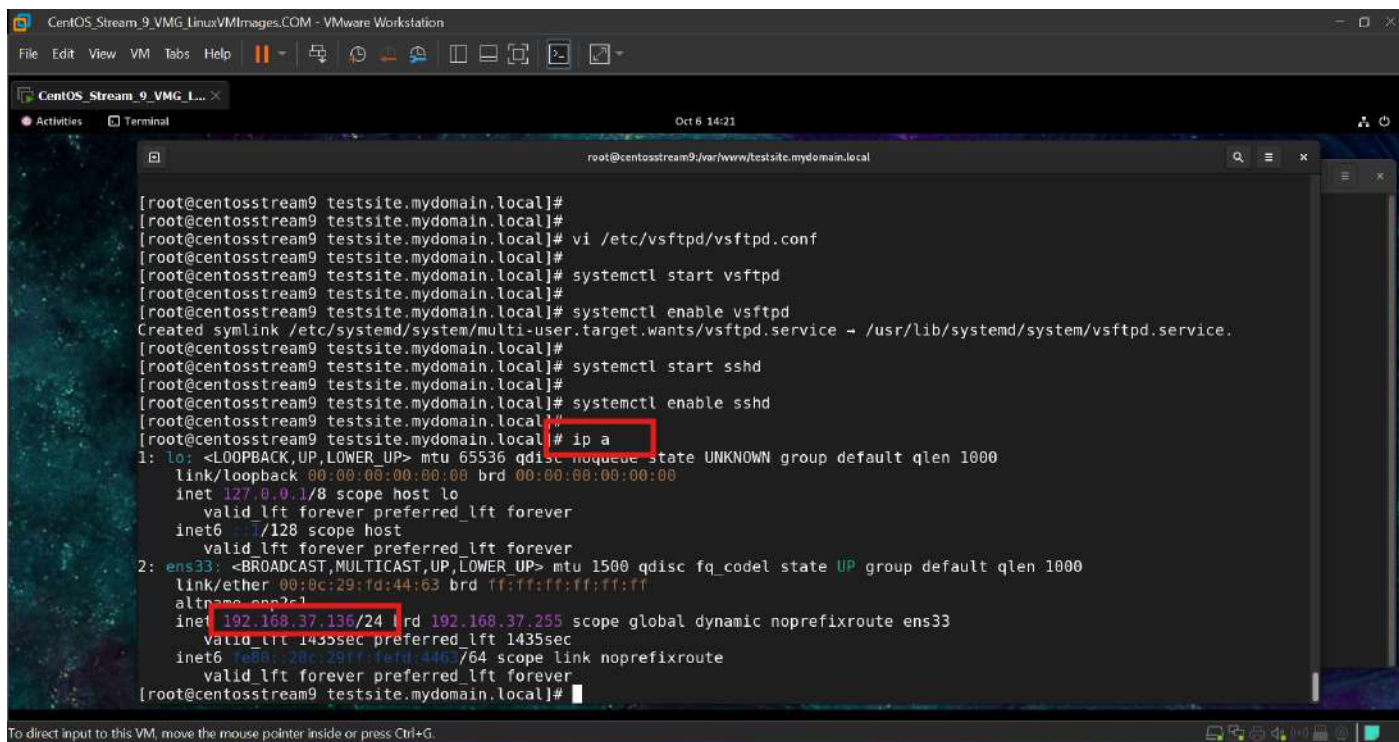
```
root@centosstream9:/var/www/testsite.mydomain.local

[root@centosstream9 testsite.mydomain.local]# touch /reports/101.log
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# chcon unconfined_u:object_r:httpd_sys_content_t:s0 /reports/101.log
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# firefox http://testsite.mydomain.local
^C
[root@centosstream9 testsite.mydomain.local]# firefox http://testsite.mydomain.local/reports
^CExiting due to channel error.
Exiting due to channel error.
Exiting due to channel error.
Exiting due to channel error.
Crash Annotation GraphicsCriticalError: [[C0][GFX1-]: CompositorBridgeChild receives IPC close with reason=AbnormalShutdown
(t=488.261) Exiting due to channel error.

[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# vi /etc/vsftpd/vsftpd.conf
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# systemctl start vsftpd
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# systemctl enable vsftpd
Created symlink /etc/systemd/system/multi-user.target.wants/vsftpd.service - /usr/lib/systemd/system/vsftpd.service.
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# systemctl start sshd
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# systemctl enable sshd
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]#
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

1.43 To check the IP of CentOS using the command: **ip a**
The IP of CentOS is 192.168.37.136



CentOS_Stream_9_VMG_LinuxVMImages.COM - VMware Workstation

File Edit View VM Tabs Help

CentOS_Stream_9_VMG_L... x

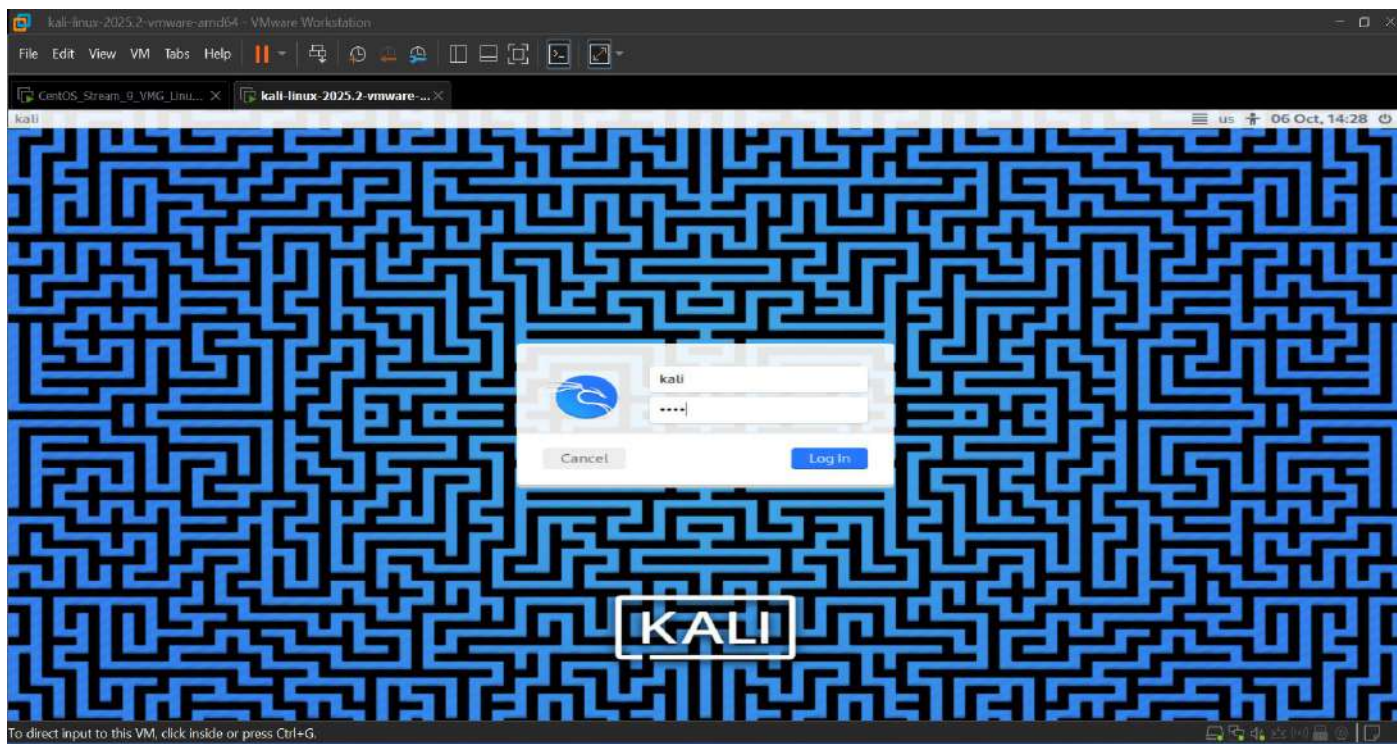
Activities Terminal Oct 6 14:21

```
root@centosstream9:/var/www/testsite.mydomain.local

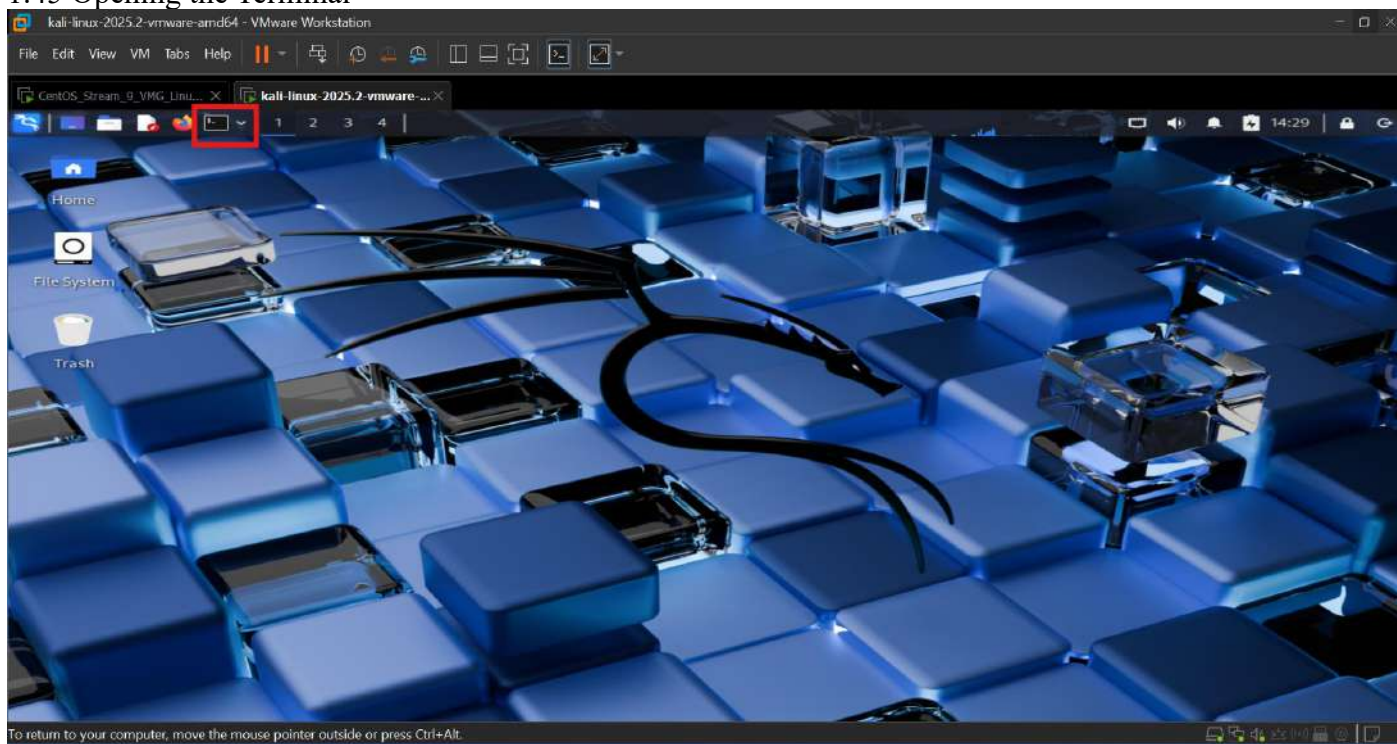
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# vi /etc/vsftpd/vsftpd.conf
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# systemctl start vsftpd
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# systemctl enable vsftpd
Created symlink /etc/systemd/system/multi-user.target.wants/vsftpd.service - /usr/lib/systemd/system/vsftpd.service.
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# systemctl start sshd
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# systemctl enable sshd
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:fd:44:63 brd ff:ff:ff:ff:ff:ff
    altname ens33
    inet 192.168.37.136/24 brd 192.168.37.255 scope global dynamic noprefixroute ens33
        valid lft 1435sec preferred_lft 1435sec
    inet6 fe80::28c:29ff:fe00:4467/64 scope link noprefixroute
        valid lft forever preferred_lft forever
[root@centosstream9 testsite.mydomain.local]#
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

1.44 Starting the kali Linux machine



1.45 Opening the Terminal



1.46 Checking IP Address by: ip a

The IP Address of Kali machine is 192.167.37.132

```
kali@kali:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:cd:43:f9 brd ff:ff:ff:ff:ff:ff
    inet 192.168.37.132/24 brd 192.168.37.255 scope global dynamic noprefixroute eth0
        valid_lft 1635sec preferred_lft 1635sec
    inet6 fe80::8f30:c8d:317:36c4/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

(kali@kali)~$
```

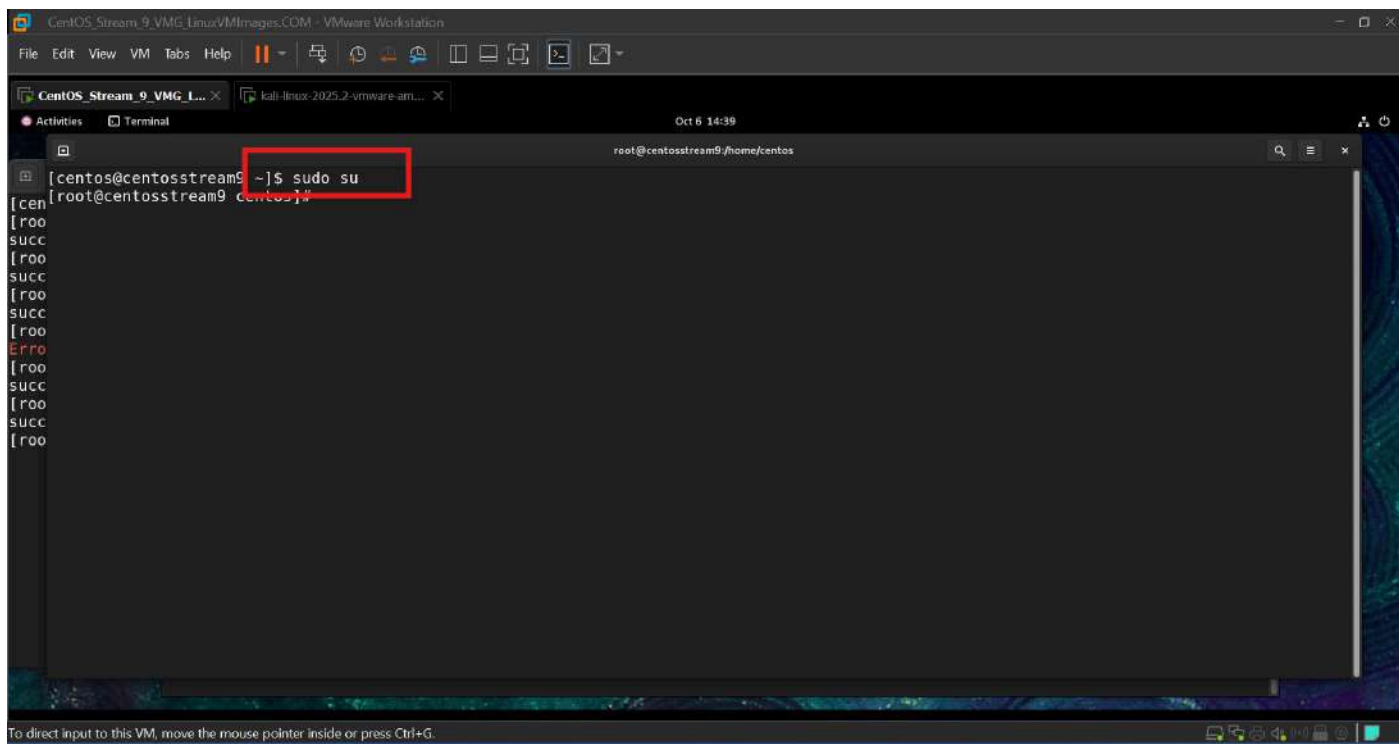
1.47 Pinging the CentOS machine by: ping 192.168.37.136

```
(kali@kali)~$ ping 192.168.37.136
PING 192.168.37.136 (192.168.37.136) 56(84) bytes of data:
64 bytes from 192.168.37.136: icmp_seq=1 ttl=64 time=34.9 ms
64 bytes from 192.168.37.136: icmp_seq=2 ttl=64 time=0.631 ms
64 bytes from 192.168.37.136: icmp_seq=3 ttl=64 time=0.685 ms
64 bytes from 192.168.37.136: icmp_seq=4 ttl=64 time=0.679 ms
64 bytes from 192.168.37.136: icmp_seq=5 ttl=64 time=0.732 ms
64 bytes from 192.168.37.136: icmp_seq=6 ttl=64 time=0.595 ms
64 bytes from 192.168.37.136: icmp_seq=7 ttl=64 time=0.623 ms
64 bytes from 192.168.37.136: icmp_seq=8 ttl=64 time=0.599 ms
64 bytes from 192.168.37.136: icmp_seq=9 ttl=64 time=0.712 ms
64 bytes from 192.168.37.136: icmp_seq=10 ttl=64 time=0.768 ms
64 bytes from 192.168.37.136: icmp_seq=11 ttl=64 time=0.705 ms
^C
--- 192.168.37.136 ping statistics ---
11 packets transmitted, 11 received, 0% packet loss, time 10183ms
rtt min/avg/max/mdev = 0.595/3.783/34.888/9.836 ms

(kali@kali)~$
```

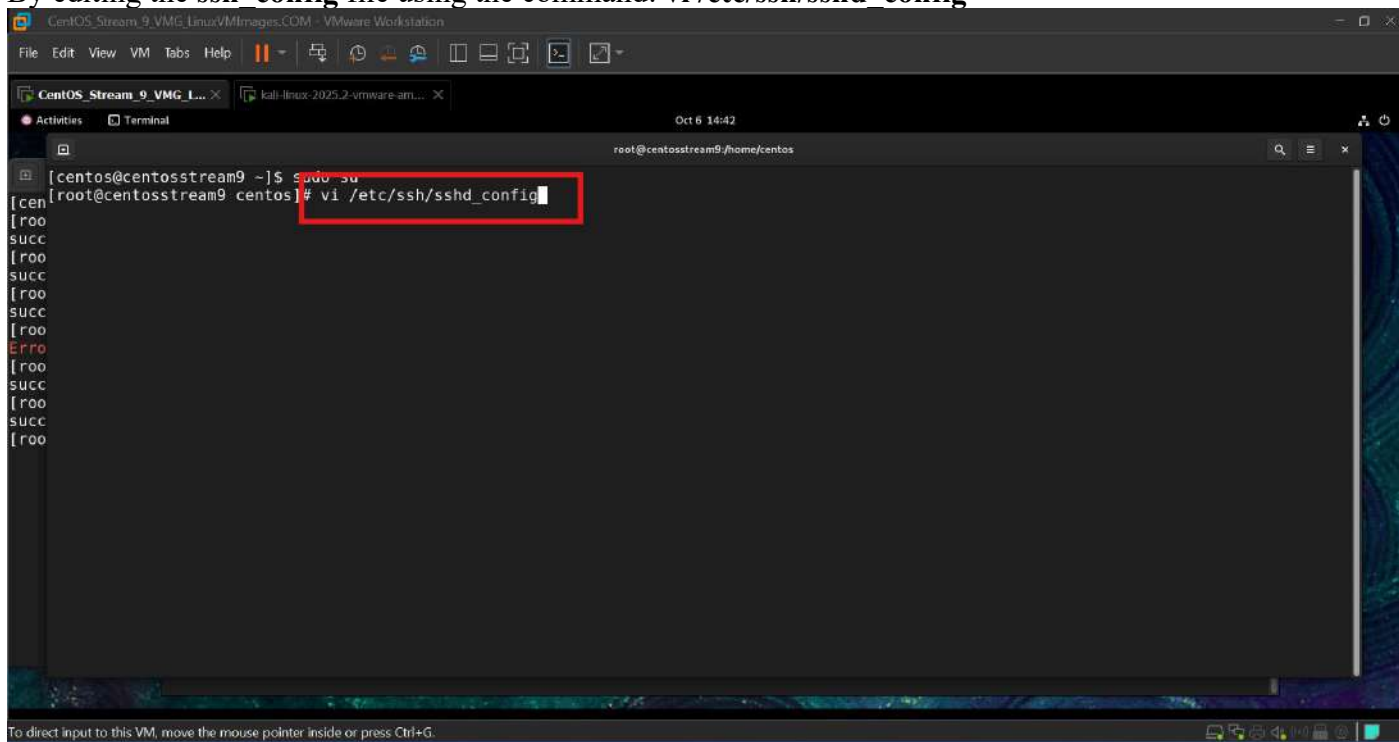
Task 2: SSH Hardening:

2.1 Navigating to CentOS and open the terminal then use the command: **sudo su -**

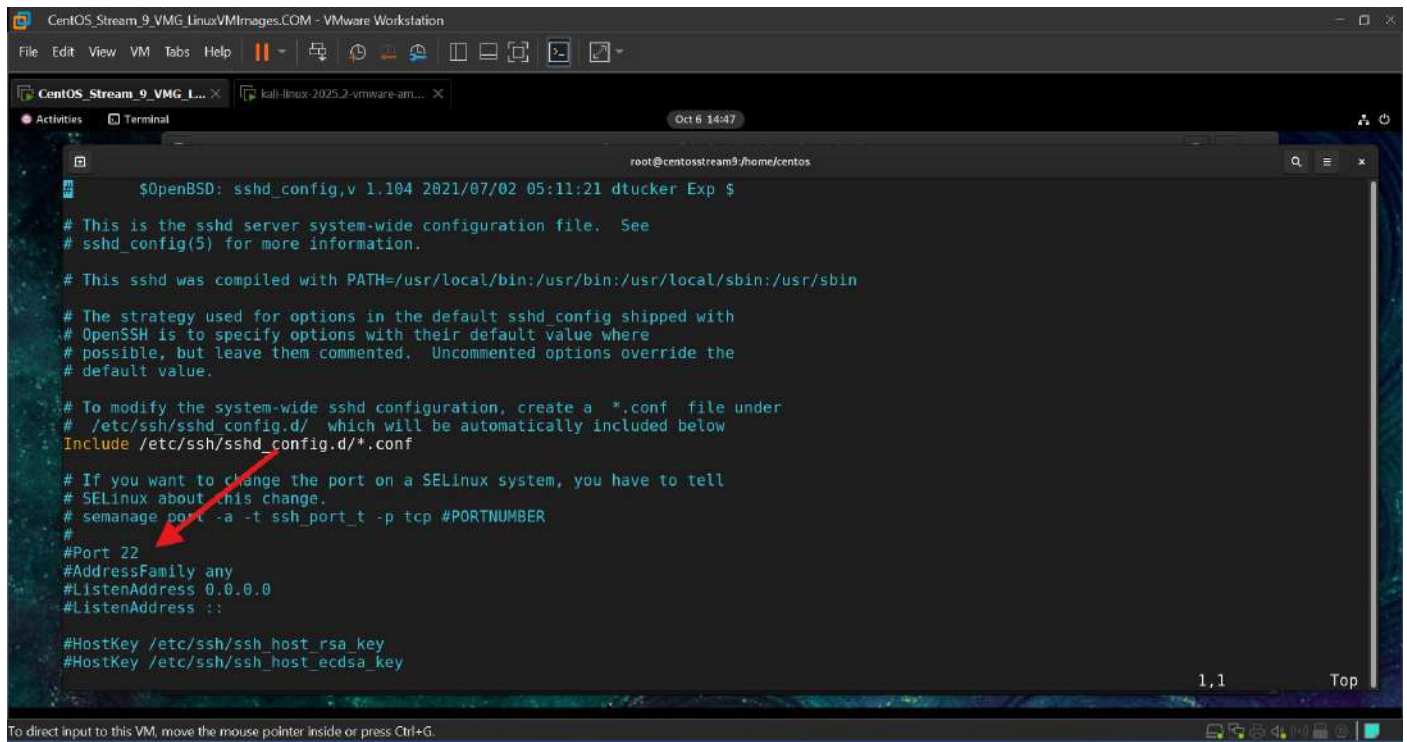


2.2 Configuring port number 2222 instead of 22

By editing the **ssh config** file using the command: **vi /etc/ssh/sshd_config**



2.3 Typed i to change “#Port22” to “Port2222”



```
CentOS_Stream_9_VMG_LinuxVMImages.COM - VMware Workstation
File Edit View VM Tabs Help
CentOS_Stream_9_VMG_L... x kali-linux-2025.2-vmware-am... x
Activities Terminal Oct 6 14:47
root@centosstream9:/home/centos

$OpenBSD: sshd_config,v 1.104 2021/07/02 05:11:21 dtucker Exp $

# This is the sshd server system-wide configuration file. See
# sshd_config(5) for more information.

# This sshd was compiled with PATH=/usr/local/bin:/usr/bin:/usr/local/sbin:/usr/sbin

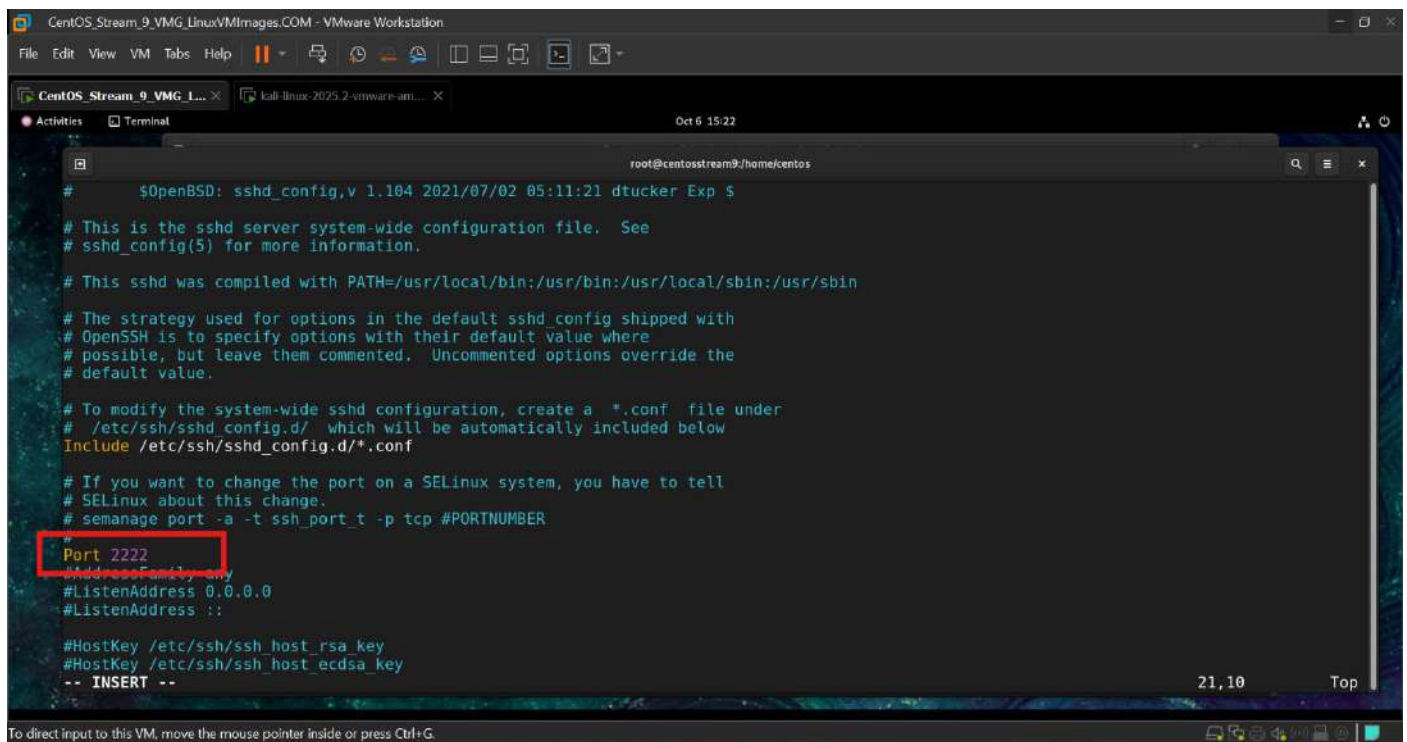
# The strategy used for options in the default sshd_config shipped with
# OpenSSH is to specify options with their default value where
# possible, but leave them commented. Uncommented options override the
# default value.

# To modify the system-wide sshd configuration, create a *.conf file under
# /etc/ssh/sshd_config.d/ which will be automatically included below
Include /etc/ssh/sshd_config.d/*.conf

# If you want to change the port on a SELinux system, you have to tell
# SELinux about this change.
# semanage port -a -t ssh_port_t -p tcp #PORTNUMBER
#
#Port 22
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.



```
CentOS_Stream_9_VMG_LinuxVMImages.COM - VMware Workstation
File Edit View VM Tabs Help
CentOS_Stream_9_VMG_L... x kali-linux-2025.2-vmware-am... x
Activities Terminal Oct 6 15:22
root@centosstream9:/home/centos

# $OpenBSD: sshd_config,v 1.104 2021/07/02 05:11:21 dtucker Exp $

# This is the sshd server system-wide configuration file. See
# sshd_config(5) for more information.

# This sshd was compiled with PATH=/usr/local/bin:/usr/bin:/usr/local/sbin:/usr/sbin

# The strategy used for options in the default sshd_config shipped with
# OpenSSH is to specify options with their default value where
# possible, but leave them commented. Uncommented options override the
# default value.

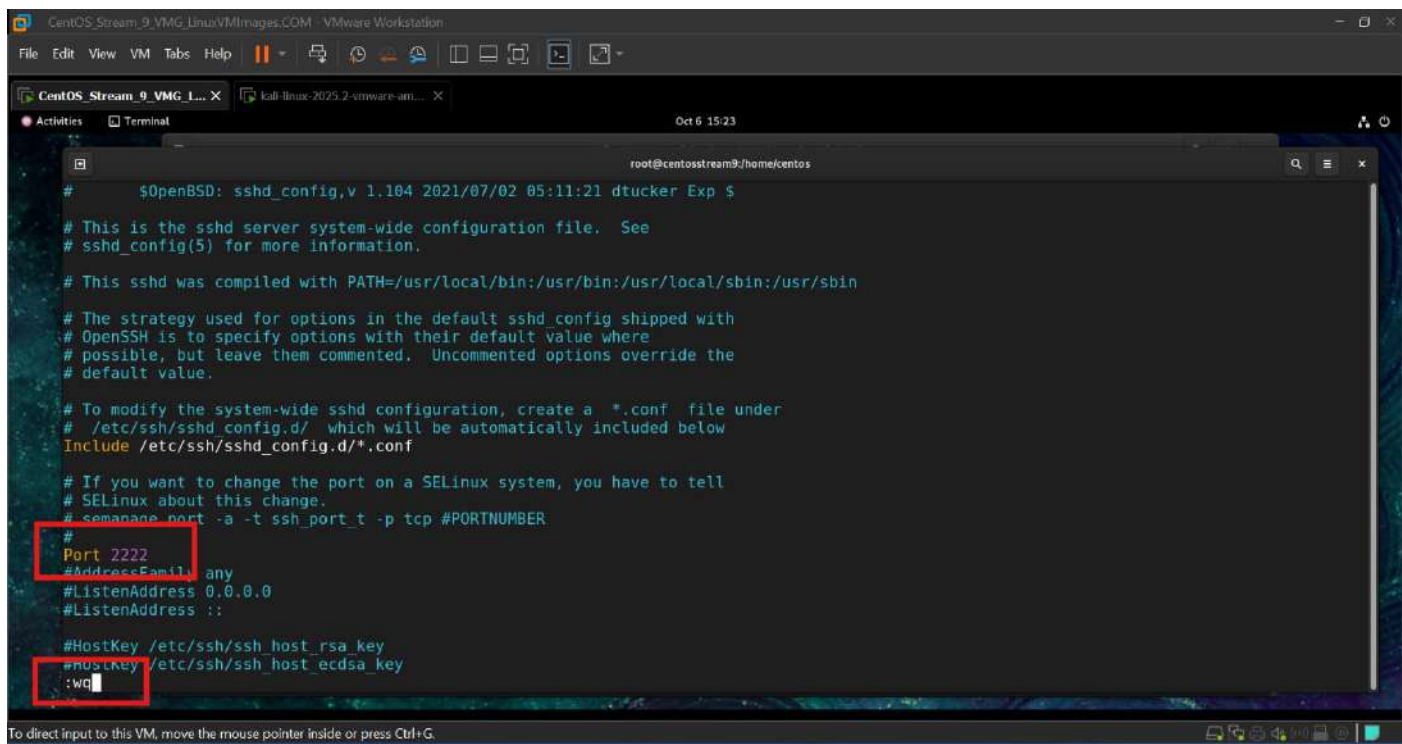
# To modify the system-wide sshd configuration, create a *.conf file under
# /etc/ssh/sshd_config.d/ which will be automatically included below
Include /etc/ssh/sshd_config.d/*.conf

# If you want to change the port on a SELinux system, you have to tell
# SELinux about this change.
# semanage port -a -t ssh_port_t -p tcp #PORTNUMBER
#
#Port 2222
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa key
-- INSERT --
```

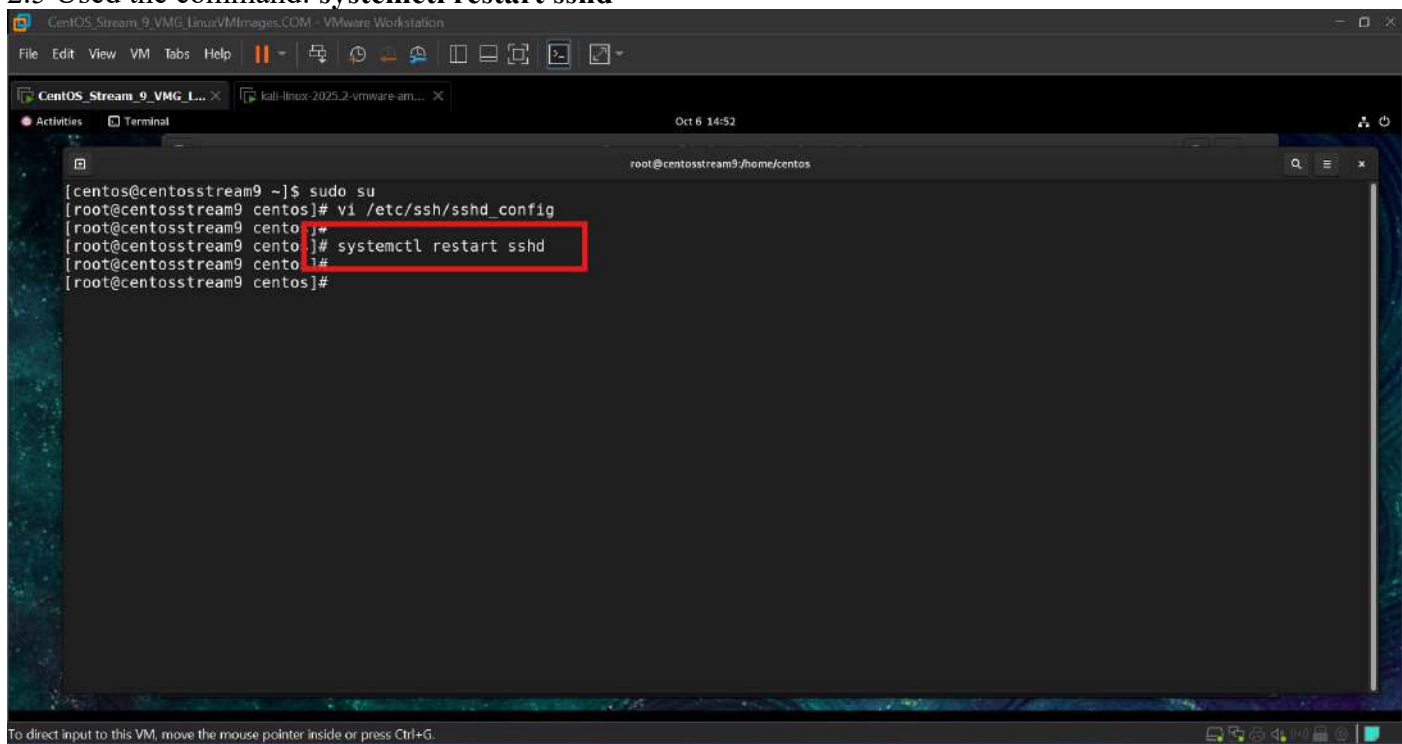
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

2.4 Saved and Exited the File by Pressing Esc, Shift + Colon (:), Typing wq, and then Enter



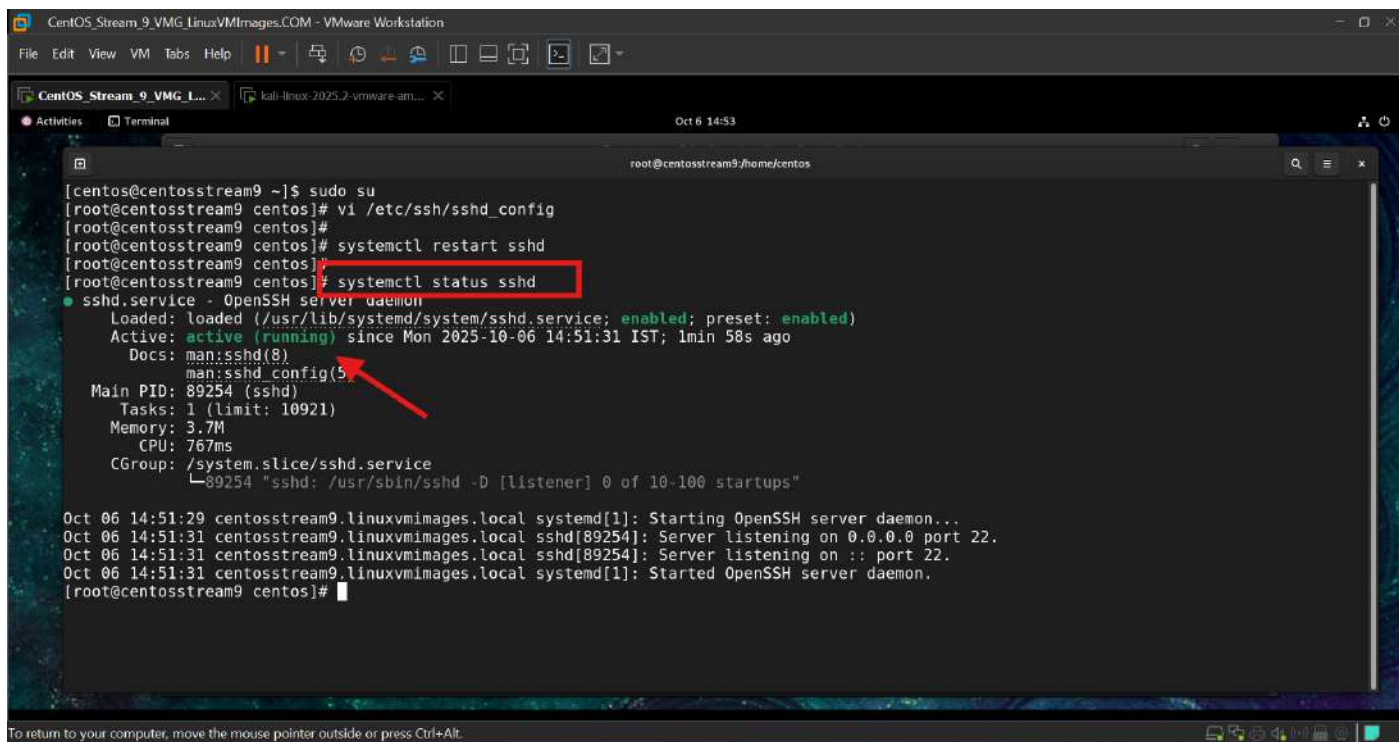
```
CentOS_Stream_9_VMG_LinuxVMImages.COM - VMware Workstation
File Edit View VM Tabs Help
CentOS_Stream_9_VMG_1... X kali-linux-2025.2-vmware-am... X
Activities Terminal Oct 6 15:23
root@centosstream9:/home/centos
# $OpenBSD: sshd_config,v 1.104 2021/07/02 05:11:21 dtucker Exp $
# This is the sshd server system-wide configuration file. See
# sshd_config(5) for more information.
# This sshd was compiled with PATH=/usr/local/bin:/usr/bin:/usr/local/sbin:/usr/sbin
# The strategy used for options in the default sshd_config shipped with
# OpenSSH is to specify options with their default value where
# possible, but leave them commented. Uncommented options override the
# default value.
# To modify the system-wide sshd configuration, create a *.conf file under
# /etc/ssh/sshd_config.d/ which will be automatically included below
Include /etc/ssh/sshd_config.d/*.conf
# If you want to change the port on a SELinux system, you have to tell
# SELinux about this change.
# semanage port -a -t ssh_port_t -p tcp #PORTNUMBER
#
Port 2222
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::
#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
:wg
```

2.5 Used the command: **systemctl restart sshd**



```
CentOS_Stream_9_VMG_LinuxVMImages.COM - VMware Workstation
File Edit View VM Tabs Help
CentOS_Stream_9_VMG_1... X kali-linux-2025.2-vmware-am... X
Activities Terminal Oct 6 14:52
root@centosstream9:/home/centos
[centos@centosstream9 ~]$ sudo su
[root@centosstream9 centos]# vi /etc/ssh/sshd_config
[centos@centosstream9 ~]$ systemctl restart sshd
```

2.6 Used the command: **systemctl status sshd**



CentOS_Stream_9_VMG_LinuxVMImages.COM - VMware Workstation

File Edit View VM Tabs Help

CentOS_Stream_9_VMG_L... kali-linux-2025.2-vmware-am...

Activities Terminal

Oct 6 14:53

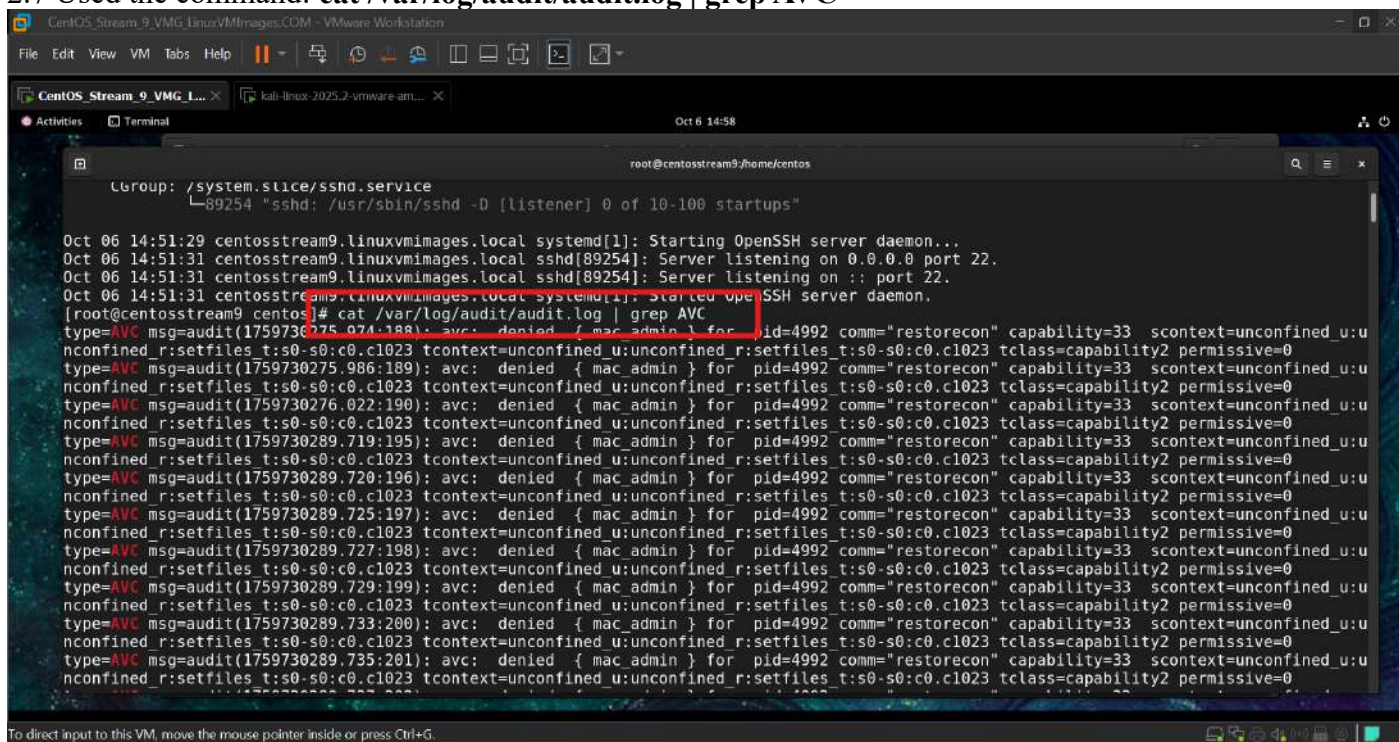
root@centosstream9:/home/centos

```
[centos@centosstream9 ~]$ sudo su
[root@centosstream9 centos]# vi /etc/ssh/sshd_config
[root@centosstream9 centos]#
[root@centosstream9 centos]# systemctl restart sshd
[root@centosstream9 centos]# systemctl status sshd
● sshd.service - OpenSSH server daemon
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Mon 2025-10-06 14:51:31 IST; 1min 58s ago
     Docs: man:sshd(8)
           man:sshd_config(5)
   Main PID: 89254 (sshd)
      Tasks: 1 (limit: 10921)
     Memory: 3.7M
        CPU: 767ms
    CGroup: /system.slice/ssh.service
            └─89254 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Oct 06 14:51:29 centosstream9.linuxvmimages.local systemd[1]: Starting OpenSSH server daemon...
Oct 06 14:51:31 centosstream9.linuxvmimages.local sshd[89254]: Server listening on 0.0.0.0 port 22.
Oct 06 14:51:31 centosstream9.linuxvmimages.local sshd[89254]: Server listening on :: port 22.
Oct 06 14:51:31 centosstream9.linuxvmimages.local systemd[1]: Started OpenSSH server daemon.
[root@centosstream9 centos]#
```

To return to your computer, move the mouse pointer outside or press Ctrl+Alt.

2.7 Used the command: `cat /var/log/audit/audit.log | grep AVC`



CentOS_Stream_9_VMG_LinuxVMImages.COM - VMware Workstation

File Edit View VM Tabs Help

CentOS_Stream_9_VMG_L... kali-linux-2025.2-vmware-am...

Activities Terminal

Oct 6 14:58

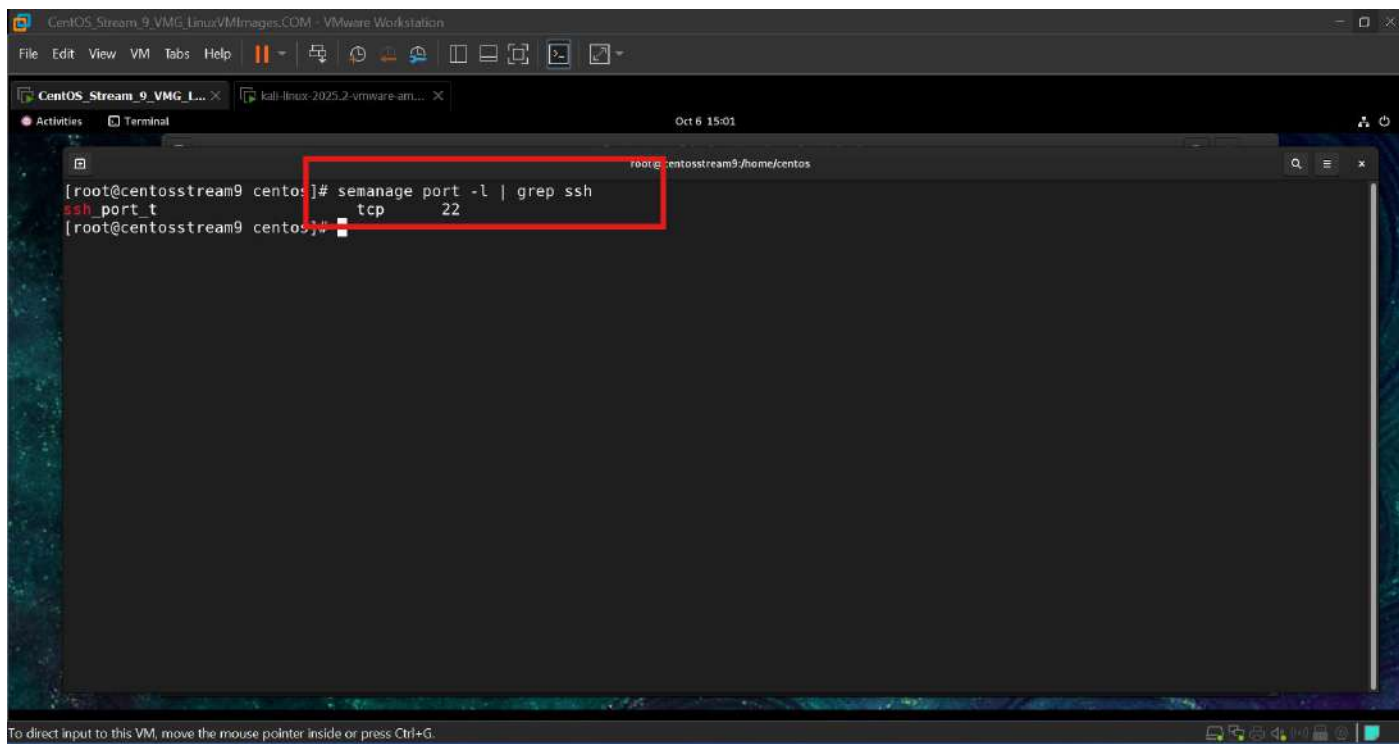
root@centosstream9:/home/centos

```
CGroup: /system.slice/ssh.service
└─89254 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Oct 06 14:51:29 centosstream9.linuxvmimages.local systemd[1]: Starting OpenSSH server daemon...
Oct 06 14:51:31 centosstream9.linuxvmimages.local sshd[89254]: Server listening on 0.0.0.0 port 22.
Oct 06 14:51:31 centosstream9.linuxvmimages.local sshd[89254]: Server listening on :: port 22.
Oct 06 14:51:31 centosstream9.linuxvmimages.local systemd[1]: Started OpenSSH server daemon.
[root@centosstream9 centos]# cat /var/log/audit/audit.log | grep AVC
type=AVC msg=audit(1759730275.074:188): avc: denied { mac_admin } for pid=4992 comm="restorecon" capability=33 scontext=unconfined_u:u
nconfined_r:setfiles t:s0-s0:c0.c1023 tcontext=unconfined_u:unconfined_r:setfiles t:s0-s0:c0.c1023 tclass=capability2 permissive=0
type=AVC msg=audit(1759730275.086:189): avc: denied { mac_admin } for pid=4992 comm="restorecon" capability=33 scontext=unconfined_u:u
nconfined_r:setfiles t:s0-s0:c0.c1023 tcontext=unconfined_u:unconfined_r:setfiles t:s0-s0:c0.c1023 tclass=capability2 permissive=0
type=AVC msg=audit(1759730276.022:190): avc: denied { mac_admin } for pid=4992 comm="restorecon" capability=33 scontext=unconfined_u:u
nconfined_r:setfiles t:s0-s0:c0.c1023 tcontext=unconfined_u:unconfined_r:setfiles t:s0-s0:c0.c1023 tclass=capability2 permissive=0
type=AVC msg=audit(1759730289.719:195): avc: denied { mac_admin } for pid=4992 comm="restorecon" capability=33 scontext=unconfined_u:u
nconfined_r:setfiles t:s0-s0:c0.c1023 tcontext=unconfined_u:unconfined_r:setfiles t:s0-s0:c0.c1023 tclass=capability2 permissive=0
type=AVC msg=audit(1759730289.720:196): avc: denied { mac_admin } for pid=4992 comm="restorecon" capability=33 scontext=unconfined_u:u
nconfined_r:setfiles t:s0-s0:c0.c1023 tcontext=unconfined_u:unconfined_r:setfiles t:s0-s0:c0.c1023 tclass=capability2 permissive=0
type=AVC msg=audit(1759730289.725:197): avc: denied { mac_admin } for pid=4992 comm="restorecon" capability=33 scontext=unconfined_u:u
nconfined_r:setfiles t:s0-s0:c0.c1023 tcontext=unconfined_u:unconfined_r:setfiles t:s0-s0:c0.c1023 tclass=capability2 permissive=0
type=AVC msg=audit(1759730289.727:198): avc: denied { mac_admin } for pid=4992 comm="restorecon" capability=33 scontext=unconfined_u:u
nconfined_r:setfiles t:s0-s0:c0.c1023 tcontext=unconfined_u:unconfined_r:setfiles t:s0-s0:c0.c1023 tclass=capability2 permissive=0
type=AVC msg=audit(1759730289.729:199): avc: denied { mac_admin } for pid=4992 comm="restorecon" capability=33 scontext=unconfined_u:u
nconfined_r:setfiles t:s0-s0:c0.c1023 tcontext=unconfined_u:unconfined_r:setfiles t:s0-s0:c0.c1023 tclass=capability2 permissive=0
type=AVC msg=audit(1759730289.733:200): avc: denied { mac_admin } for pid=4992 comm="restorecon" capability=33 scontext=unconfined_u:u
nconfined_r:setfiles t:s0-s0:c0.c1023 tcontext=unconfined_u:unconfined_r:setfiles t:s0-s0:c0.c1023 tclass=capability2 permissive=0
type=AVC msg=audit(1759730289.735:201): avc: denied { mac_admin } for pid=4992 comm="restorecon" capability=33 scontext=unconfined_u:u
nconfined_r:setfiles t:s0-s0:c0.c1023 tcontext=unconfined_u:unconfined_r:setfiles t:s0-s0:c0.c1023 tclass=capability2 permissive=0
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

2.8 Used the command: `semanage port -l | grep ssh`



CentOS_Stream_9_VMG_LinuxVMImages.COM - VMware Workstation

File Edit View VM Tabs Help

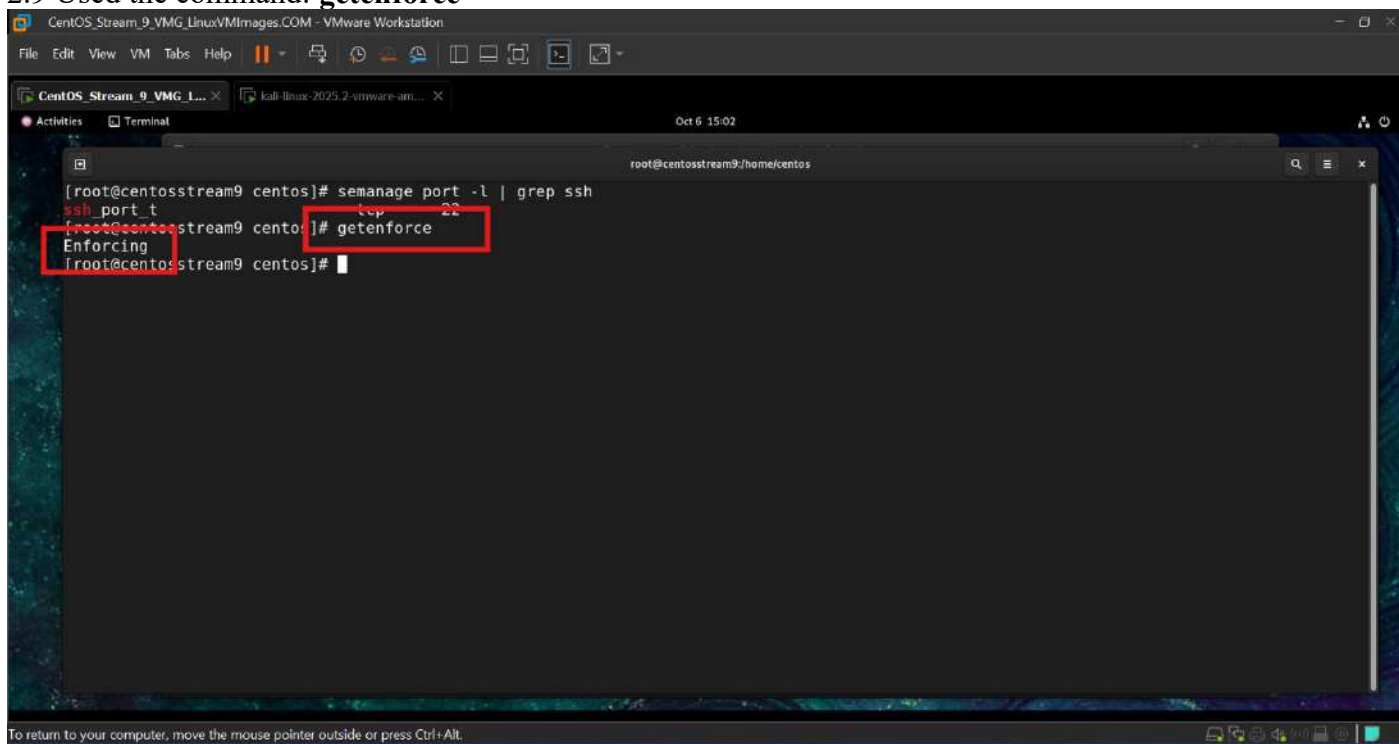
CentOS_Stream_9_VMG_L... X kali-linux-2025.2-vmware-am... X

Activities Terminal Oct 6 15:01

```
root@centosstream9:/home/centos
[root@centosstream9 centos]# semanage port -l | grep ssh
ssh_port_t tcp 22
[root@centosstream9 centos]#
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

2.9 Used the command: **getenforce**



CentOS_Stream_9_VMG_LinuxVMImages.COM - VMware Workstation

File Edit View VM Tabs Help

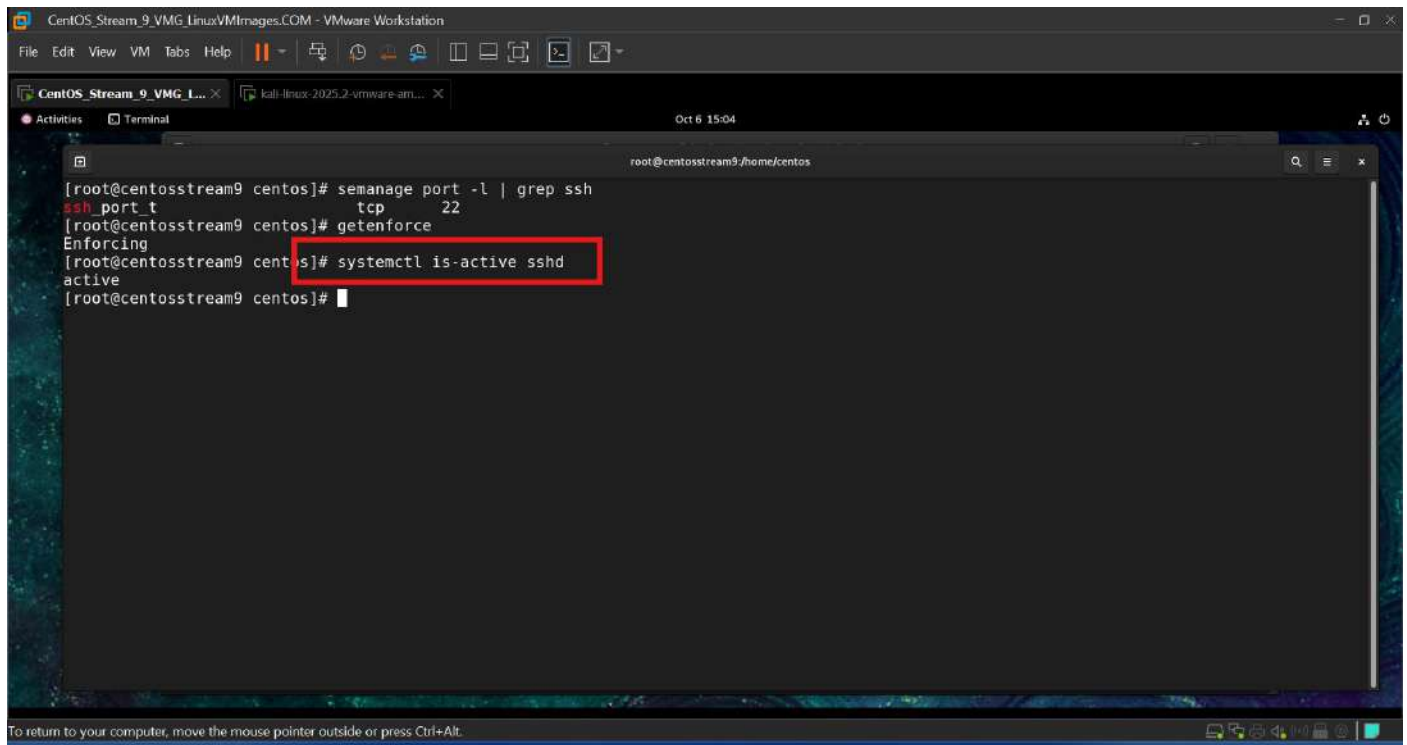
CentOS_Stream_9_VMG_L... X kali-linux-2025.2-vmware-am... X

Activities Terminal Oct 6 15:02

```
root@centosstream9:/home/centos
[root@centosstream9 centos]# semanage port -l | grep ssh
ssh_port_t tcp 22
[root@centosstream9 centos]# getenforce
Enforcing
[root@centosstream9 centos]#
```

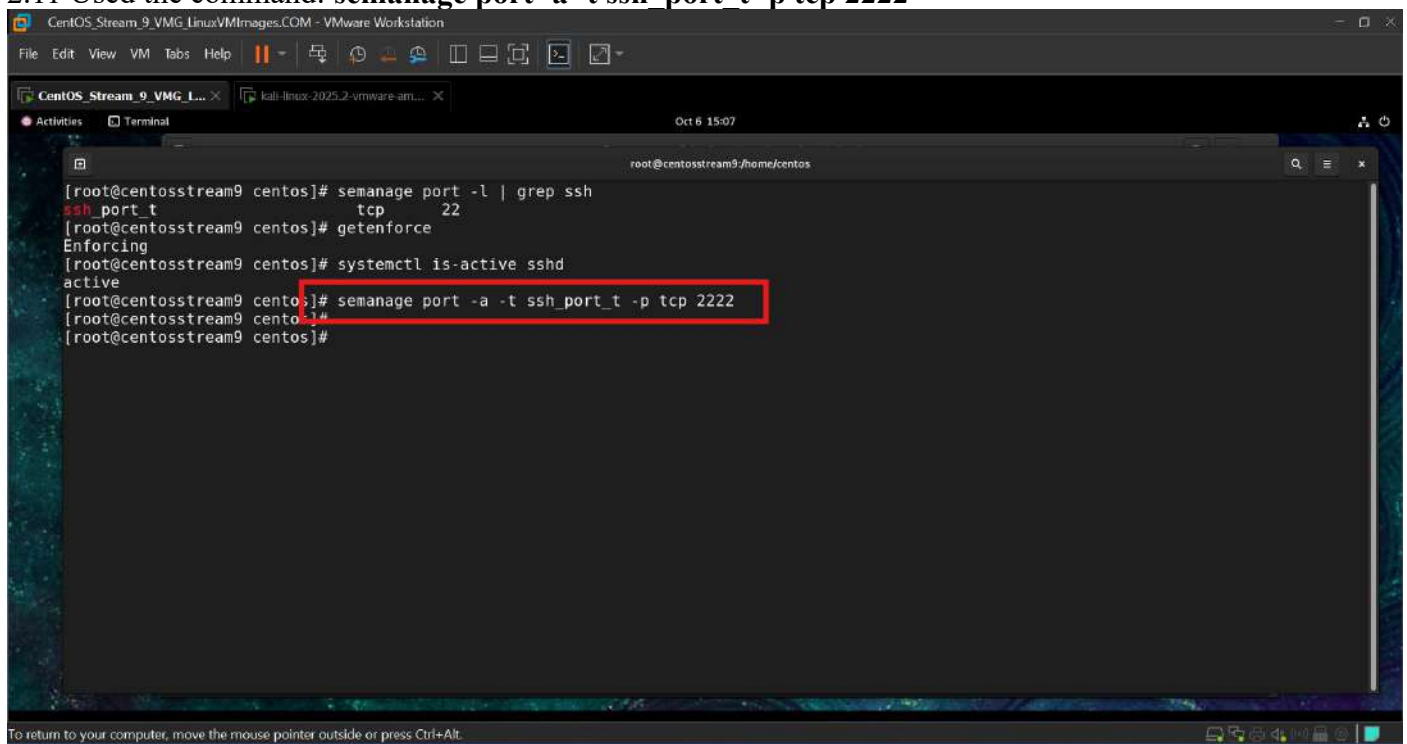
To return to your computer, move the mouse pointer outside or press Ctrl+Alt.

2.10 Used the command: **systemctl is-active sshd**



```
[root@centosstream9 centos]# semanage port -l | grep ssh
ssh_port_t                tcp                22
[root@centosstream9 centos]# getenforce
Enforcing
[root@centosstream9 centos]# systemctl is-active sshd
active
[root@centosstream9 centos]#
```

2.11 Used the command: **semanage port -a -t ssh_port_t -p tcp 2222**



```
[root@centosstream9 centos]# semanage port -l | grep ssh
ssh_port_t                tcp                22
[root@centosstream9 centos]# getenforce
Enforcing
[root@centosstream9 centos]# systemctl is-active sshd
active
[root@centosstream9 centos]# semanage port -a -t ssh_port_t -p tcp 2222
[root@centosstream9 centos]#
```

2.12 Used the command: **systemctl restart sshd**

```
CentOS_Stream_9_VMG_LinuxVMImages.COM - VMware Workstation
File Edit View VM Tabs Help
CentOS_Stream_9_VMG_L... X kali-linux-2025.2-vmware-am... X
Activities Terminal
Oct 6 15:08
root@centosstream9:/home/centos
[root@centosstream9 centos]# semanage port -l | grep ssh
ssh_port_t                tcp                22
[root@centosstream9 centos]# getenforce
Enforcing
[root@centosstream9 centos]# systemctl is-active sshd
active
[root@centosstream9 centos]# semanage port -a -t ssh_port_t -p tcp 2222
[root@centosstream9 centos]#
[root@centosstream9 centos]# systemctl restart sshd
[root@centosstream9 centos]#
```

2.13 Used the command: **systemctl status sshd**

```
CentOS_Stream_9_VMG_LinuxVMImages.COM - VMware Workstation
File Edit View VM Tabs Help
CentOS_Stream_9_VMG_L... X kali-linux-2025.2-vmware-am... X
Activities Terminal
Oct 6 15:09
root@centosstream9:/home/centos
[root@centosstream9 centos]# semanage port -l | grep ssh
ssh_port_t                tcp                22
[root@centosstream9 centos]# getenforce
Enforcing
[root@centosstream9 centos]# systemctl is-active sshd
active
[root@centosstream9 centos]# semanage port -a -t ssh_port_t -p tcp 2222
[root@centosstream9 centos]#
[root@centosstream9 centos]# systemctl restart sshd
[root@centosstream9 centos]#
[root@centosstream9 centos]# systemctl status sshd
● sshd.service - OpenSSH server daemon
   Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: enabled)
   Active: active (running) since Mon 2025-10-06 15:08:33 IST; 36s ago
     Docs: man:sshd(8)
           man:sshd_config(5)
  Main PID: 89430 (sshd)
    Tasks: 1 (limit: 10921)
  Memory: 1.5M
    CPU: 94ms
  CGroup: /system.slice/sshd.service
          └─89430 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Oct 06 15:08:33 centosstream9.linuxvmimages.local systemd[1]: Starting OpenSSH server daemon...
Oct 06 15:08:33 centosstream9.linuxvmimages.local sshd[89430]: Server listening on 0.0.0.0 port 22.
Oct 06 15:08:33 centosstream9.linuxvmimages.local sshd[89430]: Server listening on :: port 22.
Oct 06 15:08:33 centosstream9.linuxvmimages.local systemd[1]: Started OpenSSH server daemon.
[root@centosstream9 centos]#
```

2.14 Navigated to Kali Linux and trying to connect with ssh

Used the command: **ssh -p 2222 [military@192.168.37.136](#)** and password.


```
kali-linux-2025.2-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
CentOS_Stream_9_VMG_Linux... X kali-linux-2025.2-vmware-... X
kali@kali: ~
File Actions Edit View Help
(kali@kali)-[~]
$ ssh -p 22 militry@192.168.37.136
+-----+
+-----+ LINUXVMIMAGES.COM +-----+
+-----+ User Name: centos +-----+
+-----+ Password: centos (sudo su -) +-----+
militry@192.168.37.136's password:
(kali@kali)-[~]
$ ssh -p 2222 militry@192.168.37.136
+-----+
+-----+ LINUXVMIMAGES.COM +-----+
+-----+ User Name: centos +-----+
+-----+ Password: centos (sudo su -) +-----+
militry@192.168.37.136's password:
Permission denied, please try again.
militry@192.168.37.136's password:
Permission denied, please try again.
militry@192.168.37.136's password:
militry@192.168.37.136: Permission denied (publickey,gssapi-keyex,gssapi-with-mic,password).
(kali@kali)-[~]
$
```

2.15 Using the command: `ssh -p 2222 kaliya@192.168.37.136` and password.

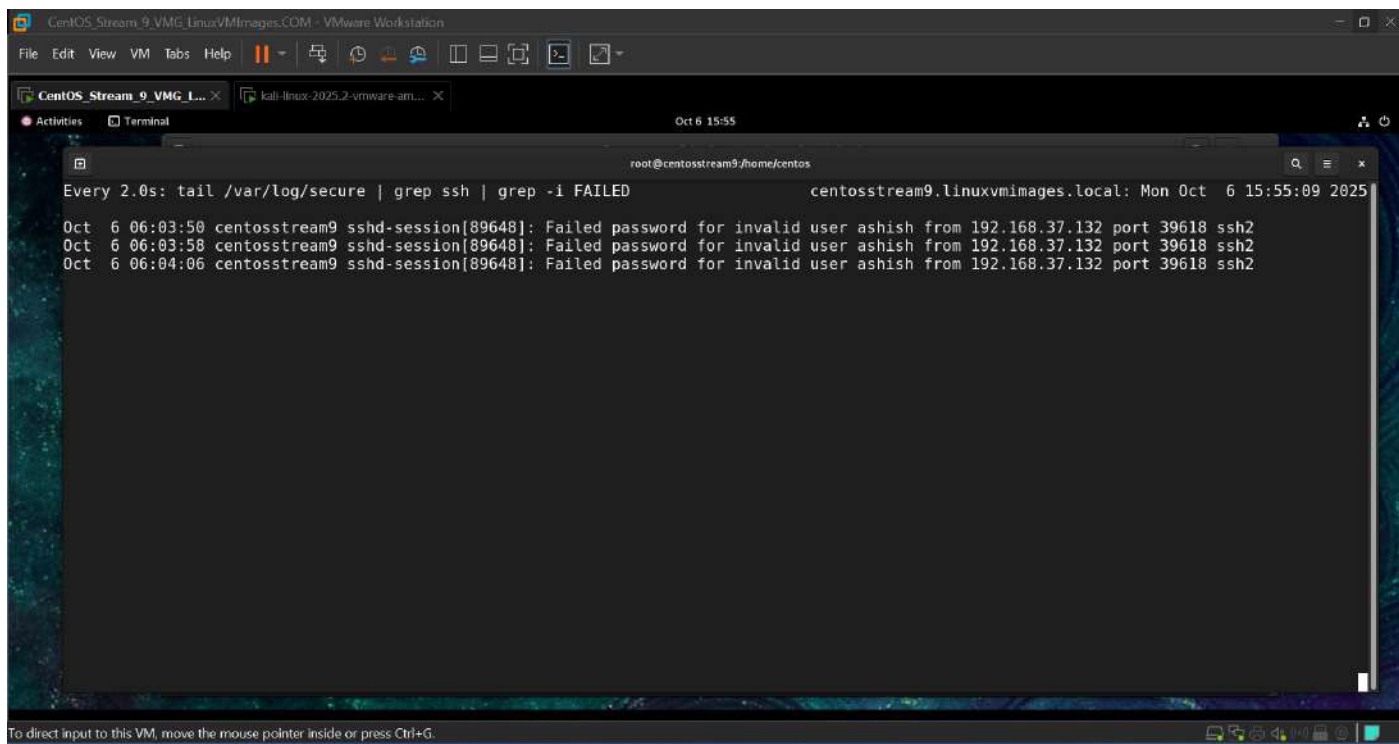
```
kali-linux-2025.2-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
CentOS_Stream_9_VMG_Linux... X kali-linux-2025.2-vmware-... X
kali@kali: ~
File Actions Edit View Help
LINUXVMIMAGES.COM
+-----+
+-----+ User Name: centos +-----+
+-----+ Password: centos (sudo su -) +-----+
militry@192.168.37.136's password:
Permission denied, please try again.
militry@192.168.37.136's password:
Permission denied, please try again.
militry@192.168.37.136's password:
militry@192.168.37.136: Permission denied (publickey,gssapi-keyex,gssapi-with-mic,password).
(kali@kali)-[~]
$ ssh -p 2222 kaliya@192.168.37.136
+-----+
+-----+ LINUXVMIMAGES.COM +-----+
+-----+ User Name: centos +-----+
+-----+ Password: centos (sudo su -) +-----+
kaliya@192.168.37.136's password:
Permission denied, please try again.
kaliya@192.168.37.136's password:
Permission denied, please try again.
kaliya@192.168.37.136's password:
kaliya@192.168.37.136: Permission denied (publickey,gssapi-keyex,gssapi-with-mic,password).
(kali@kali)-[~]
$
```

2.16 Using the command: `ssh -p 2222 ashish@192.168.37.136` and password.

```
kali@kali:~$ ssh -p 2222 ashish@192.168.37.136
kali@kali:~$
```

2.17 Now going to see how many failed login attempts were done on ssh.
Used the command: **watch 'tail /var/log/secure | grep ssh | grep -i FAILED'**

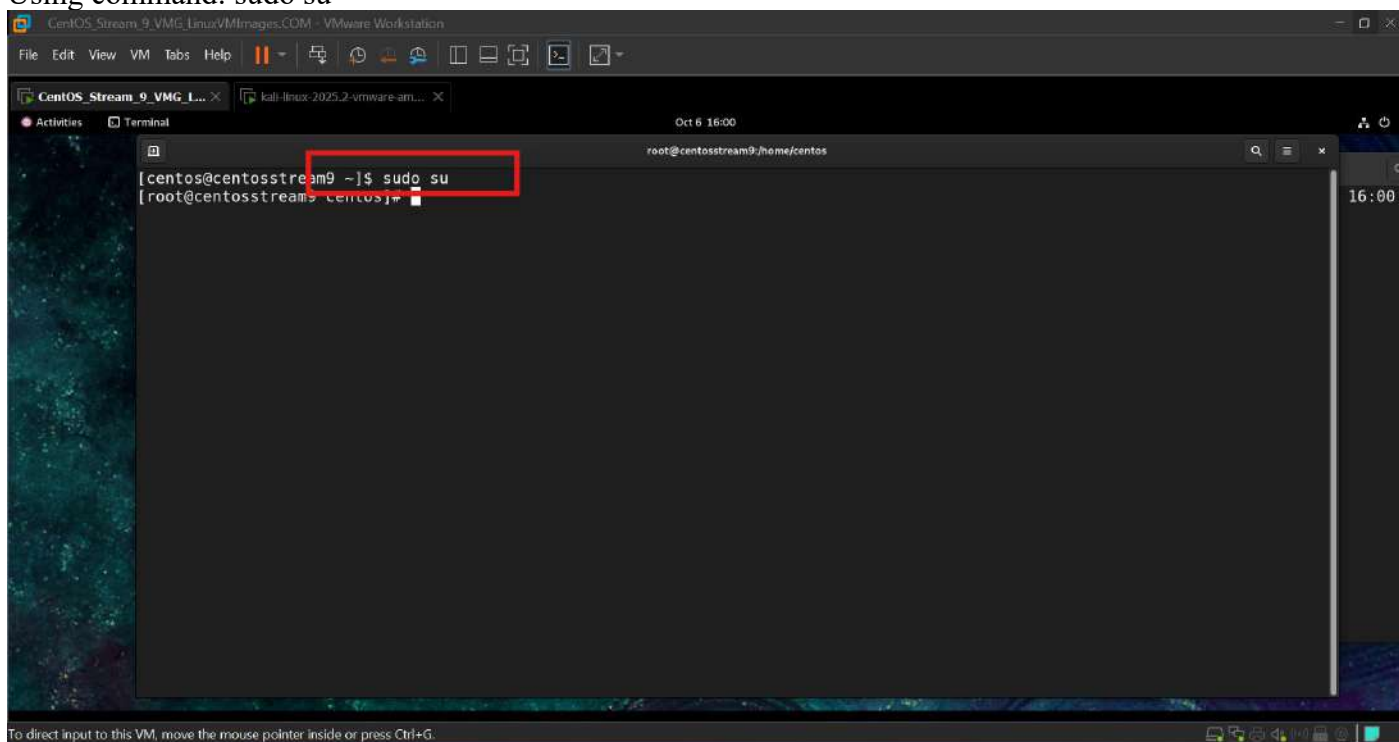
```
[root@centosstream9 ~]# watch 'tail /var/log/secure | grep ssh | grep -i FAILED'
```



The screenshot shows a terminal window within a VMware Workstation environment. The terminal is running the command `tail /var/log/secure | grep ssh | grep -i FAILED` every 2.0 seconds. The output shows three failed SSH login attempts for the user 'ashish' from IP address 192.168.37.132 on port 39618. The terminal prompt is `root@centosstream9:/home/centos`. The window title is 'CentOS_Stream_9_VMG_LinuxVMImages.COM - VMware Workstation'.

```
Every 2.0s: tail /var/log/secure | grep ssh | grep -i FAILED          centosstream9.linuxvmimages.local: Mon Oct  6 15:55:09 2025
Oct  6 06:03:50 centosstream9 sshd-session[89648]: Failed password for invalid user ashish from 192.168.37.132 port 39618 ssh2
Oct  6 06:03:58 centosstream9 sshd-session[89648]: Failed password for invalid user ashish from 192.168.37.132 port 39618 ssh2
Oct  6 06:04:06 centosstream9 sshd-session[89648]: Failed password for invalid user ashish from 192.168.37.132 port 39618 ssh2
```

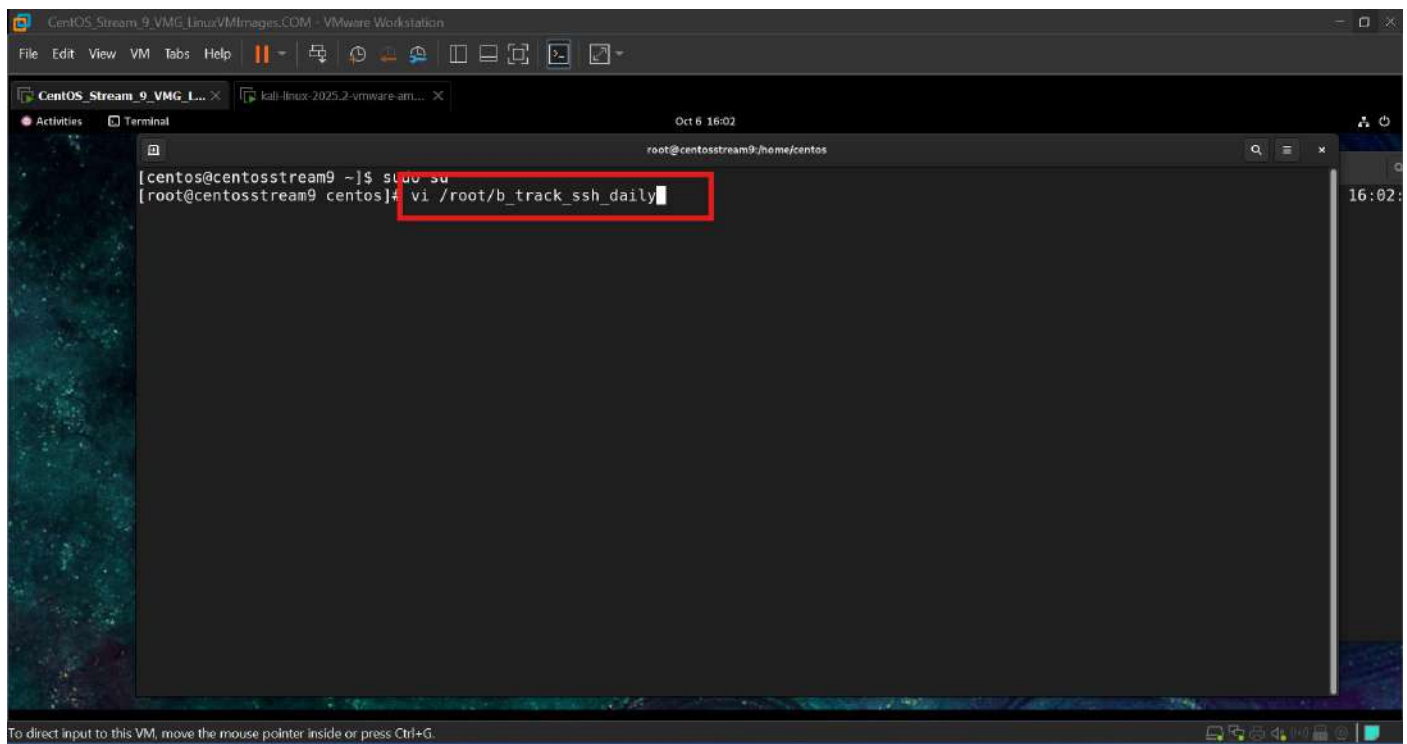
2.18 Creating a script to log all IP address which were used to attack ssh Using command: `sudo su`



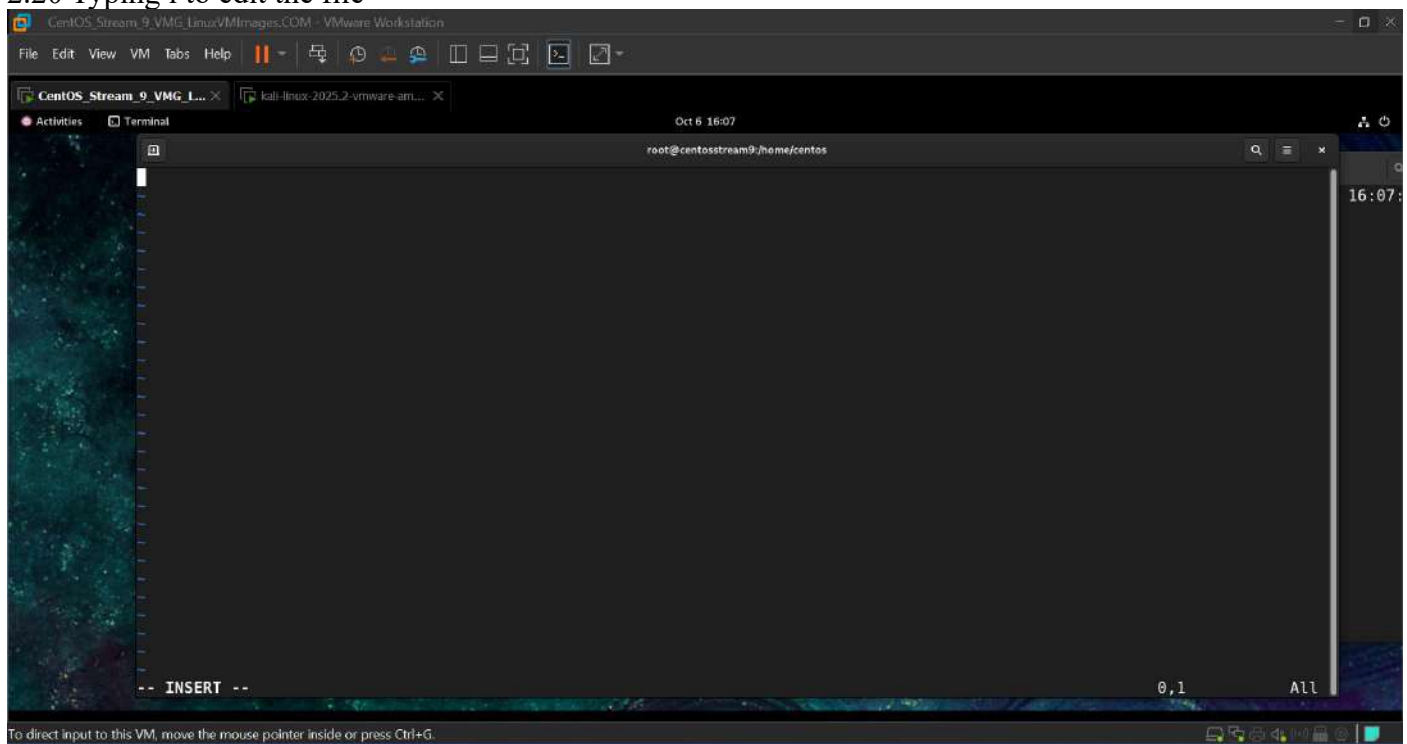
The screenshot shows a terminal window within a VMware Workstation environment. The terminal is running the command `sudo su`. The output shows the prompt changing from `[centos@centosstream9 ~]$` to `[root@centosstream9 ~]#`. The terminal prompt is `root@centosstream9:/home/centos`. The window title is 'CentOS_Stream_9_VMG_LinuxVMImages.COM - VMware Workstation'.

```
[centos@centosstream9 ~]$ sudo su
[root@centosstream9 ~]#
```

2.19 Editing the `b_track_ssh_daily` file using the command: `vi /root/b_track_ssh_daily`



2.20 Typing i to edit the file



2.21 Writing this script

```
#!/bin/bash
```

```
# Remove previous log file
```

```
rm -f ~/ssh_failed_sortlisted_logins.txt
```

```
# Extract unique failed SSH login IPs from logs
```

```
find /var/log -name "secure*" -exec cat {} \; | grep ssh | grep -i "Failed" | grep -E -o "([0-9]{1,3} [\.\ ]){3} [0-9]{1,3}" | sort -u > ~/ssh_failed_sortlisted_logins.txt
```

```

# Remove temporary files
rm -f ~/1.txt ~/2.txt ~/3.htm

# Process each IP address
while read -r ip; do
    whois_info=$(whois "$ip" | grep -i "country" | tr "a-z" "A-Z" | sort -u)

    echo "$ip" >> ~/1.txt
    if [[ -n "$whois_info" ]]; then
        echo "$whois_info" >> ~/1.txt
    else
        echo "COUNTRY: UNKNOWN" >> ~/1.txt
    fi
done < ~/ssh_failed_sortlisted_logins.txt

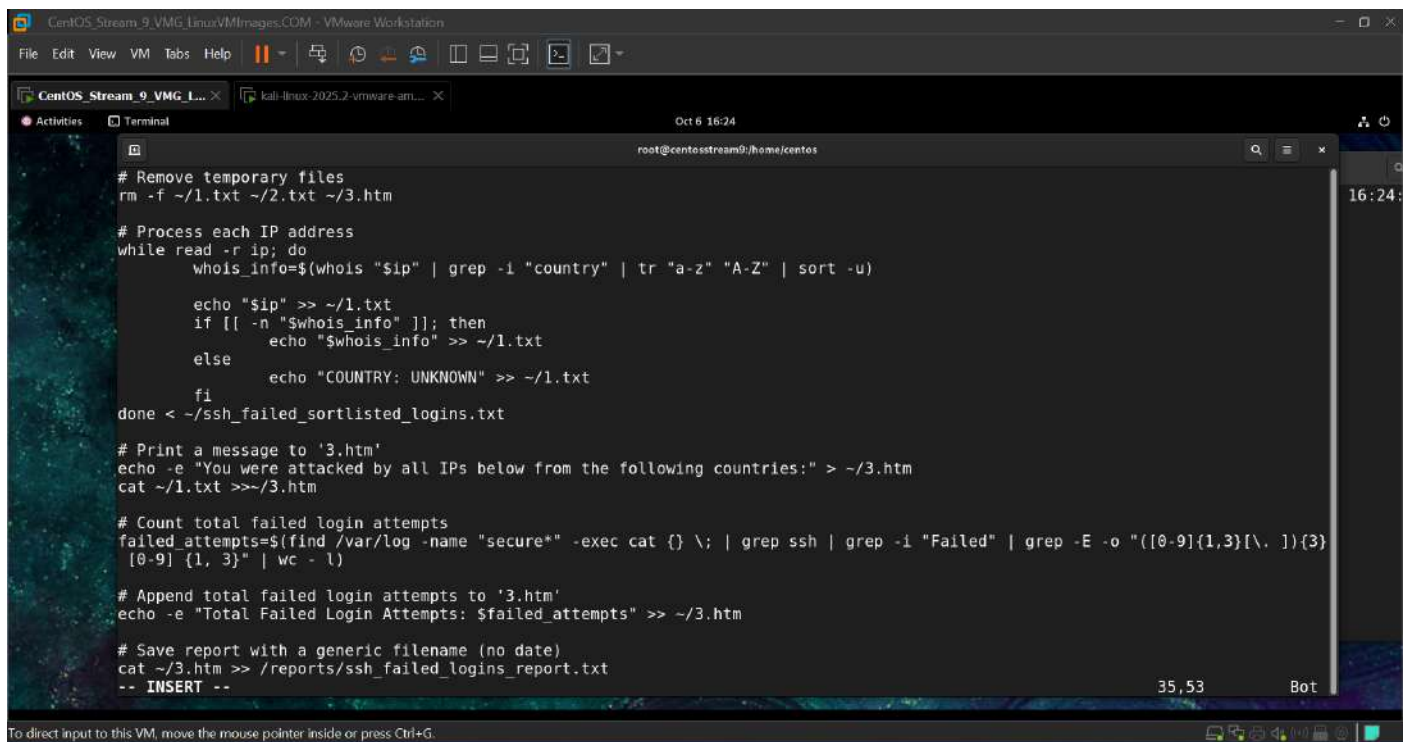
# Print a message to '3.htm'
echo -e "You were attacked by all IPs below from the following countries:" > ~/3.htm
cat ~/1.txt >> ~/3.htm

# Count total failed login attempts
failed_attempts=$(find /var/log -name "secure*" -exec cat {} \; | grep ssh | grep -i "Failed" | grep -E -o "([0-9]{1,3}[\. ]){3} [0-9] {1, 3}" | wc -l)

# Append total failed login attempts to '3.htm'
echo -e "Total Failed Login Attempts: $failed_attempts" >> ~/3.htm

# Save report with a generic filename (no date)
cat ~/3.htm >> /reports/ssh_failed_logins_report.txt

```



```

CentOS_Stream_9_VMG_LinuxVMImages.COM - VMware Workstation
File Edit View VM Tabs Help
CentOS_Stream_9_VMG_L... x kali-linux-2025.2-vmware-am... x
Activities Terminal
Oct 6 16:24
root@centosstream9:/home/centos

# Remove temporary files
rm -f ~/1.txt ~/2.txt ~/3.htm

# Process each IP address
while read -r ip; do
    whois_info=$(whois "$ip" | grep -i "country" | tr "a-z" "A-Z" | sort -u)

    echo "$ip" >> ~/1.txt
    if [[ -n "$whois_info" ]]; then
        echo "$whois_info" >> ~/1.txt
    else
        echo "COUNTRY: UNKNOWN" >> ~/1.txt
    fi
done < ~/ssh_failed_sortlisted_logins.txt

# Print a message to '3.htm'
echo -e "You were attacked by all IPs below from the following countries:" > ~/3.htm
cat ~/1.txt >> ~/3.htm

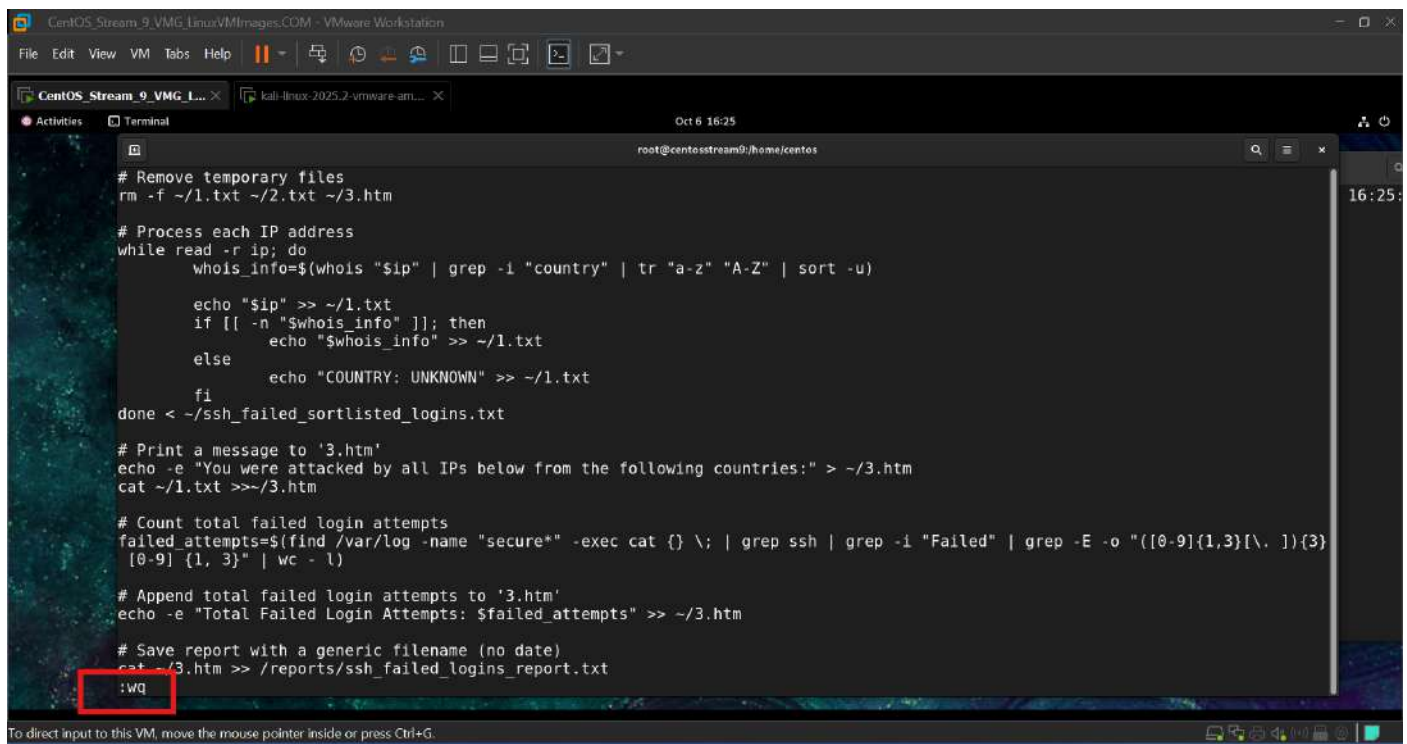
# Count total failed login attempts
failed_attempts=$(find /var/log -name "secure*" -exec cat {} \; | grep ssh | grep -i "Failed" | grep -E -o "([0-9]{1,3}[\. ]){3} [0-9] {1, 3}" | wc -l)

# Append total failed login attempts to '3.htm'
echo -e "Total Failed Login Attempts: $failed_attempts" >> ~/3.htm

# Save report with a generic filename (no date)
cat ~/3.htm >> /reports/ssh_failed_logins_report.txt
-- INSERT --
35,53 Bot

```

2.22 Saved and Exited the File by Pressing **Esc**, **Shift + Colon (:)**, Typing **wq**, and then **Enter**



```
CentOS_Stream_9_VMG_LinuxVMImages.COM - VMware Workstation
File Edit View VM Tabs Help
CentOS_Stream_9_VMG_L... X kali-linux-2025.2-vmware-am... X
Activities Terminal
Oct 6 16:25
root@centosstream9:/home/centos

# Remove temporary files
rm -f ~/1.txt ~/2.txt ~/3.htm

# Process each IP address
while read -r ip; do
    whois_info=$(whois "$ip" | grep -i "country" | tr "a-z" "A-Z" | sort -u)

    echo "$ip" >> ~/1.txt
    if [[ -n "$whois_info" ]]; then
        echo "$whois_info" >> ~/1.txt
    else
        echo "COUNTRY: UNKNOWN" >> ~/1.txt
    fi
done < ~/ssh_failed_sortlisted_logins.txt

# Print a message to '3.htm'
echo -e "You were attacked by all IPs below from the following countries:" > ~/3.htm
cat ~/1.txt >> ~/3.htm

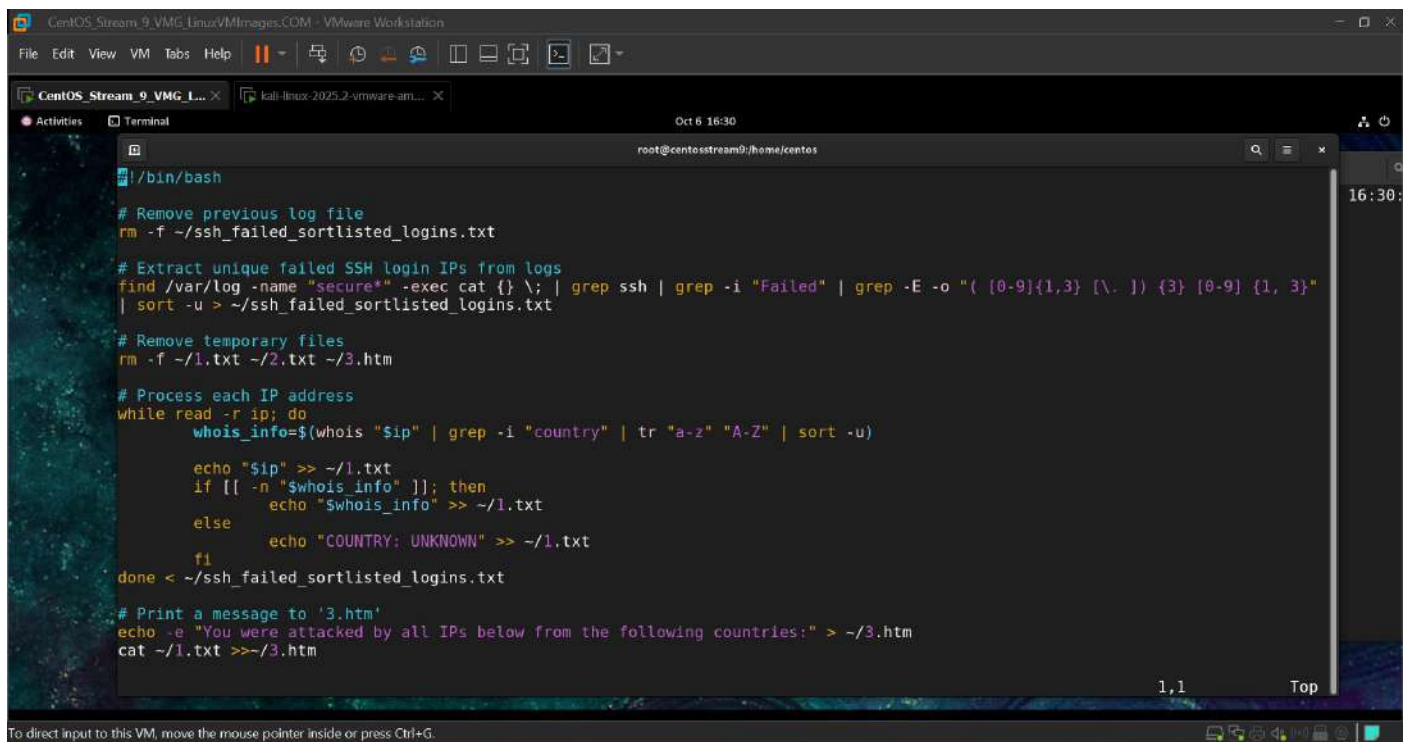
# Count total failed login attempts
failed_attempts=$(find /var/log -name "secure*" -exec cat {} \; | grep ssh | grep -i "Failed" | grep -E -o "([0-9]{1,3}[\. ]){3}([0-9]{1, 3})" | wc -l)

# Append total failed login attempts to '3.htm'
echo -e "Total Failed Login Attempts: $failed_attempts" >> ~/3.htm

# Save report with a generic filename (no date)
cat ~/3.htm >> /reports/ssh_failed_logins_report.txt

:WQ
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.



```
CentOS_Stream_9_VMG_LinuxVMImages.COM - VMware Workstation
File Edit View VM Tabs Help
CentOS_Stream_9_VMG_L... X kali-linux-2025.2-vmware-am... X
Activities Terminal
Oct 6 16:30
root@centosstream9:/home/centos

!/bin/bash

# Remove previous log file
rm -f ~/ssh_failed_sortlisted_logins.txt

# Extract unique failed SSH login IPs from logs
find /var/log -name "secure*" -exec cat {} \; | grep ssh | grep -i "Failed" | grep -E -o "([0-9]{1,3}[\. ]){3}([0-9]{1, 3})" | sort -u > ~/ssh_failed_sortlisted_logins.txt

# Remove temporary files
rm -f ~/1.txt ~/2.txt ~/3.htm

# Process each IP address
while read -r ip; do
    whois_info=$(whois "$ip" | grep -i "country" | tr "a-z" "A-Z" | sort -u)

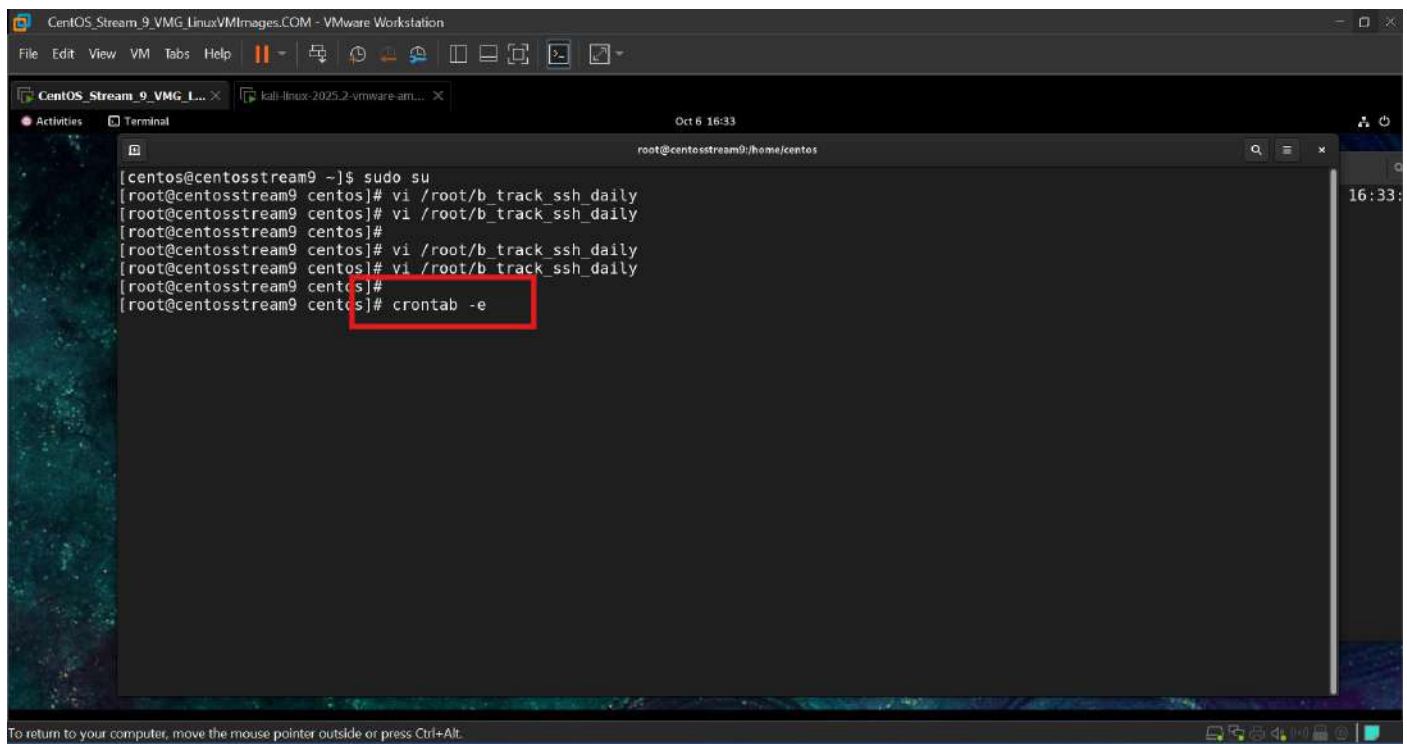
    echo "$ip" >> ~/1.txt
    if [[ -n "$whois_info" ]]; then
        echo "$whois_info" >> ~/1.txt
    else
        echo "COUNTRY: UNKNOWN" >> ~/1.txt
    fi
done < ~/ssh_failed_sortlisted_logins.txt

# Print a message to '3.htm'
echo -e "You were attacked by all IPs below from the following countries:" > ~/3.htm
cat ~/1.txt >> ~/3.htm
```

1,1 Top

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

2.23 Scheduled the script to run @regular interval
Used the command: **crontab -e**



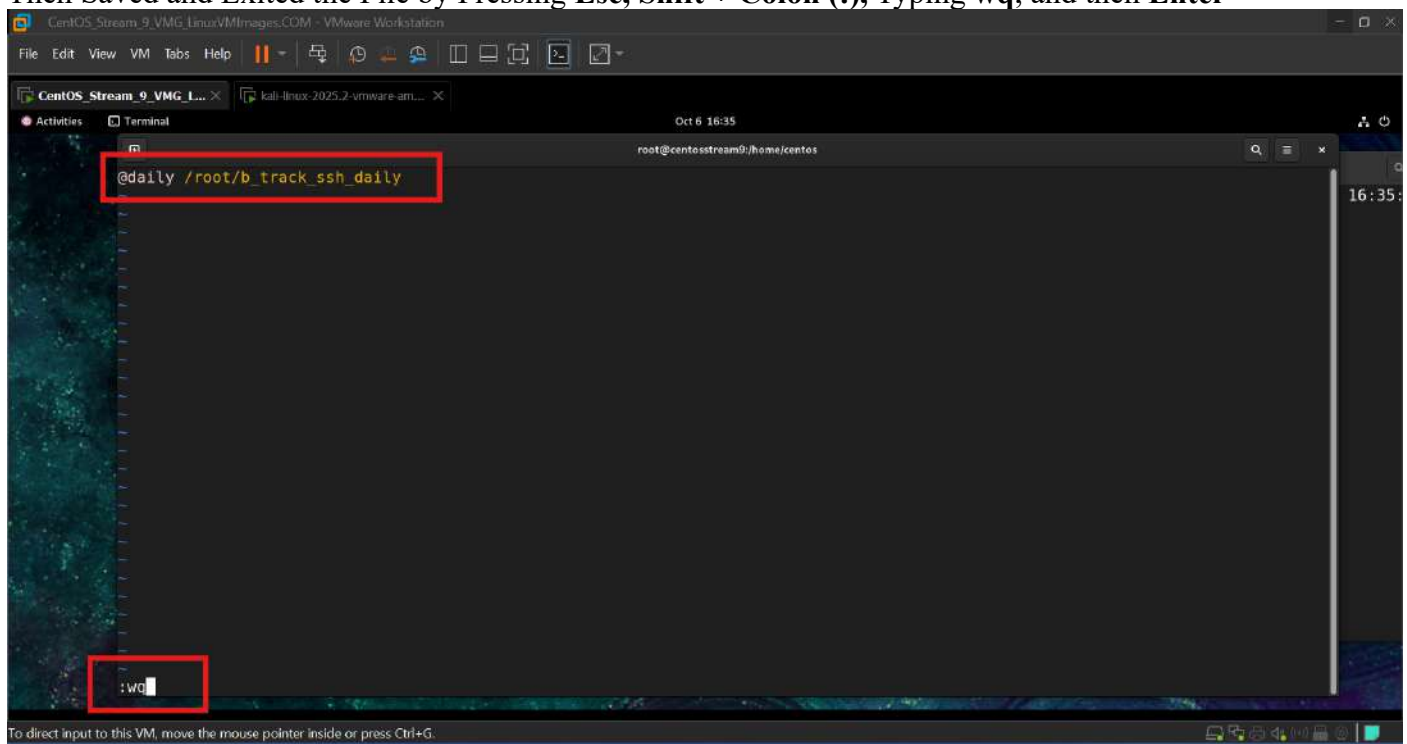
```
CentOS_Stream_9_VMG_LinuxVMImages.COM - VMware Workstation
File Edit View VM Tabs Help
CentOS_Stream_9_VMG_L... X kali-linux-2025.2-vmware-am... X
Activities Terminal
Oct 6 16:33
root@centosstream9:/home/centos

[centos@centosstream9 ~]$ sudo su
[root@centosstream9 centos]# vi /root/b_track_ssh_daily
[root@centosstream9 centos]# vi /root/b_track_ssh_daily
[root@centosstream9 centos]#
[root@centosstream9 centos]# vi /root/b_track_ssh_daily
[root@centosstream9 centos]# vi /root/b_track_ssh_daily
[root@centosstream9 centos]#
[root@centosstream9 centos]# crontab -e
```

2.24 Pressed i writing the script:

@daily /root/b_track_ssh_daily

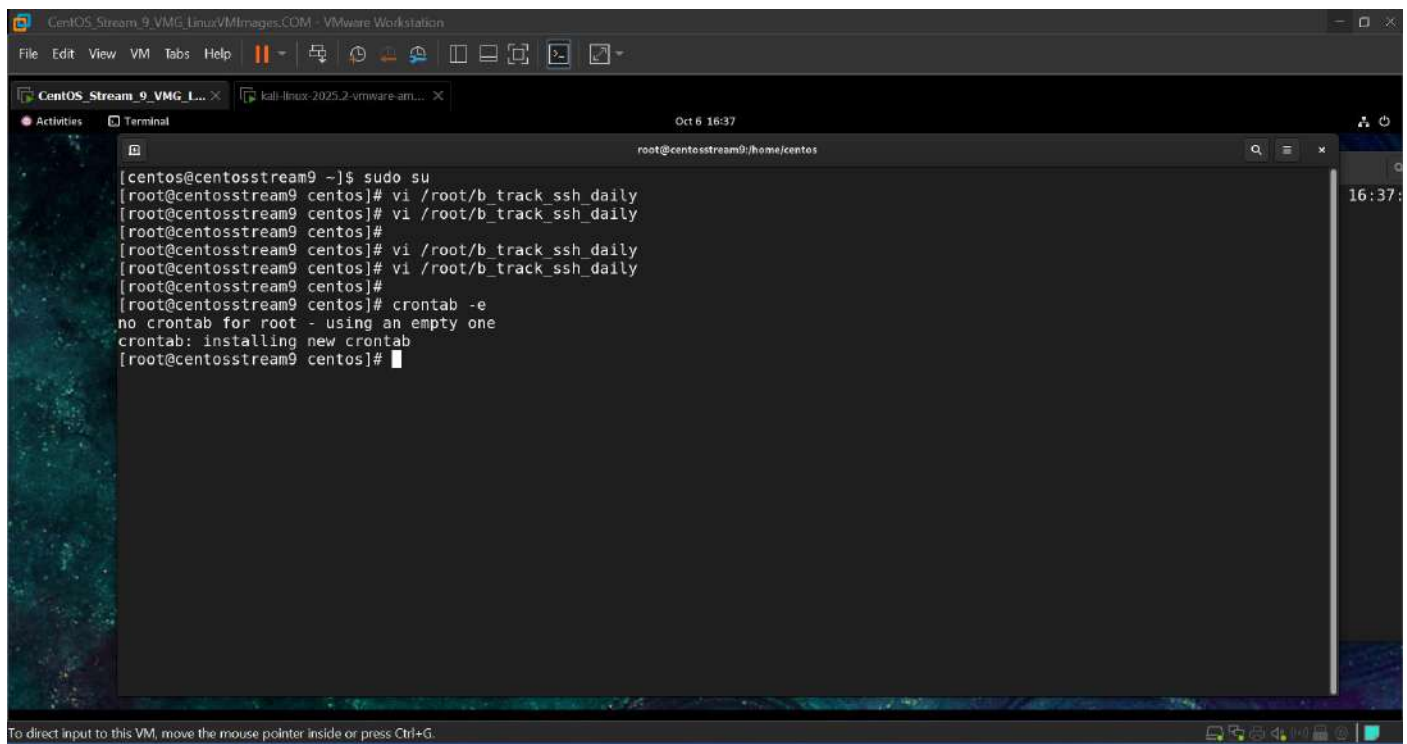
Then Saved and Exited the File by Pressing **Esc**, **Shift + Colon (:)**, Typing **wq**, and then **Enter**



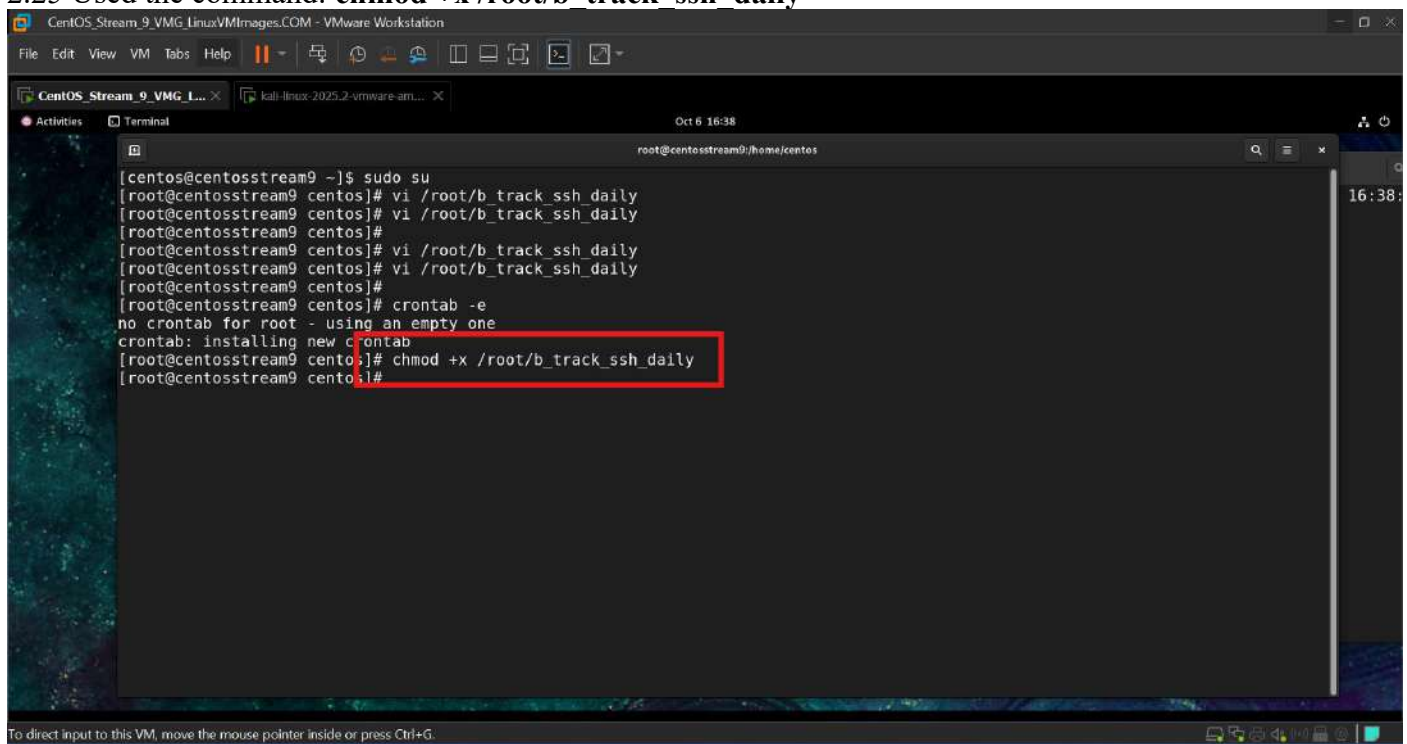
```
CentOS_Stream_9_VMG_LinuxVMImages.COM - VMware Workstation
File Edit View VM Tabs Help
CentOS_Stream_9_VMG_L... X kali-linux-2025.2-vmware-am... X
Activities Terminal
Oct 6 16:35
root@centosstream9:/home/centos

@daily /root/b_track_ssh_daily

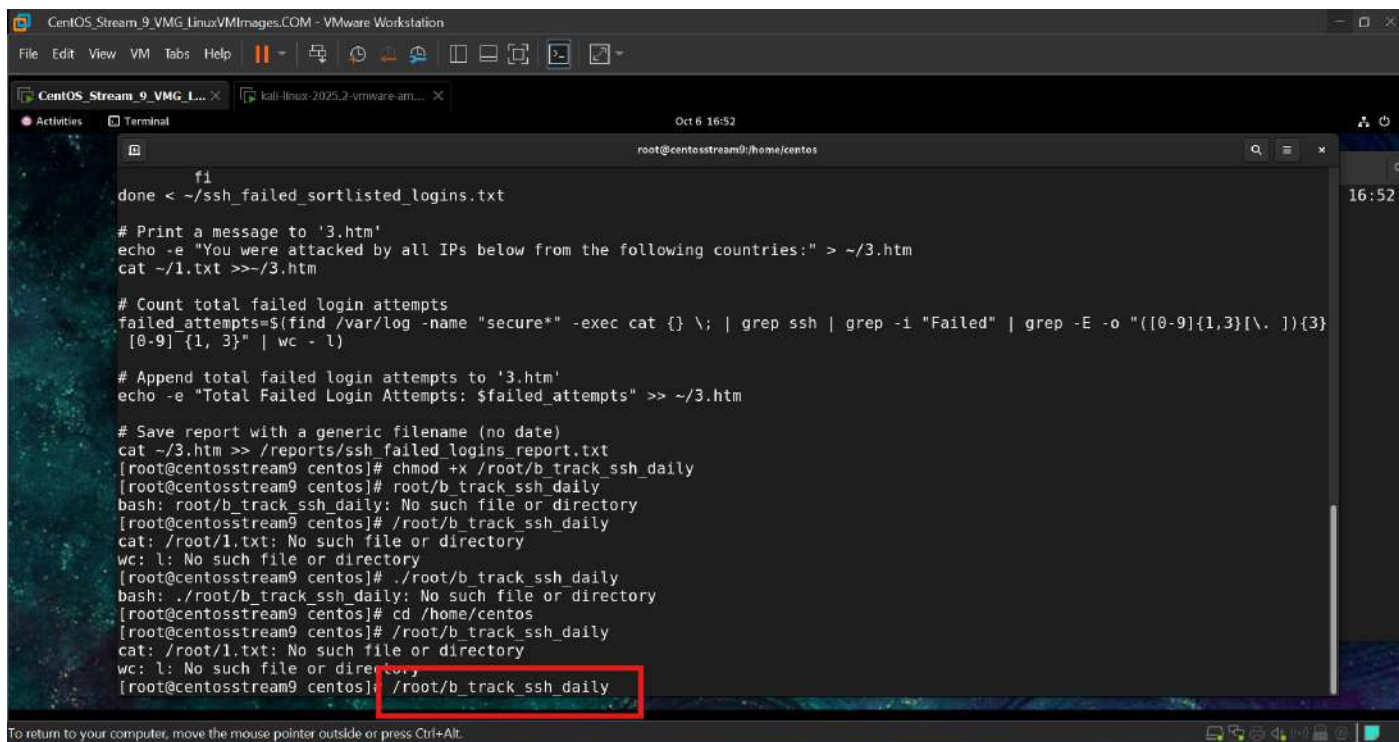
:wq
```



2.25 Used the command: **chmod +x /root/b_track_ssh_daily**



2.26 Used the command: **/root/b_track_ssh_daily**



```
CentOS_Stream_9_VMG_LinuxVMImages.COM - VMware Workstation
File Edit View VM Tabs Help
CentOS_Stream_9_VMG_L... kali-linux-2025.2-vmware-am...
Activities Terminal
Oct 6 16:52
root@centosstream9:/home/centos

f1
done < ~/ssh_failed_sortlisted_logins.txt

# Print a message to '3.htm'
echo -e "You were attacked by all IPs below from the following countries:" > ~/3.htm
cat ~/1.txt >>~/3.htm

# Count total failed login attempts
failed_attempts=$(find /var/log -name "secure*" -exec cat {} \; | grep ssh | grep -i "Failed" | grep -E -o "([0-9]{1,3}[\. ]){3}[0-9]{1, 3}" | wc - l)

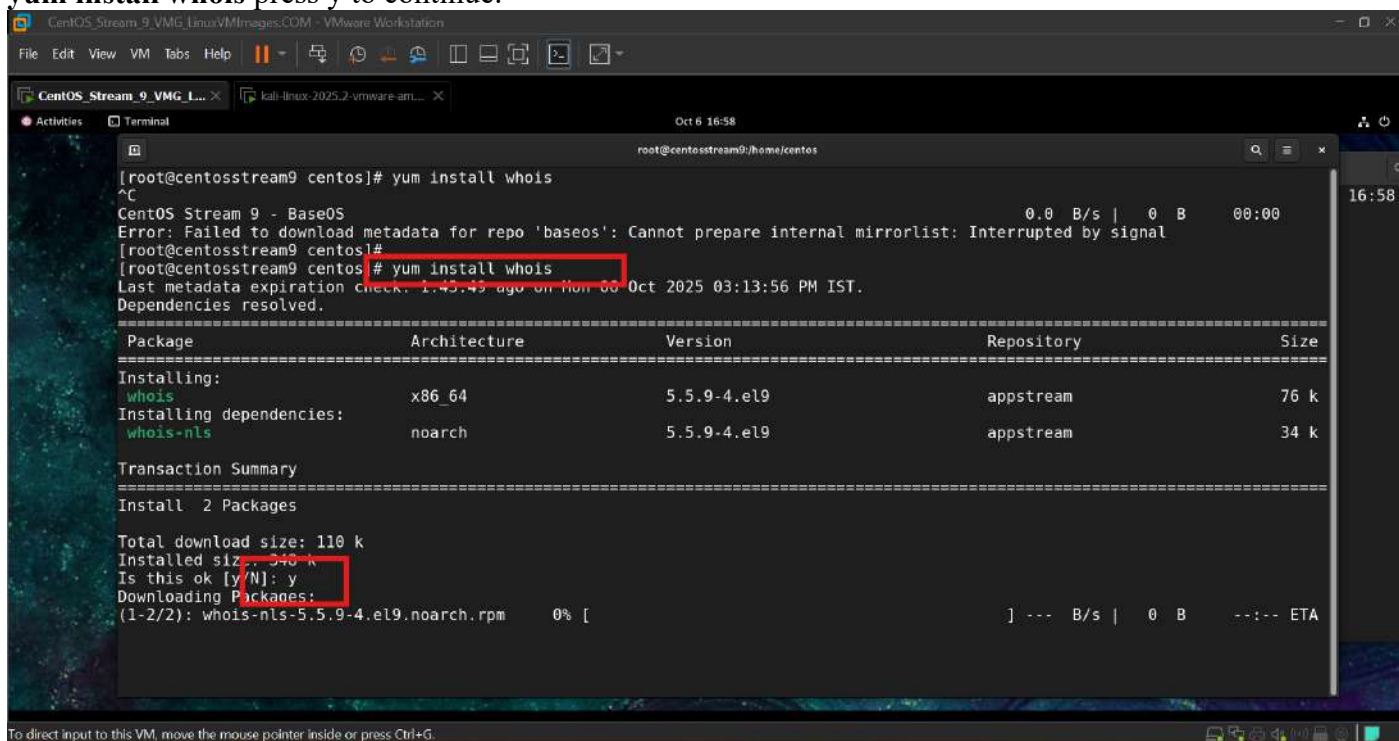
# Append total failed login attempts to '3.htm'
echo -e "Total Failed Login Attempts: $failed_attempts" >> ~/3.htm

# Save report with a generic filename (no date)
cat ~/3.htm >> /reports/ssh_failed_logins_report.txt
[root@centosstream9 centos]# chmod +x /root/b_track_ssh_daily
[root@centosstream9 centos]# root/b_track_ssh_daily
bash: root/b_track_ssh_daily: No such file or directory
[root@centosstream9 centos]# /root/b_track_ssh_daily
cat: /root/1.txt: No such file or directory
wc: l: No such file or directory
[root@centosstream9 centos]# ./root/b_track_ssh_daily
bash: ./root/b_track_ssh_daily: No such file or directory
[root@centosstream9 centos]# cd /home/centos
[root@centosstream9 centos]# /root/b_track_ssh_daily
cat: /root/1.txt: No such file or directory
wc: l: No such file or directory
[root@centosstream9 centos]# /root/b_track_ssh_daily
```

Task 3: Implementing IP Blocking:

Use of this script: If someone uses the wrong password 5 times, their IP address will not be blocked, but if someone tries 6 or more times or uses the brute force method, their IP address will be automatically blocked.

3.1 Opened the terminal in **CentOS** and install the **whois** using the command:
yum install whois press y to continue.

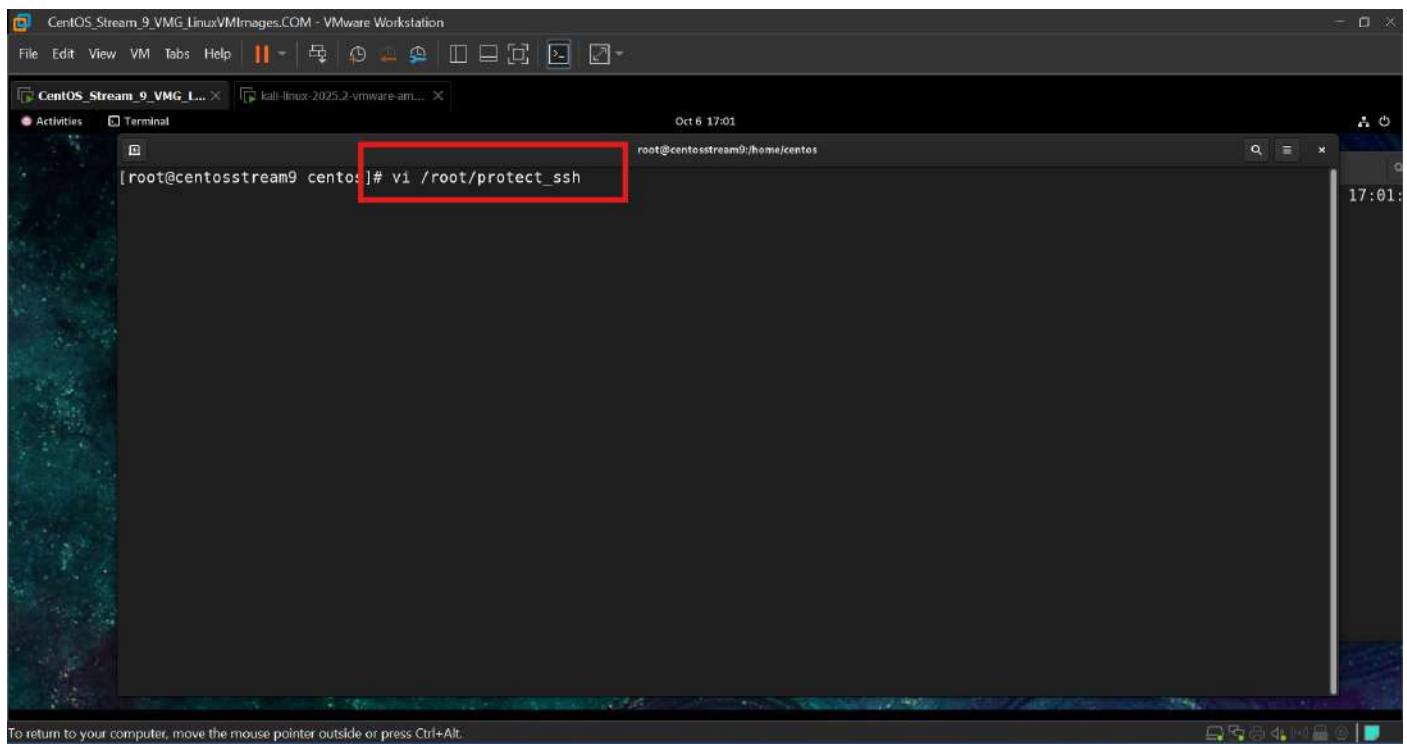


```
CentOS_Stream_9_VMG_LinuxVMImages.COM - VMware Workstation
File Edit View VM Tabs Help
CentOS_Stream_9_VMG_L... kali-linux-2025.2-vmware-am...
Activities Terminal
Oct 6 16:58
root@centosstream9:/home/centos

[root@centosstream9 centos]# yum install whois
^C
CentOS Stream 9 - BaseOS                                0.0 B/s | 0 B    00:00
Error: Failed to download metadata for repo 'baseos': Cannot prepare internal mirrorlist: Interrupted by signal
[root@centosstream9 centos]#
[root@centosstream9 centos]# yum install whois
Last metadata expiration check: 1:43:43 ago on Mon 06 Oct 2025 03:13:56 PM IST.
Dependencies resolved.
=====
Package                Architecture      Version           Size           Repository
=====
Installing:
whois                  x86_64            5.5.9-4.el9      76 k           appstream
Installing dependencies:
whois-nls              noarch            5.5.9-4.el9      34 k           appstream
=====
Transaction Summary
=====
Install 2 Packages

Total download size: 110 k
Installed size: 340 k
Is this ok [y/N]: y
Downloading Packages:
(1-2/2): whois-nls-5.5.9-4.el9.noarch.rpm    0% [
```

3.2 Edited the file using the command: **vi /root/protect_ssh**



3.3 Typed **i** to write this script:

```
#!/bin/bash
```

```
# Extract IPs that failed SSH login attempts
```

```
grep ssh /var/log/secure | grep -i Failed | grep -E -o "([0-9]{1,3}[.]){3}[0-9]{1,3}" | sort -u > /tmp/blacklist
```

```
while read ip; do
```

```
    # Count failed attempts for each IP
```

```
    attempts=$(grep -c "$ip" /var/log/secure)
```

```
    if [ "$attempts" -ge 10 ]; then
```

```
        echo "Blocking IP: $ip"
```

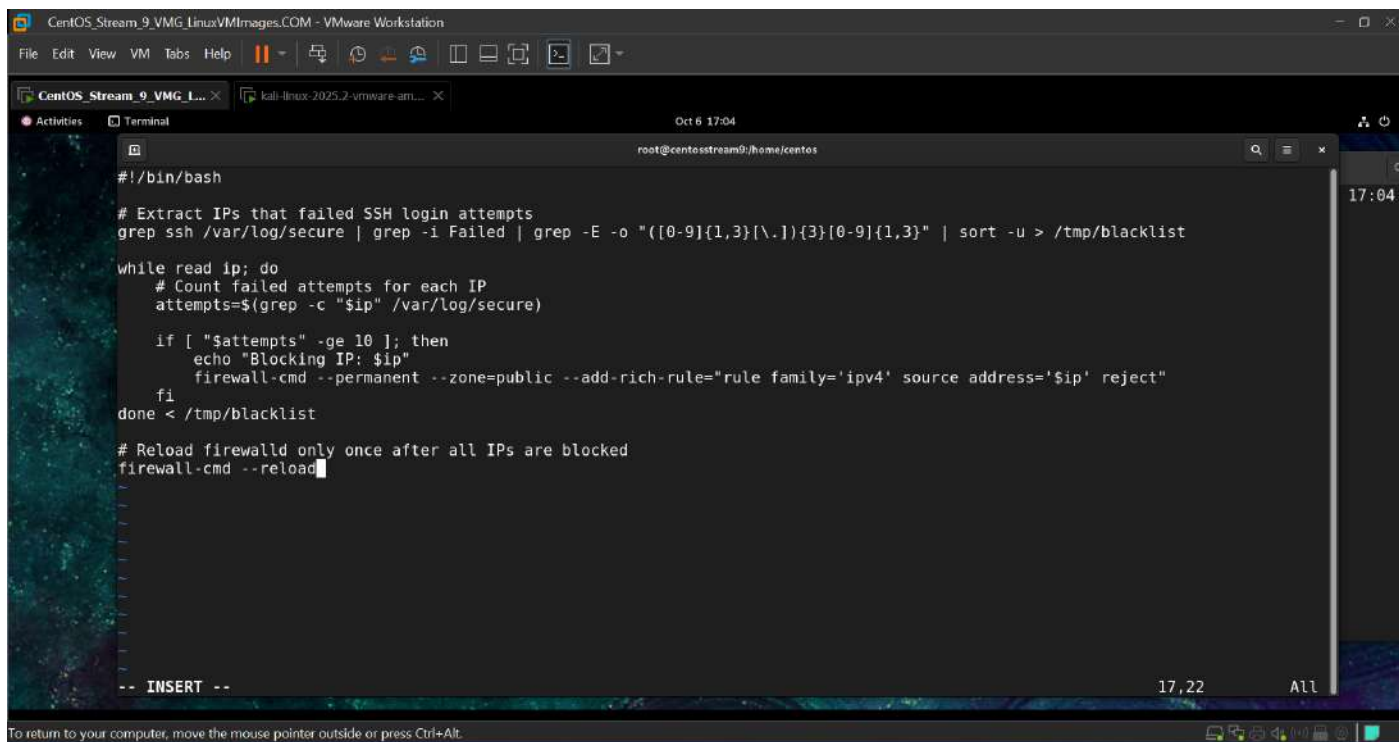
```
        firewall-cmd --permanent --zone=public --add-rich-rule="rule family='ipv4' source address='$ip' reject"
```

```
    fi
```

```
done < /tmp/blacklist
```

```
# Reload firewalld only once after all IPs are blocked
```

```
firewall-cmd --reload
```



The screenshot shows a terminal window titled "CentOS_Stream_9_VMG_LinuxVMImages.COM - VMware Workstation". The terminal is running a script to block SSH login attempts. The script uses `grep` to extract failed login attempts from `/var/log/secure` and sorts them into a file `/tmp/blacklist`. It then uses a `while` loop to read each IP from the blacklist, counts the failed attempts, and if there are 10 or more, it adds a firewall rule to reject the IP. Finally, it reloads the firewall.

```
#!/bin/bash

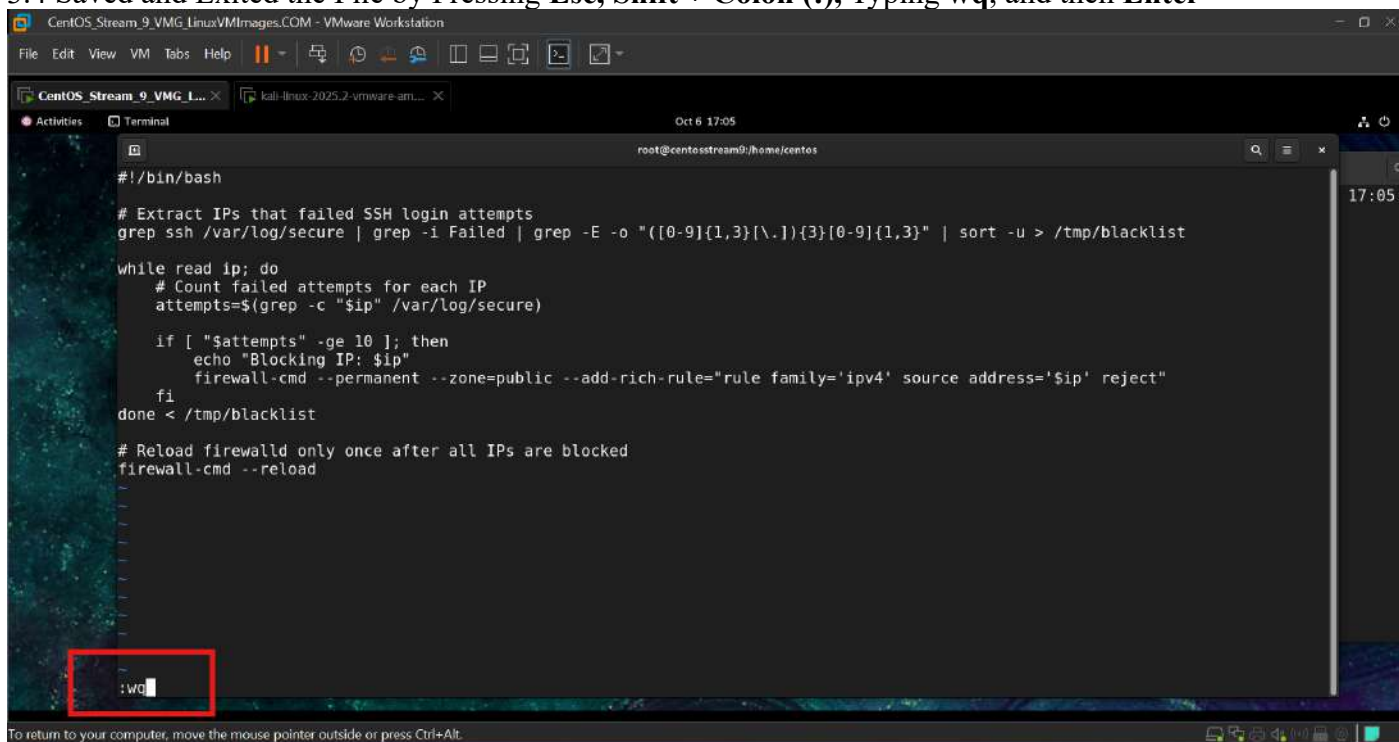
# Extract IPs that failed SSH login attempts
grep ssh /var/log/secure | grep -i Failed | grep -E -o "([0-9]{1,3}[.]){3}[0-9]{1,3}" | sort -u > /tmp/blacklist

while read ip; do
    # Count failed attempts for each IP
    attempts=$(grep -c "$ip" /var/log/secure)

    if [ "$attempts" -ge 10 ]; then
        echo "Blocking IP: $ip"
        firewall-cmd --permanent --zone=public --add-rich-rule="rule family='ipv4' source address='$ip' reject"
    fi
done < /tmp/blacklist

# Reload firewalld only once after all IPs are blocked
firewall-cmd --reload
```

3.4 Saved and Exited the File by Pressing **Esc**, **Shift + Colon (:)**, Typing **wq**, and then **Enter**



This screenshot shows the same terminal window as the previous one, but now the script has finished executing. The user has pressed `Esc`, `Shift + Colon (:)`, typed `wq`, and pressed `Enter` to save the file and exit the editor. The prompt is now `~ :wq`, which is highlighted with a red box.

```
#!/bin/bash

# Extract IPs that failed SSH login attempts
grep ssh /var/log/secure | grep -i Failed | grep -E -o "([0-9]{1,3}[.]){3}[0-9]{1,3}" | sort -u > /tmp/blacklist

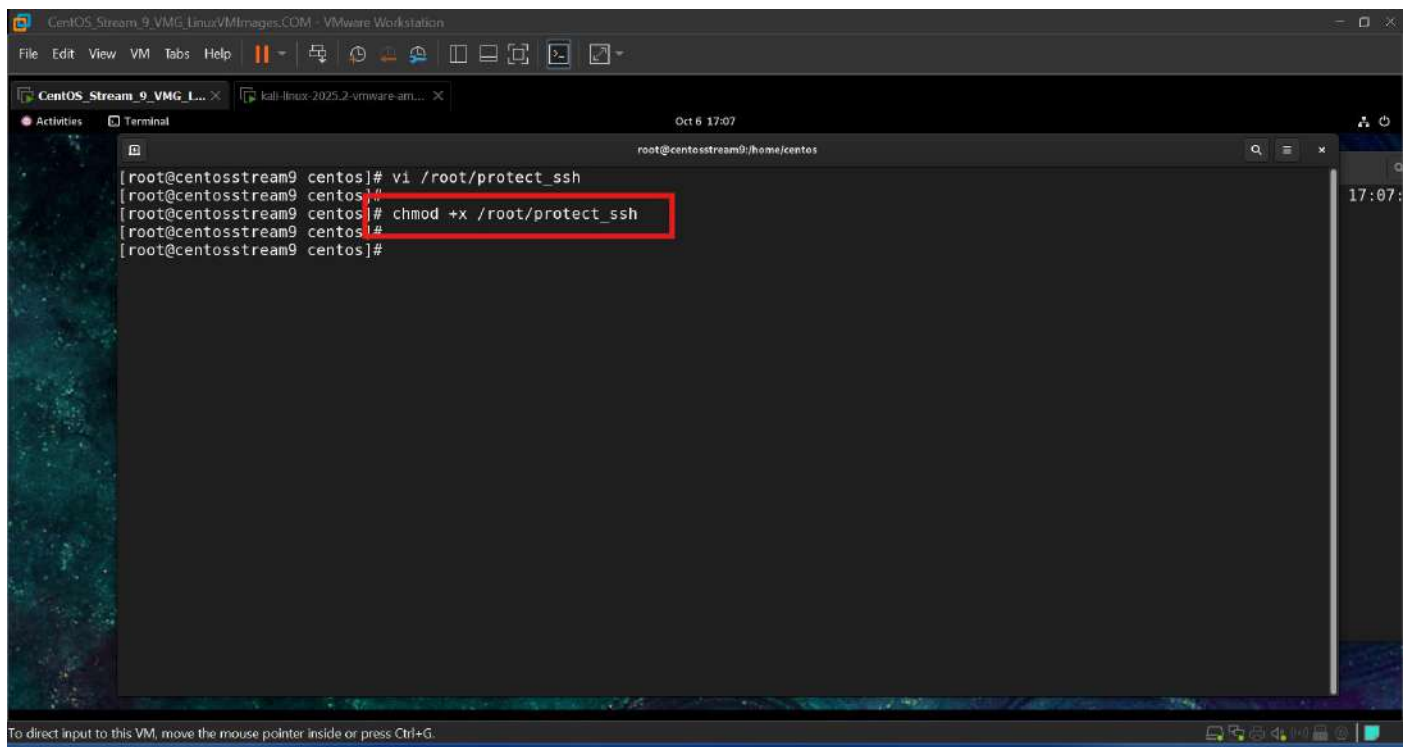
while read ip; do
    # Count failed attempts for each IP
    attempts=$(grep -c "$ip" /var/log/secure)

    if [ "$attempts" -ge 10 ]; then
        echo "Blocking IP: $ip"
        firewall-cmd --permanent --zone=public --add-rich-rule="rule family='ipv4' source address='$ip' reject"
    fi
done < /tmp/blacklist

# Reload firewalld only once after all IPs are blocked
firewall-cmd --reload

~ :wq
```

3.5 Used the command: **chmod +x /root/protect_ssh**



CentOS_Stream_9_VMG_LinuxVMImages.COM - VMware Workstation

File Edit View VM Tabs Help

CentOS_Stream_9_VMG_L... X kali-linux-2025.2-vmware-am... X

Activities Terminal

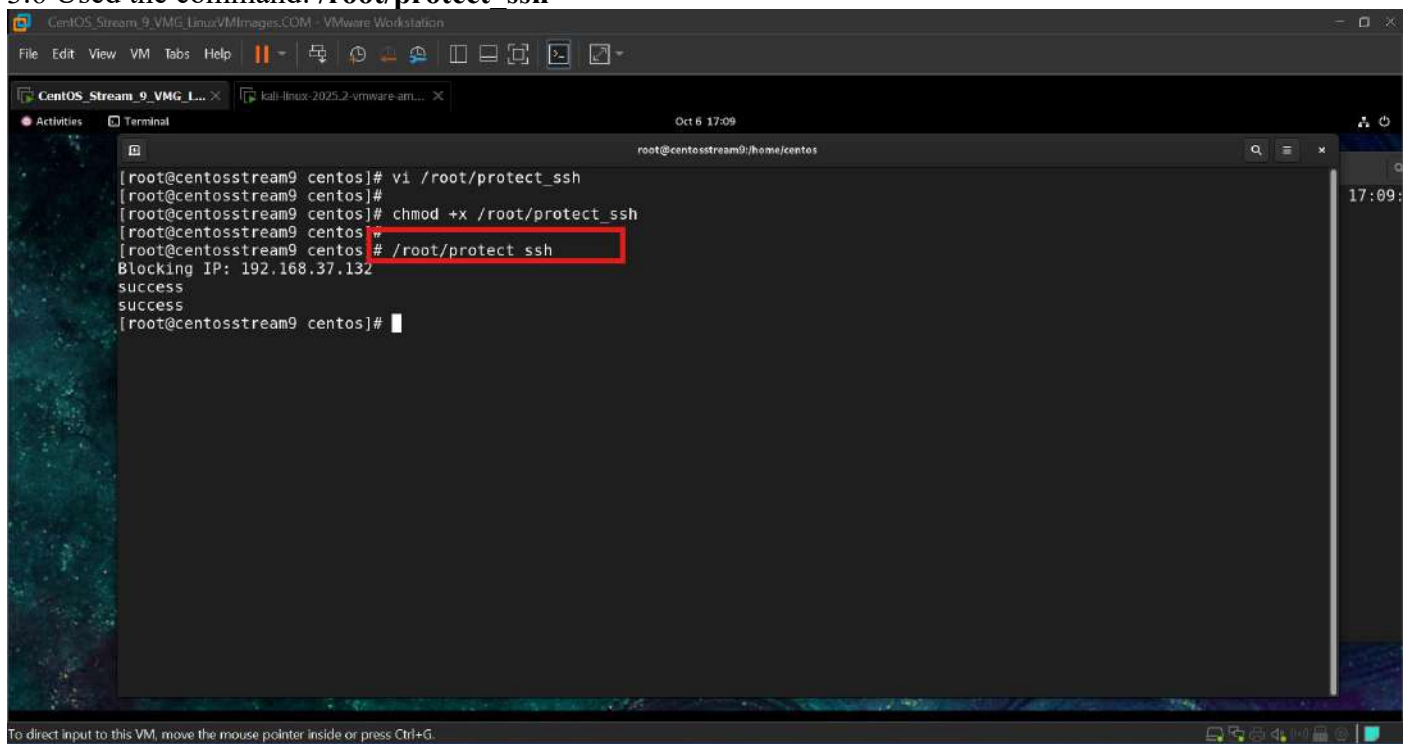
Oct 6 17:07

root@centosstream9:/home/centos

```
[root@centosstream9 centos]# vi /root/protect_ssh
[root@centosstream9 centos]#
[root@centosstream9 centos]# chmod +x /root/protect_ssh
[root@centosstream9 centos]#
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

3.6 Used the command: `/root/protect_ssh`



CentOS_Stream_9_VMG_LinuxVMImages.COM - VMware Workstation

File Edit View VM Tabs Help

CentOS_Stream_9_VMG_L... X kali-linux-2025.2-vmware-am... X

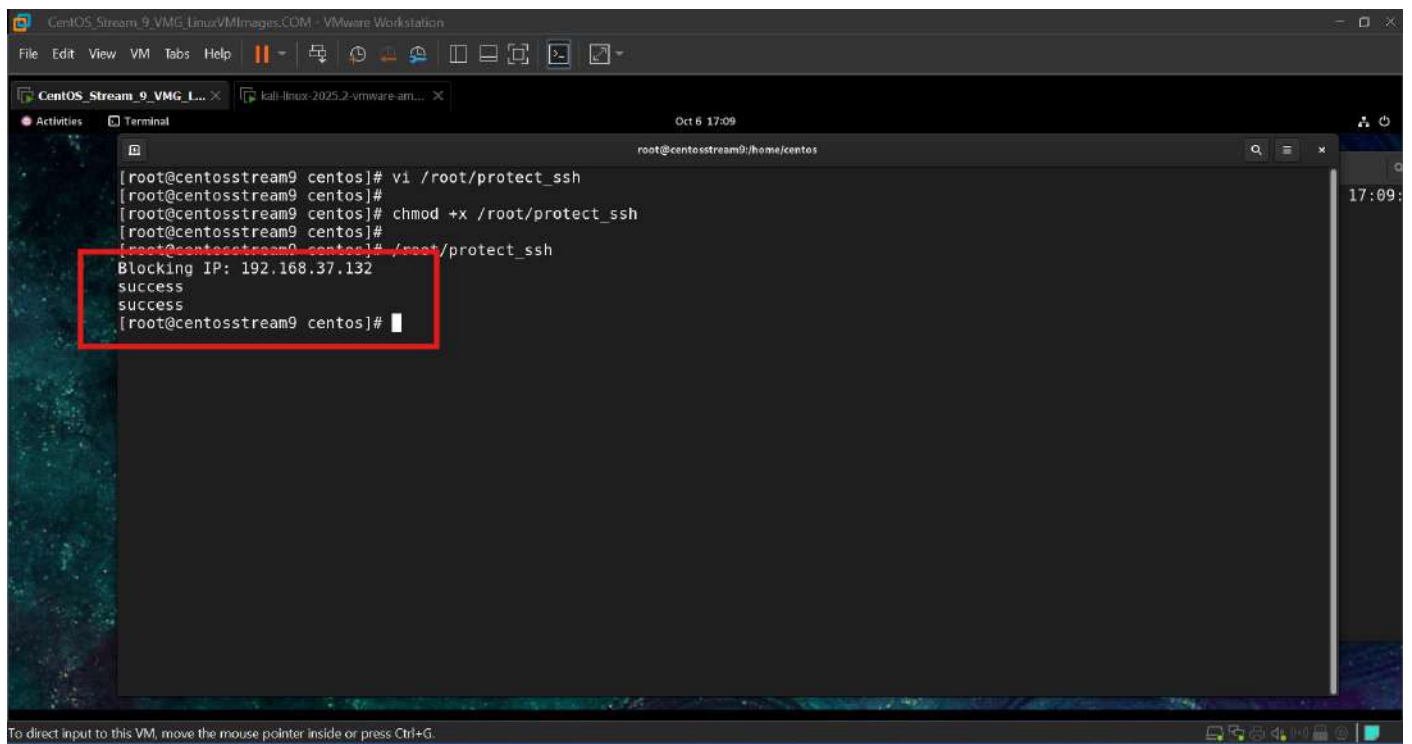
Activities Terminal

Oct 6 17:09

root@centosstream9:/home/centos

```
[root@centosstream9 centos]# vi /root/protect_ssh
[root@centosstream9 centos]#
[root@centosstream9 centos]# chmod +x /root/protect_ssh
[root@centosstream9 centos]# /root/protect_ssh
Blocking IP: 192.168.37.132
success
success
[root@centosstream9 centos]#
```

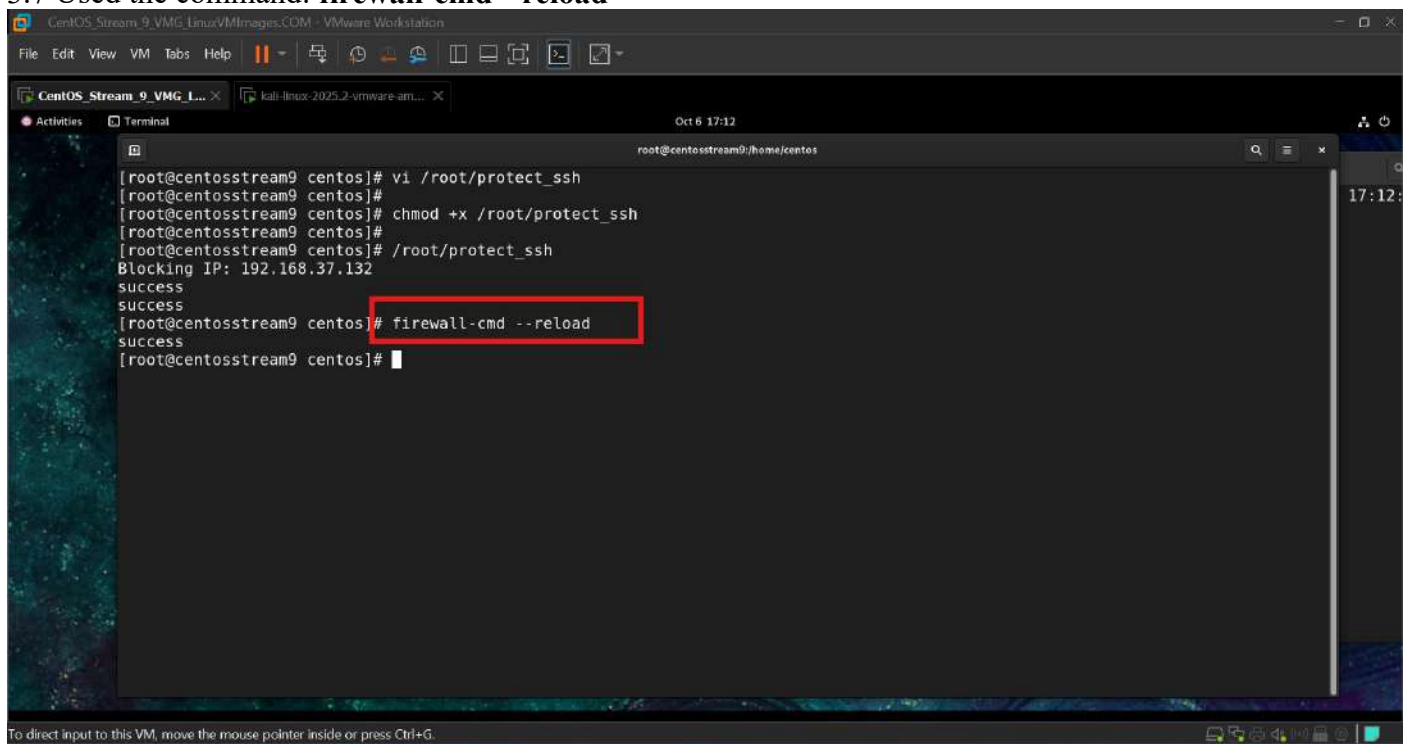
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.



The screenshot shows a terminal window within a VMware Workstation environment. The terminal is running a CentOS VM. The user is at the root prompt and has executed the following commands: `vi /root/protect_ssh`, `chmod +x /root/protect_ssh`, and `/root/protect_ssh`. The output of the script shows that the IP address 192.168.37.132 has been blocked successfully. The last line of the terminal output is `[root@centosstream9 centos]#`, which is highlighted with a red box.

```
[root@centosstream9 centos]# vi /root/protect_ssh
[root@centosstream9 centos]#
[root@centosstream9 centos]# chmod +x /root/protect_ssh
[root@centosstream9 centos]# /root/protect_ssh
Blocking IP: 192.168.37.132
success
success
[root@centosstream9 centos]#
```

3.7 Used the command: **firewall-cmd --reload**



The screenshot shows the same terminal window as the previous one. The user has executed the `firewall-cmd --reload` command, which is highlighted with a red box. The output of the command is `success`.

```
[root@centosstream9 centos]# vi /root/protect_ssh
[root@centosstream9 centos]#
[root@centosstream9 centos]# chmod +x /root/protect_ssh
[root@centosstream9 centos]# /root/protect_ssh
Blocking IP: 192.168.37.132
success
success
[root@centosstream9 centos]# firewall-cmd --reload
success
[root@centosstream9 centos]#
```

3.8 Use the command: **firewall-cmd --list-all**

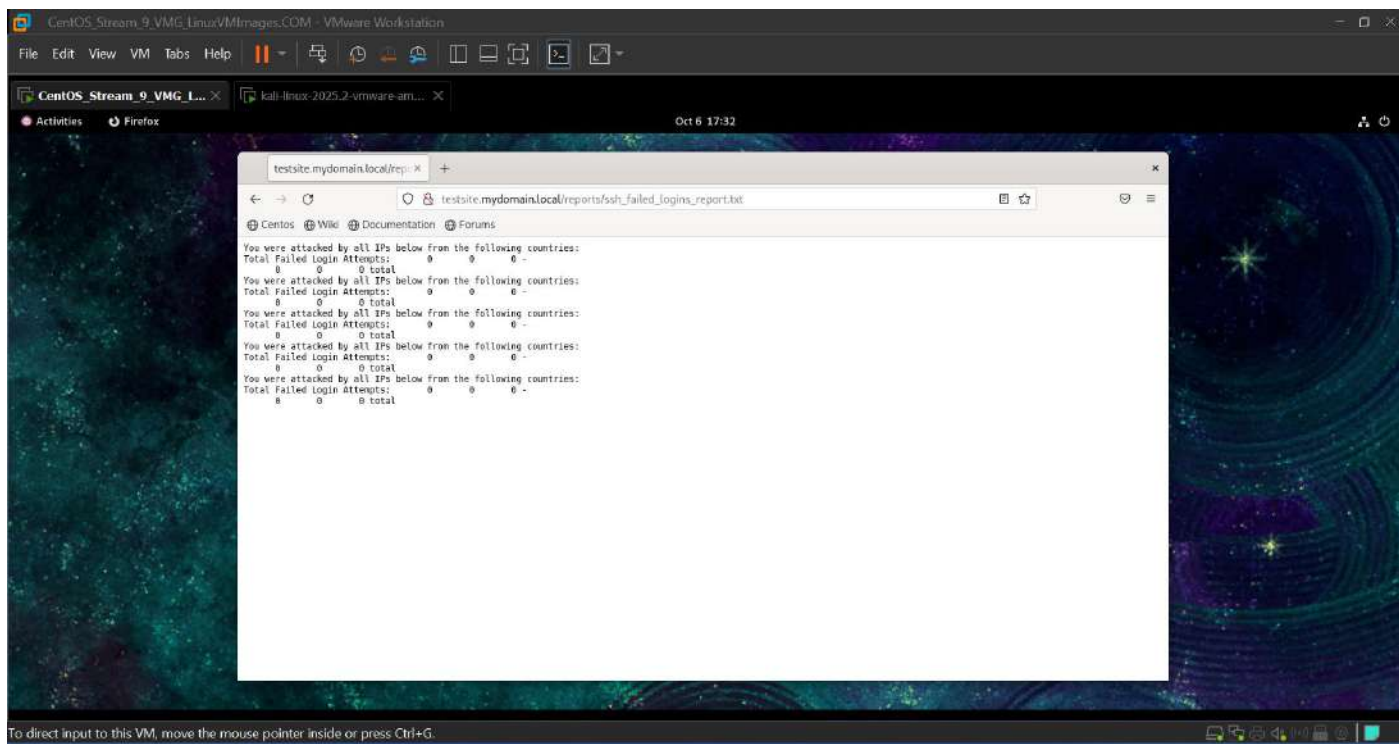
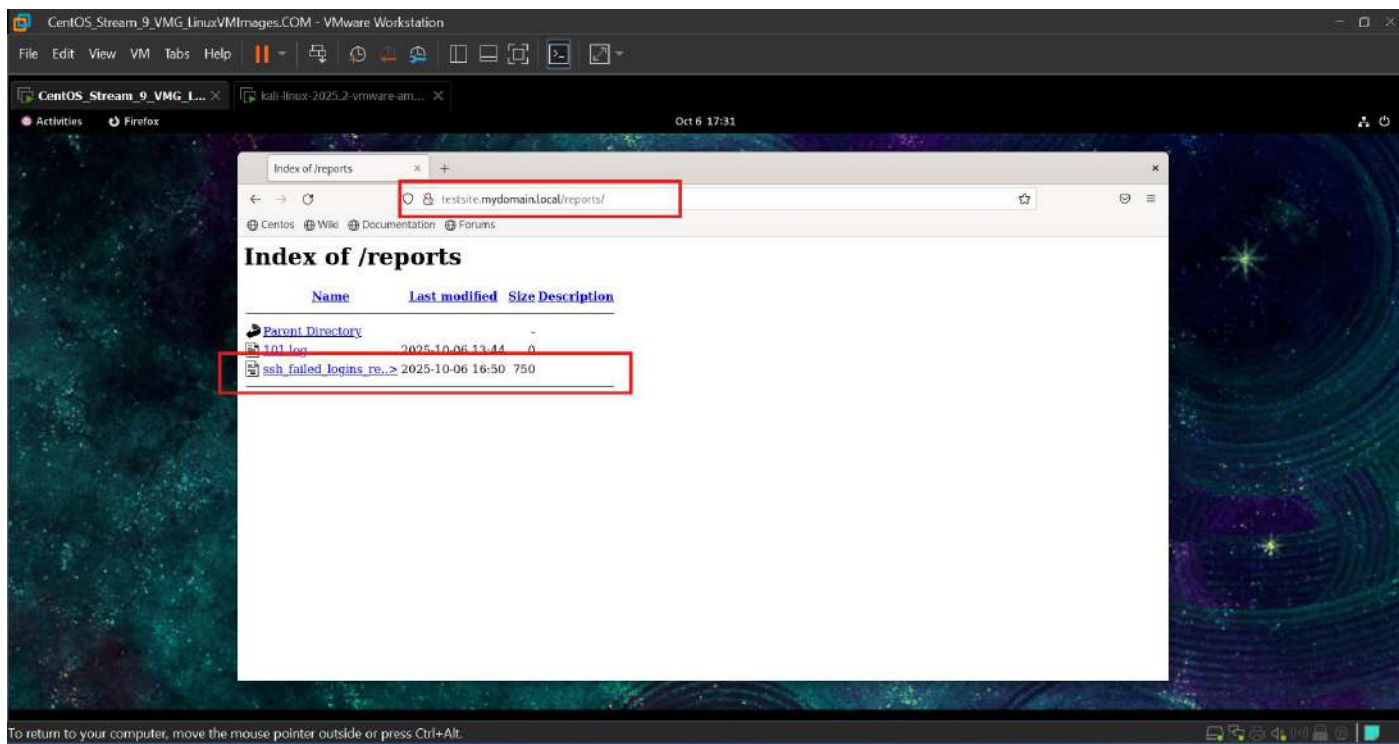
The screenshot shows a terminal window within a VMware Workstation environment. The terminal is running as root on a CentOS Stream 9 VM. The user has executed several commands to configure the firewall: `vi /root/protect_ssh`, `chmod +x /root/protect_ssh`, and `/root/protect_ssh`. The output of the last command shows that the IP 192.168.37.132 is blocked successfully. Then, the user runs `firewall-cmd --reload` and `firewall-cmd --list-all`. The output of `firewall-cmd --list-all` is highlighted with a red box and shows the following configuration: `public (active)`, `target: default`, `icmp-block-inversion: no`, `interfaces: ens33`, `sources:`, `services: cockpit dhcpv6-client ssh`, `ports: 2222/tcp 80/tcp 20/tcp 21/tcp`, `protocols:`, `forward: yes`, `masquerade: no`, `forward-ports:`, `source-ports:`, `icmp-blocks:`, `rich rules:`, and `rule family="ipv4" source address="192.168.37.132" reject`.

```
[root@centosstream9 centos]# vi /root/protect_ssh
[root@centosstream9 centos]#
[root@centosstream9 centos]# chmod +x /root/protect_ssh
[root@centosstream9 centos]#
[root@centosstream9 centos]# /root/protect_ssh
Blocking IP: 192.168.37.132
success
success
[root@centosstream9 centos]# firewall-cmd --reload
success
[root@centosstream9 centos]# firewall-cmd --list-all
public (active)
target: default
icmp-block-inversion: no
interfaces: ens33
sources:
services: cockpit dhcpv6-client ssh
ports: 2222/tcp 80/tcp 20/tcp 21/tcp
protocols:
forward: yes
masquerade: no
forward-ports:
source-ports:
icmp-blocks:
rich rules:
rule family="ipv4" source address="192.168.37.132" reject
[root@centosstream9 centos]#
```

3.12 Opened the Firefox in centos and navigate to <http://testsite.mydomain.local/reports/> And clicked on ssh failed logins to view the blocked IP address.

The screenshot shows the same terminal window as before, but now the user has executed the command `firefox http://testsite.mydomain.local/reports/`, which is highlighted with a red box. The terminal is still running as root on the CentOS Stream 9 VM.

```
[root@centosstream9 centos]# firefox http://testsite.mydomain.local/reports/
```



3.13 Again navigating to Kali Linux and trying to connect with ssh using the command:
ssh -p 2222 military@192.168.37.136

Now we can see that the connection is refused.

